

QUANTITATIVE REASONING-I

PROGRAM: BS EARLY CHILDHOOD CARE AND EDUCATION
(BS-ECCE)

Code: 8277

Unit 1–9



Early Childhood Education & Elementary Teacher Education Department
FACULTY OF EDUCATION

ALLAMA IQBAL OPEN UNIVERSITY ISLAMABAD

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Islamabad**

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FOREWORD

Welcome to this course on quantitative reasoning where we will unlock the power of numbers and develop essential skills for making informed decisions in today's data-driven world. Quantitative reasoning is not just about using numbers, it is also about understanding them, interpreting them, and using them to solve real-world problems. In this course, the fundamental principles of quantitative reasoning will equip the students with the tools, which are needed to analyze data, identify patterns and draw meaningful conclusions.

Throughout the course, students would learn to think critically, solve the problems, communicate effectively, make informed decisions, navigate uncertainty, and utilize tools and technology. These skills would involve developing a critical mindset toward data and statistics, learning to question assumptions and evaluating the validity of arguments based on numerical evidence. The problem-solving skill involves tackling complex problems by breaking them down into manageable components and applying quantitative techniques to analyze data and formulate effective solutions. Effective communication deals with the art of communicating quantitative information clearly and persuasively, whether through written reports, presentations, or visualizations. Informed decision-making involves using quantitative reasoning to make informed decisions in various contexts, from personal finance to business strategy to public policy. Navigating the uncertainty deals with understanding the role of uncertainty in quantitative analysis and learning techniques for dealing with it such as probability theory and statistical inference. Familiarity with common tools and technologies used in quantitative analysis, such as spreadsheets, statistical software, and data visualization tools is quite helpful in quantitative reasoning.

This course would guide you in a step-by-step manner for using quantitative reasoning in diverse contexts. After completing this course, you would develop the skills to identify the patterns in the data and draw meaningful insights from the data. The skills developed through this course would not only be beneficial in your academic and professional journey but it would be beneficial in the daily life activities.

Prof. Dr. Fazal Ur Rahman
Dean, Faculty of Education

INTRODUCTION OF THE COURSE

In this course on Quantitative Reasoning-I, we embark on a journey to develop the essential skills in understanding, interpreting, and analyzing quantitative information. Quantitative reasoning is a fundamental aspect of problem-solving across the various disciplines, from mathematics and science to social sciences and beyond. It empowers individuals to make informed decisions, evaluate arguments, and navigate complex real-world scenarios with confidence.

This course consists of nine units. Throughout this course, we will delve into key concepts and techniques necessary for effective quantitative reasoning. Whether you are a novice seeking to build a strong foundation or an adept learner looking to refine your skills, this course is designed to accommodate learners at all levels of proficiency.

The objectives of this course are as follows:

1. Building a solid understanding of basic mathematical principles and operations.
2. Developing critical thinking skills to interpret and evaluate quantitative data.
3. Exploring real-world applications of quantitative reasoning in diverse fields such as finance, science, economics, and more.
4. Enhancing problem-solving abilities through practical exercises and case studies.
5. Fostering a growth mindset towards quantitative reasoning, encouraging curiosity, creativity, and resilience in tackling challenges.

Throughout the course, learning material, interactive lectures, hands-on activities, and self-assessment questions will foster an engaging and dynamic learning environment. After completing this course, you will be equipped with the tools and confidence to approach quantitative problems with clarity, precision, and insight.

Dr Mubeshera Tufail
Course Coordinator

OBJECTIVES OF THE COURSE

After the successful completion of the course, you would be able to:

1. Define the term quantitative reasoning.
2. Explain the components of quantitative reasoning.
3. Interpret the applications of quantitative reasoning in daily life.
4. Analyze real-world problems to identify the most suitable problem-solving approach based on the nature of the problem and available resources.
5. Apply analytical problem-solving techniques to break down complex problems into smaller, more manageable components, facilitating systematic problem-solving processes.
6. Understand the fundamental concepts of numbers and their significance in daily life.
7. Develop critical thinking and problem-solving abilities through numerical reasoning and analysis.
8. Create and maintain a personal budget that reflects income, expenses, and financial priorities.
9. Demonstrate proficiency in basic financial calculations, including calculating interest, budgeting for expenses, and determining savings goals.
10. Develop a fundamental understanding of geometric concepts such as points, lines, angles, and shapes.
11. Identify and classify different types of geometric figures, including polygons, circles, and solids.
12. Demonstrate proficiency in measuring and calculating geometric properties, such as length, area, perimeter, volume, and surface area.
13. Understand the fundamental concepts of sets, including elements, subsets, unions, intersections, and complements.
14. Identify and analyze patterns in algebraic expressions, including variables, constants, coefficients, and terms.

15. Understand the fundamental characteristics of statistical studies, including populations, samples, variables, and data collection methods.
16. Differentiate between the various types of statistical studies, including observational studies, experiments, surveys, and retrospective studies.
17. Understand the fundamental concepts of descriptive statistics, including measures of central tendency, variability, and distribution.
18. Calculate and interpret measures of central tendency, such as the mean, median, and mode, to summarize the typical value of a dataset.
19. Understand the importance of data visualization and the role of tables and graphs in organizing and summarizing data effectively.
20. Interpret tables to organize and present data in a clear and systematic manner.
21. Apply tables and graphs effectively to analyze and interpret data, identify patterns and trends, and communicate findings to stakeholders or audiences in a clear and concise manner.

TABLE OF CONTENTS

	<i>Page #</i>
Foreword	iv
Introduction of the Course	v
Objectives of the Course	vi
Unit–1: Quantitative Reasoning.....	1
Unit–2: Approaches to Problem Solving	13
Unit–3: Use of Numbers in Daily Life.....	37
Unit–4: Managing Money	63
Unit–5: Fundamentals of Geometry	89
Unit–6: Sets and Algebraic Expressions	101
Unit–7: Introduction to Statistical Studies	117
Unit–8: Descriptive Statistics.....	133
Unit–9: Using Tables & Graphs for Data	149

Unit–1

QUANTITATIVE REASONING

Written by: Dr. Mubeshera Tufail

Reviewed by: Ms. Nazish

INTRODUCTION

We encounter problems in daily life. These problems may involve various characteristics, dimensions, shapes, qualities and quantities. In order to solve it effectively, it is important to understand the nature of the problem and notice the key information related to it. For example, it can be the income and expenses of a person for the past month or the daily values of blood pressure of a patient for one week, or the rate of Pakistani Rupee with respect to US Dollars in the past few months. This information contains some evidence that may be useful to estimate the nature of the problem, make a decision based on this information, or give a reason for that pattern of values. For example, making a budget for a person for the coming month for his/her income and expenses, advice from a doctor about the health and diet of the patient, and estimating the value of the Pakistani Rupee for the coming month. This information can be used to make decisions, and solve or avoid a problem in the future.

The problems mentioned above either involve a quantity or a pattern of values over a period of time. If a person can learn and use this information, she/he can be in a position to make better decisions. Quantitative reasoning helps you to identify these types of information and use it for decision-making. However, there are certain misconceptions about quantitative reasoning that need to be addressed. For example, Mathematics or quantitative reasoning is useful to only a certain group of people or it is too difficult to learn. So, in this unit we will learn about the concept of quantitative reasoning, the differences between Mathematics and quantitative reasoning, and the diverse range of applications of quantitative reasoning in our daily lives.

LEARNING OUTCOMES

After completing this unit, you will be able to:

1. Define the term quantitative reasoning.
2. Explain the components of quantitative reasoning.
3. Interpret the applications of quantitative reasoning in daily life.
4. Analyze the generally perceived misconceptions about Mathematics.

1.1 Defining Quantitative Reasoning

Quantitative reasoning involves the skill and ability to interpret and reason using quantitative information. It involves the use of mathematical, statistical and logical skills and knowledge to interpret data and draw conclusions. Working with numbers and basic calculations is an important part of literacy. Literacy involves the ability to read, write, reason and work with numbers. The following definitions would be helpful to explain the concept of quantitative reasoning.

- Quantitative literacy is an aggregate of skills, knowledge, beliefs, dispositions, habits of mind, communication capabilities and problem-solving skills that people require to deal with situations involving numbers at the workplace and in daily life (International Life Skills Survey, 2000).
- Quantitative reasoning is the application of mathematical concepts and skills to solve real-world problems. It helps to read and understand quantitative data, understand quantitative evidence and apply quantitative skills to solve daily life problems (Hollins University, 2011).
- Numerical abilities equip students with the ability to understand and explain the world in quantitative terms, interpret numerical data and evaluate arguments relying on quantitative information and approaches (BYU Quantitative Reasoning Foundation Document, 2003).
- Beyond arithmetic and geometry, quantitative literacy also involves logic, data analysis and probability. It enables individuals to analyze data, read graphs, understand arguments, detect logical fallacies, understand evidence and evaluate risks. Quantitative literacy means knowing how to reason and how to think (Kolata, 1997). (Mayes, Bonilla & Peterson, n.d.)

Although quantitative reasoning applies mathematical knowledge to solve real-world problems, both are not the same, as shown in table 1.1 (Mayes, Bonilla & Peterson, 2013).

Table 1.1: Contrast of Mathematics with Quantitative Reasoning-I

Mathematics	Quantitative Reasoning
Power in abstraction	Real, authentic context
Power in generality	Specific, particular applications
Some context dependency	Heavy context dependency
Society independent	Society dependent
Apolitical	Political
Methods and algorithms	Ad hoc methods
Well-defined problems	Ill-defined problems
Approximation	Estimation is critical
Heavily disciplinary	Interdisciplinary
Problem solutions	Problem descriptions
Few practice opportunities outside the classroom	Many practice opportunities outside the classroom
Predictable	Unpredictable

Mayes, Bonilla & Peterson (2013) have proposed a quantitative reasoning framework with four components. It involves:

1. **Quantification Act (QA):** It is the process of conceptualizing the attribute of an object so that the attribute is expressed using a unit measure.
2. **Quantitative Literacy (QL):** It involves the use of fundamental mathematical concepts for describing, comparing, manipulating, and drawing conclusions from variables developed in the quantification act.
3. **Quantitative Interpretation (QI):** It is the ability to use models to discover trends and make predictions.
4. **Quantitative Modeling (QM):** Ability to create representations to explain a phenomenon and to revise them based on fit to reality.

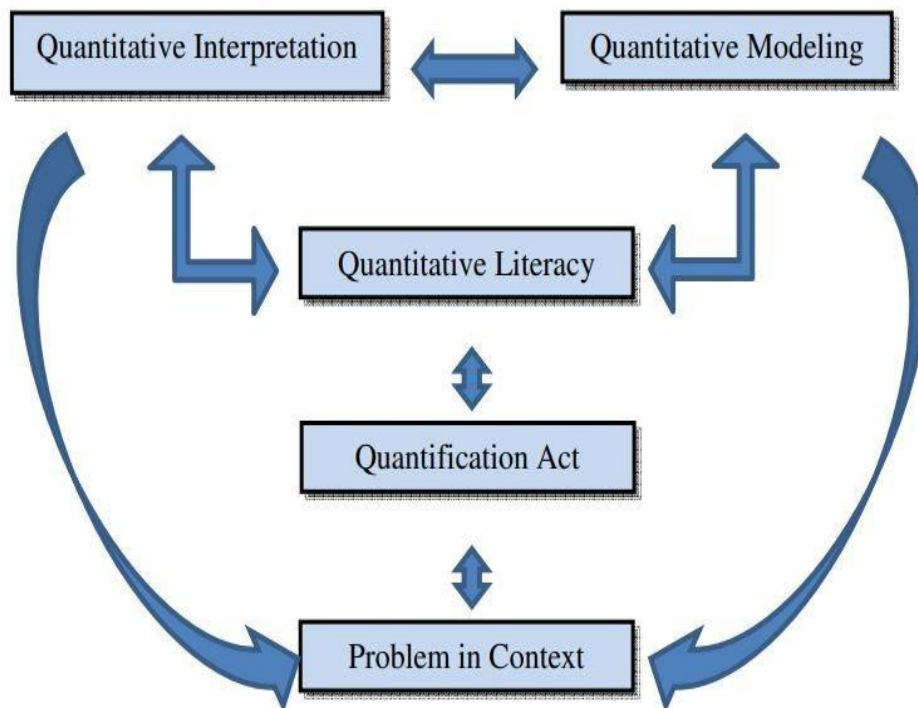


Figure 1.1. Quantitative Reasoning cycle

Quantification Act	Quantitative Literacy	Quantitative Interpretation	Quantitative Modeling
Variable Identification	Numeracy	Representations	Logic
Object	Number Sense	Tables	
Attribute	Small/large	Graphs/diagrams	Problem Solving
Measure	Numbers	Equations	Problem Formulation
	Scientific Notation	Linear	
Communication	Logic	Quadratic	Modeling
Force-dynamic		Power	Normal Distribution
Scientific discourse	Measurement	Exponential	Regression Model
Quantitative discourse	Accuracy	Statistical displays	linear
	Precision	Translation	polynomial
Context	Estimation		power
Avoids QR	Units	Science diagrams	exponential
Computation Driven		Complex systems	logarithmic
Situative view	Proportional Reasoning		Logistic Growth Model
	Fraction	Statistics & Probability	Multivariate Model
Variation	Ratio	Randomness	Simulation Model
Causation	Percent	Evaluating Risks	Scientific Diagram
Correlation	Rates/Change	Normal Distribution	Table & Graph Models
Covariation	Proportions	Statistical Plots	
	Dimensional	Correlation	Inference
	Analysis	Causality	Inference
		Z-scores	Hypothesis Testing
	Basic Prob/Stats	Confidence Intervals	Practical Significance
	Empirical Prob.		
	Counting	Logarithmic Scales	
	Central tendency		

Figure 1.2. Detail of Components of Quantitative Reasoning Framework

1.2 Applications of Quantitative Reasoning

Quantitative reasoning equips students with knowledge and skills, which can be used in the academic field, workplace, career and daily life. The knowledge of mathematical concepts plays a major role in enabling students to appreciate great work in the fields of history, culture, arts, architecture, science, music, language and literature. For example, the work of builders of the great city of Zimbabwe in Africa, the ancient Egyptians and Greeks, and modern and classical music.

At the workplace, quantitative reasoning is a very demanding skill. The knowledge and skills of using basic mathematical concepts play a positive role in seeking good career opportunities. As it helps to perform job tasks, most of the occupations demand a high level of language and mathematical skills. For example, chemical engineer, lawyer, newspaper editor, mathematician, cardiologist, weather forecaster, primary/elementary/secondary school teacher, financial analyst, journalist, webpage designer, corporate executive, salesman, travel agent, cook, corporate executive, manager, poultry farmer, personnel manager, psychologist, toll collector and retail store manager etc. The basic literacy skills include competence in handling arithmetic operations and solving day-to-day problems involving numbers and shapes.

1.3 Misconceptions about Mathematics

Keeping in view the importance of mathematical knowledge in quantitative reasoning, it is essential for a student to learn about it. Mathematics is often perceived as a dry, difficulty and/or boring subject. The attitude of students towards mathematics is affected by their perception of mathematics. The reason for this perception may be based on misconceptions about mathematics. In this section, we are looking at some misconceptions about mathematics.

1.3.1 Misconception 1: Mathematics Requires a Special Brain

One misconception about mathematics is that there are some special abilities required to learn mathematics. The reality is that everyone can learn mathematics just any other subject or skill such as learning to read or playing hockey. It requires hard work, self-confidence, self-efficacy and consistency. Individual differences among people affect their learning of a skill or subject. Therefore, different people learn mathematics at different rates and in different ways. For example, some people learn by concentrating on concrete problems, others by thinking visually, and still others by thinking abstractly. The learning style of a person may differ from others but s/he can still learn and succeed in mathematics.

1.3.2 Misconception 2: Mathematics in the Context of Modern Issues is Very Complex

Learning mathematics involves solving problems. Misconception two is that advanced mathematical concepts are too complex for an average person to understand. It is true that only a few people receive advanced training in a field such as mathematics, chemistry, music, and professional writing, however, most people can learn basic mathematical concepts to such an extent that they can understand complex issues and make reasonable informed decisions about them. The emphasis here is on learning the concepts, not mastering the advanced skills. For example, it takes a lot of training, study and hard work to become a professional writer, but most people can read a book or write a few sentences. Mathematics is no different. You can learn about mathematical concepts to such an extent that you can make daily life decisions based on available data or information.

1.3.3 Misconception 3: Mathematics Makes you Less Sensitive

Some people believe that learning mathematics will somehow make them less sensitive to the aesthetic aspects of life. However, understanding the mathematics that explains the colors of a sunset or the geometric design in a work of art can help to observe and appreciate it. Furthermore, many people find beauty and elegance in mathematics itself. People trained in mathematics have made important contributions to art, music, architecture and many other fields.

1.3.4 Misconception 4: Mathematics do not have a Room for Creativity

Some of the facts, formalisms, and skills required for mathematical proficiency are dry, but using these mathematical tools takes creativity. For example, consider designing and building a home. The task demands specific skills to lay the foundation, frame in the structure, install plumbing and wiring, and paint walls. But creativity is needed in building the home to develop the architectural design, and respond to on-the-spot problems during construction, and constraints based on budgets and building codes. The mathematical skills learned in school are like the skills of carpentry, plumbing or other technical skills. Applying mathematics is like the creative process of building a home. So, we can say that mathematics has room for creativity.

1.3.5 Misconception 5: Mathematics Provides Exact Answers

A mathematical formula yields a specific result as you solve a given problem in school. It would be marked a right or wrong. But in real life, mathematics involves more than just using a formula; it involves the understanding of given data, its context, and possible factors which may affect it. So, in real life, when you use mathematics, answers are never so and clear-cut. For example,

A bank offers simple interest of 3%, paid at the end of one year (that is, after one year the bank pays you 3% of your account balance). If you deposit \$1000 today and make no further deposits or withdrawals, how much will you have in your account after one year?

A straight mathematical calculation seems enough: 3% of Rs. 1000/- is Rs. 30/- so, you should have \$1030 at the end of a year. However other factors must be considered while calculating the exact amount. For example, to what extent, will your balance be affected by service charges or taxes on interest earned? What if the bank fails? What if the currency is devalued during this time? Choosing a bank for investing your money is a real mathematics problem that may not involve a simple or definitive solution.

1.3.6 Misconception 6: Math is Irrelevant to My Life

No matter what your path in college, career, and life, you will find mathematics involved in many ways. A major goal of this text is to show you hundreds of examples in which mathematics applies to everyone's life. We hope you will find that mathematics is not only relevant but also interesting and enjoyable.

1.3.7 Misconception 7: It's OK if I am not Good at "Math"

Mathematics underlies nearly everything in modern society from the daily financial decisions that all of us must make to the way in which we understand and approach global issues of the economy, politics, and science. We cannot possibly hope to act

wisely if we approach mathematical ideas with a poor attitude. Moreover, it's an attitude that can easily spread to others. After all, if a child hears a respected adult saying that he or she is "bad at math," the child may be less inspired to do well.

So, think about your own attitudes toward mathematics. With a good attitude and some hard work, by the end of your course, you will not only be better at math, but you will be helping future generations by making it socially unacceptable for anyone to be "bad at math."

1.4 Introduction to Mathematics and Its Branches

The word mathematics is derived from the Greek word *mathēmatikē* which means "inclined to learn." Mathematics is about being curious, open-minded, and interested in always learning more. There are three different ways to look at Mathematics (1) as the sum of its branches (2) as a way to model the world, and (3) as a language.

1.4.1 Mathematics as the Sum of Its Branches

The common branches of mathematics are given below:

- i. Logic:** The study of principles of reasoning
- ii. Arithmetic:** Methods for operating on numbers
- iii. Algebra:** Methods for working with unknown quantities
- iv. Geometry:** The study of size and shape
- v. Trigonometry:** The study of triangles and their uses
- vi. Probability:** The study of chance
- vii. Statistics:** Methods for analyzing data
- viii. Calculus:** The study of quantities that change

The branches of Mathematics are itself a comprehensive and abstract body of knowledge. However, in this course, we will focus on the use of Mathematics and its branches in quantitative thinking and critical reasoning.

1.4.2 Mathematics as a Way to Model the World

Mathematics can be used to create models and to view the patterns in a real phenomenon. For example, an ice-cream company keeps track of the quantity of various flavors of ice creams sold in the summer season based on previous sales. The weather forecast and physical health facilities available for the public are two other examples where mathematical models can be helpful to understand, represent and forecast events. Mathematical models are used in various disciplines to understand and solve problems.

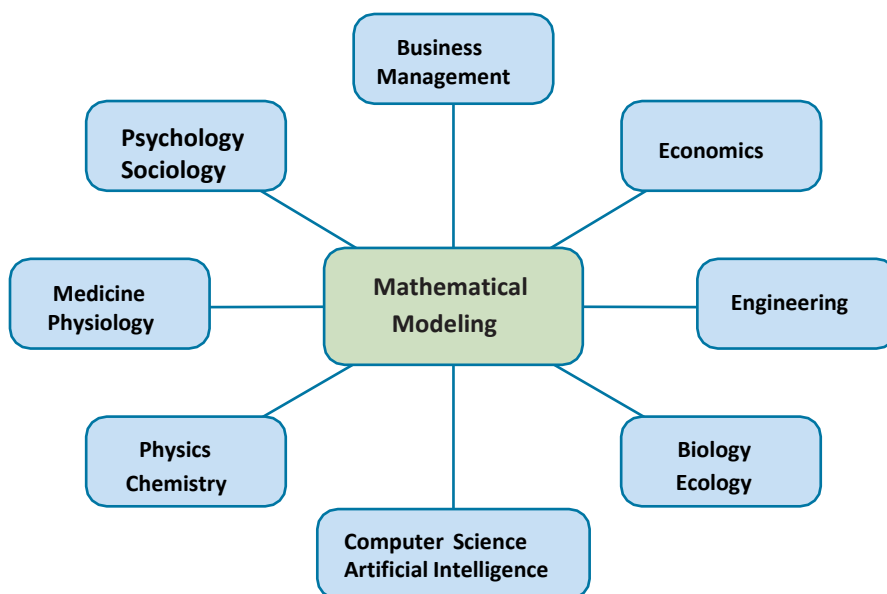


Figure 1.3. Use of Mathematical Modeling in various disciplines

1.4.3 Mathematics as a Language

Mathematics, as a language, has its own vocabulary and grammar. Learning the language of Mathematics is help for using it in daily life to understand and solve problems. Just like in a language, different degrees of competencies are possible. Getting familiarity with the mathematical language will be helpful to lean Mathematics. From this point of view, quantitative literacy is the level of competency to use mathematical vocabulary and concepts for working successfully in daily life. Table 1.2 compares ‘learning the language of Mathematics’ to ‘learning a language’.

Table 1.2: ‘Learning a Language’ vs ‘Learning the Language of Mathematics’

S #	Learning a Language	Learning the language of Mathematics
1.	Learn many styles of speaking and writing, such as essays, poetry, and drama, and reading skill.	Learn techniques from many branches of mathematics, such as arithmetic, algebra, and geometry.
2.	Place literature in context through the history and social conditions under which it was created.	Place mathematics in context through its history, purposes, and applications.
3.	Learn the elements of language such as words, parts of speech (nouns, verbs, etc.), and rules of grammar and practice their proper use.	Learn the elements of mathematics such as numbers, variables, and operations and practice using them to solve simple problems.

4.	Critically analyze language in forms such as novels, short stories, essays, poems, speeches, and debates.	Critically analyze quantitative information in mathematical models, statistical studies, economic forecasts, investment strategies, and more.
5.	Use language creatively for various purposes such as writing a term paper, application, story or engaging in a debate	Use mathematics creatively to solve problems and to understand issues in the modern world.

SUMMARY

In this unit, we learned about the basic concept of quantitative reasoning, the difference between Mathematics and quantitative reasoning and misconceptions about Mathematics. Quantitative reasoning involves the knowledge, skills and dispositions to read and interpret the numerical data and use it for solving daily life problems. Mathematics can be regarded as a language learning process where you get familiar with certain operations and concepts so that you can use it for the communication of information. Quantitative reasoning is different from Mathematics because it involves the use of mathematical principles for understanding and dealing with daily life ill-defined problems. It also involves the context-based applications of mathematical principles and logic to the problems faced in the society. Its applications cover every aspect of life from academic life, to the workplace and daily life. Therefore, quantitative reasoning is a very important course for every member of the society for the effective and efficient decision-making process in daily life.

EXERCISES

Multiple Choice Questions

- What is the primary goal of quantitative reasoning in daily life?
 - Solving complex mathematical problems
 - Making informed decisions based on data and numbers**
 - Memorizing mathematical formulas
 - Conducting scientific research
- Which of the following is an example of quantitative reasoning in daily life?
 - Writing a poem
 - Painting a picture
 - Baking cookies using a recipe**
 - Singing a song

3. If you want to compare the prices of two similar products at the grocery store, what quantitative reasoning skill would you use?
 - a) Addition
 - b) Subtraction
 - c) Multiplication
 - d) **Comparison and estimation**
4. In a recipe, if you need 1 cup of milk but only have a 1/2-cup measuring cup, how many times would you need to fill the measuring cup to get 1 cup of milk?
 - a) 1 time
 - b) **2 times**
 - c) 3 times
 - d) 4 times
5. Which of the following is an example of interpreting data in daily life?
 - a) Solving quadratic equations
 - b) **Analyzing a weather forecast**
 - c) Memorizing historical dates
 - d) Composing a piece of music
6. What is the primary benefit of quantitative reasoning in daily life?
 - a) Solving complex mathematical puzzles
 - b) **Making logical decisions based on data and evidence**
 - c) Avoiding all forms of mathematics
 - d) Engaging in creative activities without constraints
7. In which discipline mathematics is used to model and analyze physical phenomena such as motion, energy, and forces?
 - a) Biology
 - b) Economics
 - c) **Physics**
 - d) History
8. In architecture and engineering, what mathematical branch is essential for designing structures that can withstand various loads and stresses?
 - a) Trigonometry
 - b) **Geometry**
 - c) Calculus
 - d) Linear algebra
9. What is the primary distinction between quantitative reasoning and mathematics?
 - a) Quantitative reasoning is purely theoretical, while mathematics is practical.
 - b) **Mathematics focuses on calculations and formulas, while quantitative reasoning emphasizes critical thinking and problem solving.**
 - c) Quantitative reasoning is a subset of mathematics.
 - d) Mathematics is limited to whole numbers, while quantitative reasoning deals with fractions and decimals.

10. Which discipline heavily depends on mathematical concepts, such as differential equations, for modeling and predicting climate patterns and weather phenomena?
- a) Geography
 - b) **Meteorology**
 - c) Literature
 - d) Music theory

Short-Answer Questions

- 1. Define quantitative reasoning.
- 2. Why do we need to study the quantitative reasoning?
- 3. Explain the scope of quantitative reasoning.
- 4. What is the difference among quantitative literacy, quantitative reasoning and numeracy?
- 5. What is the relationship between Mathematics and quantitative reasoning?

Subjective Questions

- 1. Explain the concept and importance of quantitative reasoning.
- 2. Discuss the scope of quantitative reasoning in our daily life activities.
- 3. What are the major misconceptions about Mathematics?

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Unit–2

APPROACHES TO PROBLEM SOLVING

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Reviewed by: Ms. Nazish

INTRODUCTION

Problem-solving is a fundamental skill that is essential in all aspects of life, from everyday challenges to complex professional endeavors. In this unit, we will explore various approaches to problem-solving, equipping you with the tools and techniques necessary to tackle problems effectively and efficiently. Problem-solving involves the systematic process of identifying, analyzing, and resolving challenges or obstacles. It requires critical thinking, creativity, and perseverance to navigate through the problem-solving process and arrive at viable solutions. By understanding different problem-solving approaches, you will be better equipped to approach problems with confidence and achieve successful outcomes.

In this unit, we will begin by examining the importance of problem-solving skills in various contexts, including academic, professional, and personal settings. We will explore the characteristics of effective problem solvers and identify strategies for developing and honing your problem-solving abilities. Throughout this unit, we will emphasize the importance of critical thinking and adaptability in problem-solving. By developing a repertoire of problem-solving approaches and techniques, you will be better prepared to navigate through the complexities of problem-solving and overcome obstacles with confidence.

By the end of this unit, you will have gained a deeper understanding of problem-solving approaches and acquired practical skills to apply them in various contexts. Whether you are a student, professional, or lifelong learner, the ability to effectively solve problems is a valuable skill that will serve you well in all aspects of your life. So let us start our unit.

LEARNING OUTCOMES

By completing this unit, you will be able to:

1. Define and differentiate between various problem-solving strategies including analytical, algorithmic, heuristic, and creative approaches.
2. Analyze real-world problems to identify the most suitable problem-solving approach based on the nature of the problem and available resources.
3. Apply analytical problem-solving techniques to break down complex problems into smaller, more manageable components, facilitating systematic problem-solving processes.
4. Implement algorithmic problem-solving methods to develop step-by-step procedures for solving repetitive or structured problems efficiently.
5. Utilize heuristic problem-solving strategies to generate innovative solutions in situations where traditional methods may be insufficient or impractical.
6. Foster creativity and lateral thinking skills through exercises and activities designed to challenge conventional problem-solving paradigms.

Usually, we think that the mathematical problems involve only numbers and symbols. But the mathematical problems that we encounter in other subject areas, at the workplace, and in daily life are almost always posed in words. For example, budgeting your expenses and income, and estimating the cost of travelling to your workplace per month. How much amount of a medicine is prescribed by the doctor for a patient? Which minimum quantity can be dangerous for the patient? What happens to the sea level if all the ice melts in the Antarctica? It can be answered by taking into account the total land area of Antarctica, the mean thickness of the ice in Antarctica, the total surface area of the earth's ocean, and the volume of water resulting from the melting of the ice (volume of this water would be 5/6 times of the original ice volume). These problems can give you an idea of simple and complex problems in real life where mathematical knowledge is required to understand and solve the problem.

2.1 Working with Units

Numbers in our daily life represent the amount or quantity of an object, property or something. For example, the number 2 can represent 2 hours, 2nd position in class, 2 rupees, 2 bananas, 2 books, 2 buses, 2 glasses etc. The words that describe what we are measuring or counting such as books, rupees, hours, and position are called the units associated with the number. The technique of working with units to solve problems is called unit analysis (or dimensional analysis).

Units: Units of a quantity describe what that quantity measures or counts. For example, gram, kilogram and pounds for weight, kilometers and miles for distance, and hours, minutes and seconds for time.

Unit Analysis: Unit analysis is the process of working with units to help solve problems.

Units provide context to the numbers. For example, “weight of apples is 5” does not convey a clear meaning on whether it is about grams, pounds or kilograms. Similarly, for a distance, we use miles or kilometers. Suppose a car covers 100 miles of distance in 2 hours, then what is the speed of the car per hour? We divide the 100-mile distance driven by the 2 hours of time it took, showing the units as we work.

$$\begin{aligned} 100 \text{ mi} \div 2 \text{ hr} &= 100 \text{ mi}/2 \text{ hr} \\ &= 50 \text{ mi/hr} \quad [\text{read as miles per hour}] \end{aligned}$$

We notice three things here:

- i. It is easy to do division or multiplication by writing the units with quantities.
- ii. Final units are miles divided by hours which we read as miles per hour.
- iii. We notice the abbreviating (short form) units ‘mi’ for miles and ‘hr’ for hours. The abbreviating unit is the same for singular and plural. We use ‘mi’ for mile and miles, and ‘hr’ for hour and hours.

There are two important keywords “per” and “of” in mathematical problems. The average speed example shows that the word ‘per’ hour (which means “for every” hour) is a key word in mathematical problems because it tells us to divide. A second important key word is of, which usually implies multiplication. For example, if you buy 10 mangoes at a price of Rs. 8/- per mango, the total price you will pay is:

$$10 \text{ ~~mangoes~~ } \times \frac{\text{Rs.8/-}}{\text{mango}} = \text{Rs.80/-}$$

We notice three important steps in this example:

- i. We multiply we see the word “of”.
- ii. We wrote the price of \$2 per apple with division (as a fraction).
- iii. Just as we may cancel a number that appears in the numerator and denominator of a fraction, we may do the same cancellation of units.

[Note: in a fraction ‘10/2’, 10 is the numerator and 2 is the denominator.]

Self-Assessment Questions

1. What is the total distance travelled when a person takes 8 rounds of an 800-meter-long track?
2. How many oranges are there in 25 boxes, if each box carries 30 oranges?

2.2 Common Fractions

A fraction can be expressed in three basic ways: as a common fraction such as $\frac{1}{2}$, as a fraction in decimal form such as 0.5, and as a percentage such as 50%. Common fractions represent division and are written in the form $\frac{a}{b}$, where a and b can be any numbers as long as b is not zero. The number on top is the numerator and the number on the bottom is the denominator.

$$\begin{array}{cc} \text{Numerator} & \underline{a} \\ \text{Denominator} & b \end{array}$$

2.2.1 Adding and Subtracting Fractions

If two fractions have the same denominator (a common denominator), we can add or subtract them by adding or subtracting their numerators. For example:

$$\frac{2}{6} + \frac{3}{6} = \frac{2+3}{6} = \frac{5}{6}$$

For subtracting or adding two fractions with different denominators, we must rewrite the fractions with the same denominator before adding or subtracting. We will get the same denominator by multiplying the denominators of both fractions. For example,

$$\frac{2}{3} + \frac{1}{2} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6}$$

Multiplying Fractions

To multiply fractions, we multiply the numerators and denominators separately. For example:

$$\frac{3}{3} \times \frac{1}{4} = \frac{3 \times 1}{3 \times 4} = \frac{3}{12}$$

Sometimes we can simplify fractions at the same time we multiply them by canceling terms that occur in both the numerator and the denominator. For example:

$$\frac{5}{3} \times \frac{2}{5} = \frac{5 \times 2}{3 \times 5} = \frac{2}{3}$$

2.2.2 Reciprocals and Division

Two (nonzero or numbers other than zero) numbers are reciprocals if their product is 1. For example:

Examples:

1) 3 and $\frac{1}{3}$ are reciprocals because $3 \times \frac{1}{3} = 3 \times \frac{1}{3} = 1$

(Note: Integer has a denominator of 1, as shown in the case of integer “3” in the above example)

2) $\frac{5}{6}$ and $\frac{6}{5}$ are reciprocals because $\frac{5}{6} \times \frac{6}{5} = \frac{5}{6} \times \frac{6}{5} = 1$

The word integer originated from the Latin word “Integer” which means whole or intact. Integers are a special set of numbers comprising zero, positive numbers and negative numbers.

Examples of Integers: – 1, -12, 6, 15.

The symbol ‘Z’ represents integers.

$Z = \{\dots\dots-8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, \dots\dots\}$

If we have two fractions or one integer and one fraction with a ‘÷’ division sign between them, we can replace division with a multiplication sign by inverting and multiplying (this inverting of the fraction is called reciprocal). For example,

$$\begin{array}{l} 1) \quad 9 \div \frac{1}{3} = 9 \times \frac{3}{1} = 27 \\ 2) \quad \frac{7}{3} \div \frac{2}{5} = \frac{7}{3} \times \frac{5}{2} = \frac{7 \times 5}{3 \times 2} = \frac{35}{6} \end{array}$$

Table 2.1: Summary of Rules for Fractions

S#	Operation	Detail of Rule
1.	Addition/subtraction (in case of the same denominator)	$\frac{4}{3} + \frac{2}{3} = \frac{4+2}{3} = \frac{6}{3}$ OR $\frac{4}{3} - \frac{2}{3} = \frac{4-2}{3} = \frac{2}{3}$
2.	Multiplication (multiply numerator with numerator and denominator with denominator)	$\frac{4}{5} \times \frac{2}{6} = \frac{4 \times 2}{5 \times 6} = \frac{8}{30}$
3.	Division (invert and multiply)	$\frac{2}{3} \div \frac{5}{7} = \frac{2}{3} \times \frac{7}{5} = \frac{2 \times 7}{3 \times 5} = \frac{14}{15}$

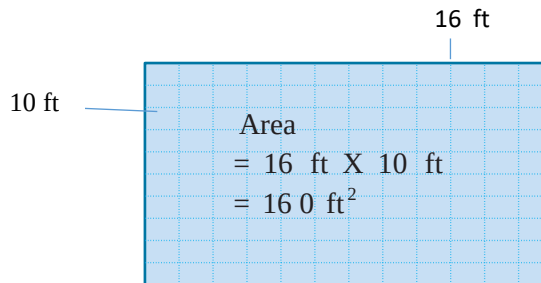
2.3 Squares, Cubes and Hyphens

2.3.1 Square

To find the area of a room, multiply the length of the room by its width. For example, if the length of the room is 16 feet (ft) and its width is 10 feet (ft), then the area of the room will be:

$$16 \text{ ft} \times 10 \text{ ft} = 160 \text{ ft} \times \text{ft} = 160 \text{ ft}^2$$

We write ft^2 here because the unit of length and width is feet so we keep track of both in the answer. If the area of a room is 160 ft^2 , it means that the floor of the room can fit in 160 boxes of 1 ft^2 .



2.3.2 Cubes

To calculate the volume of a room, multiply the length, width and height of the room. If the room is 5 meters long, 4 meters wide and 3 meters high, the volume of the room is 60 m^3 (read as 60 cubic meters), as given below.

$$5 \text{ m} \times 4 \text{ m} \times 3 \text{ m} = 60 \text{ m} \times \text{m} \times \text{m} = 60 \text{ m}^3$$

2.3.3 Hyphens

Hyphen (-) is a key symbol that is written with units. For example, if you look at the electricity bill, it will probably state electricity usage in units of “kilowatt-hours”. The hyphen means multiplication. For example, if a fridge uses 0.5-kilowatt energy for a time of 8 hours, its energy usage is:

$$0.5 \text{ kilowatt} \times 8 \text{ hr} = 4 \text{ kilowatt} \times \text{hr} = 4 \text{ kilowatt-hr}$$

(read as “kilowatt-hours”)

Self-Assessment Questions

1. If you purchase 5 liters of petrol for Rs. 3000/-, what is the price of 1 liter of petrol? Or if the cost of 1 liter of petrol is Rs. 300/-, then what is the price of 7 liters of petrol?
2. What is the volume of a box that is 7 inches (in) long, 5 inches wide and 6 inches tall? [‘in’ is used to represent inch.]
3. What is the area of a room that is 6 meters (m) long and 7 meters wide? [‘m’ is used to represent meter.]

2.3.4 Unit Conversions

Many everyday problems require converting numbers from one unit to another, such as from miles to kilometers or quarts to cups. As a simple example, suppose we want to convert 2 feet to inches. Because 1 foot is the same as 12 inches, we do the conversion as follows:

$$2 \text{ ft} = 2 \text{ ft} \times \frac{12 \text{ in}}{1 \text{ ft}} = 24 \text{ in}$$

Because $12 \text{ in} = 1 \text{ ft}$

Notice that 12 inches and 1 foot are equal. This idea extends to all unit conversions, which always require an appropriate way of multiplying by 1 so we do not change the meaning of the original expression. For example, the following are all different ways of writing 1:

$$1 = \frac{1}{1} = \frac{8}{8} = \frac{1}{4} = \frac{1 \text{ kilogram}}{1000 \text{ grams}} = \frac{1 \text{ week}}{7 \text{ days}} = \frac{12 \text{ inches}}{1 \text{ foot}}$$

We can denote the conversion of units in three ways:

$$12 \text{ in} = 1 \text{ ft} \quad \text{or} \quad \frac{12 \text{ in}}{1 \text{ ft}} = 1 \quad \text{or} \quad \frac{1 \text{ ft}}{12 \text{ in}} = 1$$

Self-Assessment Questions

1. Convert a length of 27 inches into feet.
2. A box is 3 feet long. What is its length in inches?

2.4 Decimal Fractions

For a fraction in decimal form, each digit corresponds to a certain place value, which is always a power of 10 (such as 10, 100, 1000...). The following example shows values for the decimal places in the number **3.141**.

3	.	1	4	1
Unit	Decimal point	tenths	hundredths	thousandths
اکائی	اعشاریہ	دہائی	سیکڑا	ہزار

2.4.1 Converting Decimal Fraction to Common Form

Converting a fraction from decimal to common form requires recognizing the value of the last digit in the decimal. For example:

1. $0.6 = \frac{6}{10}$
2. $5.67 = \frac{567}{100}$
3. $0.056 = \frac{56}{1000}$

2.4.2 Converting Common Fraction to Decimal Form

To convert a common fraction into decimal form, we perform the division implied by the fraction. For example:

1. $\frac{1}{2} = 0.5$
2. $\frac{3}{5} = 0.6$
3. $\frac{1}{7} = 0.14285714$

In daily life, we round off the repeated numbers after decimal such as in the example (3), we can write it as 0.14.

2.5 International Metric System and Metric Conversion

The international metric system was invented in France late in the 18th century for two primary reasons: (1) to replace many customary units with just a few basic units and (2) to simplify conversions through the use of a decimal (base 10) system.

The basic units of length, mass, time, and volume in the metric system are:

- meter for length, abbreviated m
- kilogram for mass, abbreviated kg
- second for time, abbreviated s
- liter for volume, abbreviated L

These basic units can be combined with a prefix that indicates multiplication by a power of 10. For example, kilo means 1000 so a kilometer is 1000 meters, and micro means one-millionth so a microgram is a millionth of a gram. Figure 2.1 lists common metric prefixes.

Figure 2.2 lists common metric conversion values which can be helpful in daily life activities.

Small Values			Large Values		
Prefix	Abbrev.	Value	Prefix	Abbrev.	Value
deci	d	10^{-1} (one-tenth)	deca	da	10^1 (ten)
centi	c	10^{-2} (one-hundredth)	hecto	h	10^2 (hundred)
milli	m	10^{-3} (one-thousandth)	kilo	k	10^3 (thousand)
micro	μ or mc*	10^{-6} (one-millionth)	mega	M	10^6 (million)
nano	n	10^{-9} (one-billionth)	giga	G	10^9 (billion)
pico	p	10^{-12} (one-trillionth)	tera	T	10^{12} (trillion)

* Micro is usually abbreviated with μ (the Greek letter mu), but in medical fields it is common to use "mc" instead.

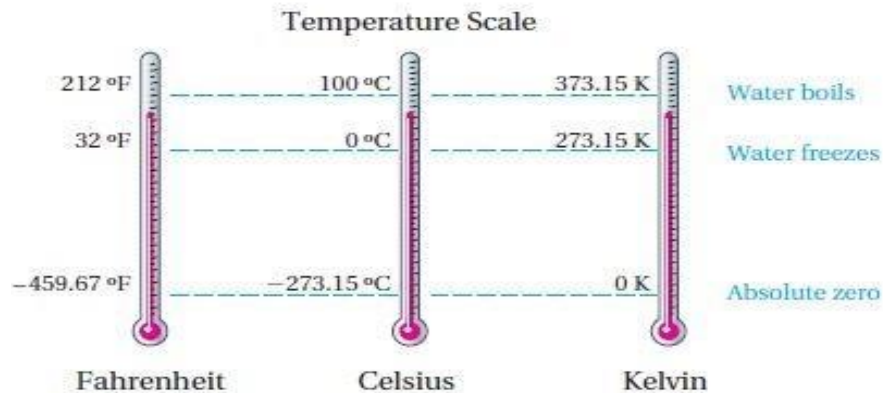
Figure 2.1. Common Metric Prefix

USCS to Metric	Metric to USCS
1 in = 2.540 cm	1 cm = 0.3937 in
1 ft = 0.3048 m	1 m = 3.28 ft
1 yd = 0.9144 m	1 m = 1.094 yd
1 mi = 1.6093 km	1 km = 0.6214 mi
1 lb = 0.4536 kg	1 kg = 2.205 lb
1 fl oz = 29.574 mL	1 mL = 0.03381 fl oz
1 qt = 0.9464 L	1 L = 1.057 qt
1 gal = 3.785 L	1 L = 0.2642 gal

Figure 2.2. Common metric conversions

Another important set of standardized units are the units we use to measure temperature. Three commonly used temperature scales are:

- The Fahrenheit scale, commonly used in the United States, is defined so water freezes at 32°F and boils at 212°F.
- The rest of the world uses the Celsius scale, which places the freezing point of water at 0°C and the boiling point at 100°C.
- For the measurement of temperature, we use the Kelvin scale, which is the same as the Celsius scale except for its zero point, which corresponds to -273.15°C. A temperature of 0 K is known as absolute zero, because it is the coldest possible temperature. (The degree symbol [°] is not used on the Kelvin scale.



The following figure 2.3 guides us about the conversion of value from one unit to the other unit of temperature.

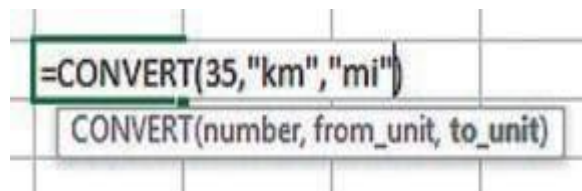
Temperature Conversions

The conversions are given both in words and with formulas in which C , F , and K are Celsius, Fahrenheit, and Kelvin temperatures, respectively.

To Convert from	Conversion in Words	Conversion Formula
Celsius to Fahrenheit	Multiply by 1.8 (or $\frac{9}{5}$). Then add 32.	$F = 1.8C + 32$
Fahrenheit to Celsius	Subtract 32. Then divide by 1.8, which is $\frac{5}{9}$, or equivalently multiply by $\frac{5}{9}$.	$C = \frac{F - 32}{1.8}$
Celsius to Kelvin	Add 273.15.	$K = C + 273.15$
Kelvin to Celsius	Subtract 273.15.	$C = K - 273.15$

For metric or unit conversion, you can try to use MS Excel or Google. In MS Excel, there is a function CONVERT by entering the number along with the correct unit abbreviations in quotes. Google can also give you the exact answer when you type what you want to convert into the Google search box.

Microsoft Excel



Google search

50 liters in gallons

Volume

50 = 13.2086

Liter US gal

Figure 2.3 Metric or unit conversion through excel and Google search

Interesting Information

There are standardized units for measuring various properties such as length, volume and weight. The image given below showed the units according to the US customary system of measurement.

LENGTHS	1 inch (in) = 2.54 centimeters	1 furlong = 40 rods = $\frac{1}{8}$ mile	
	1 foot (ft) = 12 inches	1 mile (mi) = 1760 yards = 5280 feet	
	1 yard (yd) = 3 feet	1 nautical mile = 1.852 km $\approx$ 6076.1 feet	
	1 rod = 5.5 yards	1 league (marine) = 3 nautical miles	
	1 fathom = 6 feet		
WEIGHTS	Avoirdupois	Troy	Apothecary
	1 grain = 0.0648 gram	1 grain = 0.0648 gram	1 grain = 0.0648 gram
	1 ounce (oz) = 437.5 grains	1 carat = 0.2 gram = 3.086 grains	1 scruple = 20 grains
	1 pound (lb) = 16 ounces	1 pennyweight = 24 grains	1 dram = 3 scruples
	1 ton = 2000 pounds	1 troy ounce = 480 grains	1 apoth. ounce = 8 drams
	1 long ton = 2240 pounds	1 troy pound = 12 troy ounces	1 apoth. pound = 12 ounces
VOLUMES	Liquid Measures		Dry Measures
	1 tablespoon (tbsp or T) = 3 teaspoons (tsp or t)		1 in ³ $\approx$ 16.387 cm ³
	1 fluid ounce (fl oz) = 2 tablespoons = 1.805 in ³		1 ft ³ = 1728 in ³ = 7.48 gallons
	1 cup (c) = 8 fluid ounces		1 yd ³ = 27 ft ³
	1 pint (pt) = 16 fluid ounces = 28.88 in ³		1 dry pint (pt) = 33.60 in ³
	1 quart (qt) = 2 pints = 57.75 in ³		1 dry quart (qt) = 2 dry pints = 67.2 in ³
	1 gallon (gal) = 4 quarts		1 peck = 8 dry quarts
	1 barrel of petroleum = 42 gallons		1 bushel = 4 pecks
	1 barrel of liquid = 31 gallons		1 cord = 128 ft ³

2.6 Powers of 10

Powers of 10 indicate how many times to multiply 10 by itself. For example:

$$10^2 = 10 \times 10 = 100$$

$$10^6 = 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 1,000,000$$

Negative powers indicate reciprocals of corresponding positive powers. For example:

$$10^{-2} \frac{1}{10^2} = \frac{1}{100} = 0.01$$

$$10^{-6} \frac{1}{10^6} = \frac{1}{1,000,000} = 0.000001$$

Notice that powers of 10 follow five basic rules:

1. A positive exponent tells how many 0s follow the 1. For example, 100 is a 1 followed by a number of 0s; 10^8 is a 1 followed by eight 0s.

2. A negative exponent tells how many places are to the right of the decimal point, including the 1. For example, $10^{-1} = 0.1$ has one place to the right of the decimal point; $10^{-6} = 0.000001$ has six places to the right of the decimal point.

3. Multiplying powers of 10 simply requires adding exponents. For example:

$$10^4 \times 10^7 = 10,000 \times 10,000,000 = 100,000,000,000$$

4. Dividing powers of 10 requires subtracting exponents. For example:

$$\frac{10^5}{10^3} = 100,000 \div 1000 = 10^{5-3} = 100$$

5. There is no shortcut for adding or subtracting powers of 10. The values must be written in longhand notation. For example:

$$10^6 + 10^2 = 1,000,000 + 100 = 1,000,100$$

$$10^8 + 10^{-3} = 100,000,000 + 0.001 = 100,000,000.001$$

$$10^6 - 10^3 = 10,000,000 - 1000 = 9,999,000$$

6. There is no shortcut for adding or subtracting powers of 10. The values must be written in longhand notation. For example:

$$10^6 + 10^2 = 1,000,000 + 100 = 1,000,100$$

$$10^8 + 10^{-3} = 100,000,000 + 0.001 = 100,000,000.001$$

$$10^6 - 10^3 = 10,000,000 - 1000 = 9,999,000$$

2.7 Currency Conversions

While travelling or making some payments outside the country, we need to convert one country's money or currency to another country's currency. To convert the currency, you must know the exact exchange rates for that currency. If we want to convert \$5000 US Dollars (USD) to PKR (Pakistani Rupee), we must know about the exchange rate from 1 USD = how many PKR. Suppose that \$1 USD is equal to PKR 280/- for today, then $\$5000 \text{ USD} = 5000 \times 280 = \text{PKR } 1400000/-$.

You can get the latest currency exchange rates by simply searching in the Google search engine (as shown in figure 2.4). You can type the exact amount you want to convert in the box in front of the currency (PKR or Euro) to get the exact exchange amount in the other currency, as shown in the figure 2.5. However, the cost of changing currency from one to another currency is affected by two factors: (i) the exchange rate and (2) fees for the exchange. While the exchange rate may be more or less the same in different shops for a given currency but the fee for the exchange may vary. Many money changers charge a fee every time when you make an exchange.

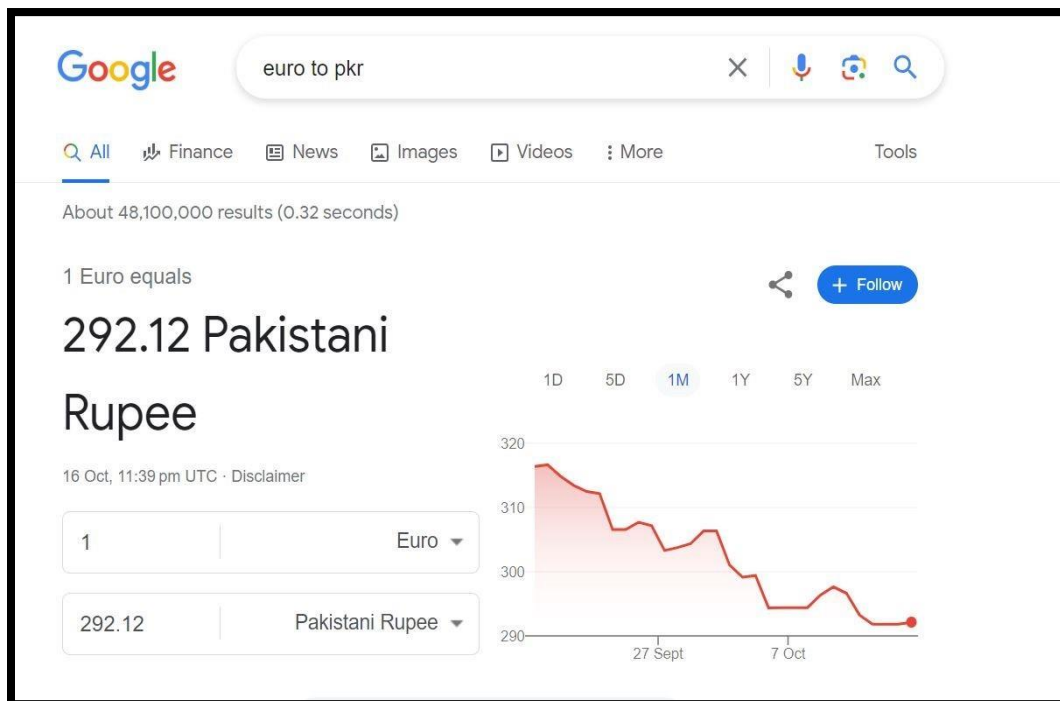


Figure 2.4. Currency exchange rate through Google search

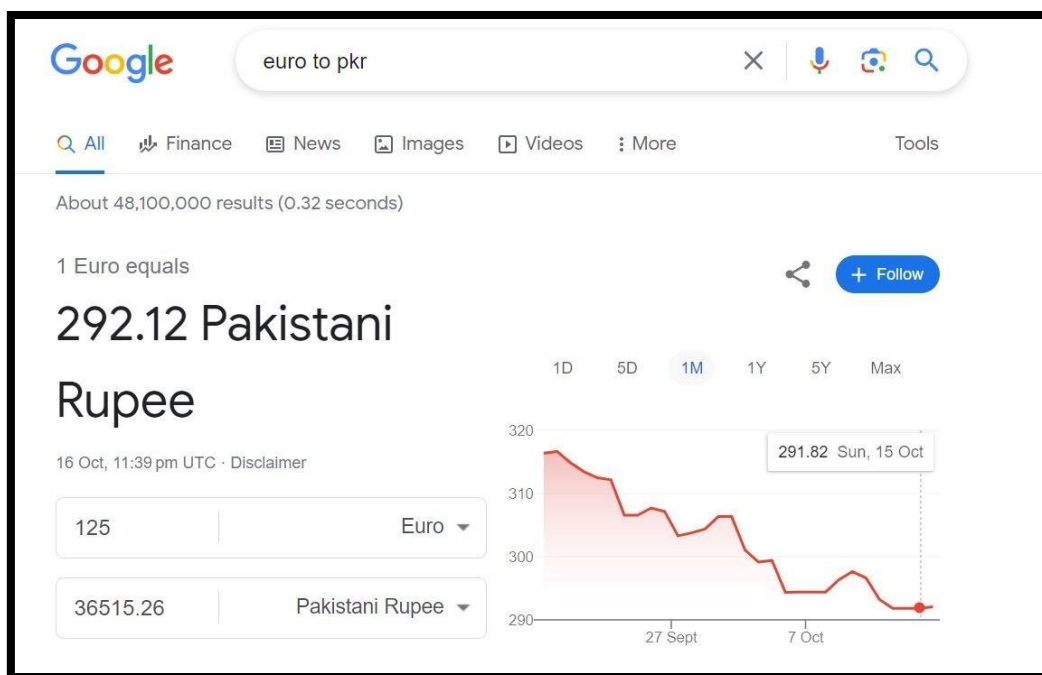


Figure 2.5. Calculation of Currency exchange rate through Google search

Self-Assessment Questions

1. The price of cereal is \$20 USD. What is the price of cereal in Pakistani Rupees?
2. A notebook costs \$8 USD. What is the price of that notebook in euros?
3. The price of a diary is 2000 PKR (Pakistani Rupees).

2.8 Problem Solving with Units

Unit analysis gives us a way to make sure that answers to problems come out with the units that we expect. The basic procedure in the unit analysis is given below:

1. Identify the units involved in the problem and the units that you expect for the answer.
2. Use the given units and the expected answer units to help you find a strategy for solving the problem. Be sure to perform all operations (such as multiplication or division) on both the numbers and their associated units.

[Remember, You cannot add or subtract numbers with different units, but you can combine different units through multiplication, division, or raising powers. (ii) It is easier to keep track of units if you replace division with multiplication by the reciprocal. For example, instead of dividing by 60 s/min, multiply by 1 min/60 s.

3. When you complete your calculations, make sure that your answer has.

Example

Problem: A car travels 25 miles every half-hour. How fast is it going?

Solution

Step 1. The “how fast” suggests that the final answer should be speed. Because the given units are miles and hours, we expect a speed in miles per hour.

Step 2. Recalling that per means divided by, the fact that we are looking for a speed in miles per hour tells us that we should divide the distance traveled by the time it takes.

$$25 \text{ mi} \div 0.5 \text{ hr} = 25 \text{ mi} \div \frac{1}{2} \text{ hr} = 25 \text{ mi} \times \frac{2}{1} \text{ hr} = 50 \text{ mi/hr}$$

Step 3. We have found that the car travels at a speed of 50 miles per hour. The answer has the units we expect. It gives us confidence that we approached the problem correctly.

Self-Assessment Question

You are buying 12 acres of land at a rate of Rs. 2,00,000/- per acre. What is the total cost of land?

2.9 Problem-Solving Process

There are four steps to solving a problem.

Step-1. Understand the problem. Be sure that you understand the nature of the problem. For example:

- Think about the context of the problem (that is, how it relates to other problems in the real world) to gain insight into its purpose.
- Make a list or table of the specific information given in the problem, including units for numerical data.
- Draw a picture or diagram to help you make sense of the problem.

- Restate the problem in different ways to clarify its question.
- Make a mental or written model of the solution, into which you can insert details as you work through the problem.

Step-2: Devise a strategy for solving the problem. Finding an appropriate strategy requires creativity, organization, and experience. In seeking a strategy, try any or all of the following:

- Obtain needed information that is not provided in the problem statement, using recall, estimation, or research.
- Make a list of possible strategies and hints that will help you select your overall strategy.
- Map out your strategy with a flow chart or diagram.

Step-3: Carry out your strategy, and revise it if necessary. In this step, you are likely to use analytical and computational tools. As you work through the mathematical details of the problem, be sure to do the following:

- Keep an organized, neat, and written record of your work, which will be helpful if you later need to review your solution.
- Double-check each step so you do not risk carrying errors through to the end of your solution.
- Constantly reevaluate your strategy as you work. If you find a flaw in your strategy, return to step 2 and create a revised strategy.

Step-4: Look back to check, interpret, and explain your result. Although you may be tempted to think you have finished after you find a result in step 3, this final step is the most important.

After all, a result is useless if it is wrong or misinterpreted or cannot be explained to others. Always do all of the following:

- Be sure that your result makes sense. For example, be sure that it has the expected units, that its numerical value is sensible, and that it is a reasonable answer to the original problem.
- Once you are sure that your result is reasonable, recheck your calculations or find an independent way of checking the result.
- Identify and understand potential sources of uncertainty in your result.
- Write your solution clearly and concisely, including a discussion of any relevant uncertainties or assumptions.
- Consider and discuss any pertinent implications of your result.

2.10 Guidelines for Solving Problems

While every problem is different, the process of problem solving has a few common features. In this section, we will examine eight general hints about problem solving.

Hint 1: There may be More than one Answer.

Most people recognize that policy questions do not have unique answers, but the same is true of many mathematical problems. For example, both $x = 4$ and $x = -4$. For example, how can we reduce the total amount of greenhouse gases emitted into the atmosphere? Or how can we control environmental pollution? These questions do not require a simple single answer. There are a number of strategies which may contribute to the strategy of solving this problem.

Hint 2: There may be More than one Strategy

Just as there may be more than one right answer, there may be more than one strategy for finding an answer. But not all strategies are equally efficient. As the following example illustrates, an efficient strategy can save a lot of time and work.

Hint 3: Use Appropriate Tools

You do not need a computer to check your tab in a restaurant, but you would not want to use an abacus to do your income tax. For any given task, there is an appropriate level of power that is needed, and it is a matter of style and efficiency neither to underestimate nor to overestimate that level. You usually will have a choice of tools to use in any problem. Choosing the tools most suited to the job will make your task much easier.

Hint 4: Consider Simpler, Similar Problems

Sometimes you are confronted with a problem that at first seems daunting. Our fourth hint is to consider a simpler, but similar, problem. The insight gained from solving the easier problem may help you understand the original problem.

Hint 5: Consider Equivalent Problems with Simpler Solutions

Replacing a problem with a similar, simpler problem can reveal essential insights about a problem, as we have just seen. However, “similar” is not good enough when we need a numerical answer. In that case, a useful approach to a difficult problem is to look for an equivalent problem. An equivalent problem will have the same numerical answer but may be easier to solve.

Hint 6: Approximations can be Useful

Another useful strategy is to make problems easier by using approximations. Most real problems involve approximate numbers, to begin with, so an approximation

often is good enough for a final answer. In other cases, an approximation will reveal the essential character of a problem, making it easier to reach an exact solution. Approximations also provide a useful check: If you come up with an “exact solution” that is not close to the approximate one, something may have gone wrong.

Hint 7: Try Alternative Patterns of Thought

Try to avoid rigid patterns of thought that tend to suggest the same ideas and methods over and over again. Instead, approach every problem with an openness that allows innovative ideas to percolate. In its most wondrous form, this approach is typified by what Martin Gardner (1914– 2010), a well-known popularizer of mathematics, called *aha!* problems. These are the problems whose best solution involves a penetrating insight that reduces the problem to its essential parts.

Hint 8: Do not spin Your Wheels

Often the best strategy in problem solving is to put a problem aside for a few hours or days. You may be amazed at what you see (and what you overlooked) when you return to it.

Self-Assessment Questions

1. It takes you 30 seconds to walk from the first (ground) floor of a building to the third floor. How long will it take to walk from the first floor to the sixth floor (at the same pace, assuming that all floors have the same height)?
2. A woman bought a horse for Rs. 50,000/- and then sold it for Rs. 60,000/-. She bought it back for Rs. 70,000/- and then sold it again for Rs. 80,000/-. How much did she gain or lose on these transactions?

SUMMARY

Throughout this unit, we have delved into various problem-solving approaches, equipping ourselves with the tools and techniques necessary to tackle challenges effectively and efficiently. In this unit, we have gone through the unit conversion methods, and solved the daily life problems through mathematical conversions of units and quantities. While it is important to understand the nature of the problem before we effectively solve it, it is also very important to follow the problem solving process and hints to get the best solution for the given problem.

Problem-solving is a critical skill that transcends all aspects of life, from academic and professional pursuits to personal endeavors. By understanding different problem-solving approaches and strategies, we have empowered ourselves to approach problems with confidence, creativity, and adaptability. By exploring analytical, creative, and systematic problem-solving approaches, we have

broadened our problem-solving toolkit, enabling us to adapt our strategies to different types of problems and situations. We have embraced the importance of flexibility and adaptability in problem-solving, recognizing that there is no one-size-fits-all approach and that success often requires a willingness to explore multiple avenues and perspectives. As we move forward, let us carry forward the knowledge and skills gained from this unit, recognizing the importance of problem-solving in all aspects of our lives.

EXERCISES

Multiple Choice Questions

1. What is the reciprocal of $\frac{5}{6}$?

- a) $\frac{5}{6}$
- b) $\frac{6}{5}$
- c) $\frac{1}{5}$
- d) 6

Answer: b) $\frac{6}{5}$

2. If 1 kg = 1000 grams, how many grams are there in 2.5 kg?

- a) 1500 grams
- b) 2000 grams
- c) 2500 grams
- d) 3000 grams

Answer: c) 2500 grams

3. How many weeks are there in 35 days?

- a) 4 weeks
- b) 5 weeks
- c) 6 weeks
- d) 7 weeks

Answer: b) 5 weeks

4. What is the area of a square with a side length of 4 inches?

- a) 8 square inches
- b) 12 square inches
- c) 16 square inches
- d) 20 square inches

Answer: c) 16 square inches

5. Convert 2.75 to a common fraction.

- a) $\frac{11}{4}$
- b) $\frac{13}{4}$
- c) $\frac{22}{7}$
- d) $\frac{25}{4}$

Answer: b) $\frac{11}{4}$

6. 1 USD is equal to 280 PKR. How much is 10 USD in PKR?

- a) 280 PKR
- b) 2800 PKR
- c) 3000 PKR
- d) 3200 PKR

Answer: b) 2800 PKR

7. What is 10 to the power of 3 (10^3)?

- a) 100
- b) 1000
- c) 10000
- d) 100000

Answer: b) 1000

8. Convert 72 inches to feet.

- a) 6 ft
- b) 8 ft
- c) 10 ft
- d) 12 ft

Answer: a) 6 ft

9. What is the volume of a cube with a side length of 3 meters?

- a) 9 cubic meters
- b) 18 cubic meters
- c) 27 cubic meters
- d) 36 cubic meters

Answer: c) 27 cubic meters

10. If 1 Euro = 320 PKR, how much is 5 Euros in PKR?

- a) 1500 PKR
- b) 1600 PKR
- c) 1800 PKR
- d) 2000 PKR

Answer: d) 2000 PKR

Short-Answer Questions

1. Define the term "reciprocal" with an example.
2. Convert 1.25 into a common fraction.
3. How many grams are there in 7.5 kilograms?
4. If a square has a side length of 8 cm, what is its area?
5. Convert 2500 grams to kilograms.
6. Write down the formula for converting days to weeks.
7. How do you calculate the area of a cube?
8. What is 100 USD in PKR if 1 USD = 280 PKR?
9. List three common units in the International Metric System.
10. Calculate the product of $\frac{3}{5} \times \frac{2}{7}$.

Subjective Questions

1. Solve the following:
 - a) Add $\frac{2}{3}$ and $\frac{3}{5}$.
 - b) Subtract $\frac{5}{8}$ from $\frac{7}{4}$.
 - c) Multiply $\frac{4}{7}$ by $\frac{3}{9}$ and simplify.
 - d) Divide $\frac{7}{6}$ by $\frac{5}{3}$.
2. A cube has a side length of 5 meters.
 - (a) Calculate the surface area of the cube.
 - (b) Calculate the volume of the cube.
3. Perform the following conversions:
 - a) Convert 8.5 kg to grams.
 - b) Convert 3.5 feet to inches.
 - c) Convert 120 hours into weeks and days.
 - d) Convert 0.875 into a fraction and simplify.
4. If a rectangle has a length of 6 inches and a width of 4 inches:
 - (a) Calculate the area of the rectangle.
 - (b) Convert the area into square feet.
5. A traveler has 200 Euros. If 1 Euro = 300 PKR, how much PKR will the traveler have after converting? If they spend 20% of this amount on shopping, how much money will be left?

6. Explain the concept of powers of 10 with examples for 10^2 , 10^3 , and 10^4 . Then, calculate the value of $(10^3 \times 10^2) \div 10$.
7. A family plans to tile a square room with a side length of 12 feet. Each tile measures 1 square foot and costs 300 PKR. Calculate:
 - a) The total number of tiles required.
 - b) The total cost of the tiles.
 - c) If the family has a budget of 45,000 PKR, will it be enough?

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Unit–3

USE OF NUMBERS IN DAILY LIFE

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INTRODUCTION

Numbers are an integral part of our daily lives, shaping the way we understand and interact with the world around us. Numbers play a crucial role in various aspects of our routine activities. Whether we are calculating time, measuring distances, managing finances, or analyzing data, our reliance on numbers is evident in almost every task we undertake.

In this unit, we will explore the significance of numbers in our daily lives and how they serve as essential tools for communication, problem-solving, and decision-making. By understanding the fundamental concepts of numbers and their applications, we can gain a deeper appreciation for their role in shaping our understanding of the world.

Through engaging activities, practical examples, and real-life scenarios, we will delve into the diverse ways in which numbers influence our everyday experiences. From basic arithmetic operations to more advanced mathematical concepts, we will uncover the relevance of numbers in various contexts, including personal finance, time management, and measurement.

By the end of this unit, students will not only develop a stronger grasp of numerical concepts but also gain valuable skills that they can apply to navigate the complexities of everyday life more effectively. So, let us embark on this journey to explore the fascinating world of numbers and discover their profound impact on our daily existence.

LEARNING OUTCOMES

After completing this unit, you would be able to:

1. Understand the fundamental concepts of numbers and their significance in daily life.
2. Develop proficiency in basic arithmetic operations such as addition, subtraction, multiplication, and division.
3. Apply numerical skills to solve practical problems encountered in everyday scenarios, including time management, budgeting, and measurement.

4. Explore real-life applications of numbers in various fields, including finance, science, and technology.
5. Enhance critical thinking and problem-solving abilities through numerical reasoning and analysis.
6. Recognize the importance of accuracy and precision in numerical calculations and measurements.
7. Cultivate an appreciation for the role of numbers in fostering effective communication and decision-making.
8. Utilize technology and mathematical tools to facilitate numerical computations and data analysis.

3.1 Percentage

Percent is expressing a quantity in the form of “divided by 100,” so the quantity (x) expressed in percentage (%) form simply means the quantity divided by 100 i.e., $x/100$. For example, 10.4% means $10.4/100$, or 0.104. Therefore, if 10.4% of eighth-graders secured an A grade in the annual examination and there are 100,000 eighth-graders, then the number who secured an A grade, is 10.4% of 100,000, or:

$$10.4\% \times 100,000 = 0.104 \times 100,000 = 10,400$$

Notice that the word “of” refers to multiply. We can see here that if 10.4% of 100,000 eighth graders secured an A grade then it means that there are 10,400 students with an A grade. The quantities in the media are frequently reported in percentage form such as news related to sports, the stock market, inflation rate, health etc.

3.2 Uses of Percentage

Percentage can be used in three different ways: as a fraction of a quantity, percentage to describe a change in a quantity or characteristic and compare the performance and cost between two products.

3.2.1 Using Percentage as Fraction

Percentage is just an alternative form of fractions (*Unit 02, 2.4: Decimal Fractions*). For example, 2.6% of 13,000 employees in the manufacturing sector lost their jobs.

$$2.6\% \times 13,000 = 0.026 \times 13,000 = 338$$

Example:

An opinion poll finds that 65% of 1050 people surveyed said that they are affected by the inflation. How many said that they were affected by the inflation?

Solution: Because of indicates multiplication, 65% of the 1069 respondents are:

$$65\% \times 1050 = 0.65 \times 1050 = 682.5 \approx 683$$

About 683 people said that they were affected by the inflation. We rounded the answer to 683 to obtain a whole number of people. (The symbol $\approx$ is used to denote quantities that are approximately equal.)

3.2.2 Using Percentage to Describe a Change

Percentages can be used to describe the change in a quantity over a certain period of time. For example, there was a 2% increase in the value of market shares of a multinational company in the last month. Another example is the increase in the population of a city over a certain period of time. The population of the town was 100,000 in the 2010 census and 150,000 in the 2023 census. We can express the change in population in two basic ways. As the population rose by 50,000 people (from 100,000 to 150,000), we say that the absolute change in the population was 50,000 people. Another way is the percentage. As the increase of 50,000 people was 50% of the starting population of 100,000, we say that the relative change in the population was 50% or 0.5. (0.5 is $50/100 = 0.5$).

3.2.3 Using Percentage for Comparison

Percentage can be used to compare two numbers. For example, we can compare the price of two phones or vehicles, and other features. The battery of 'x' phone lasts 125% longer than the 'y' phone, but it costs 150% more. If we compare the price of Suzuki Alto (Rs. 2,200,000/-) and WagonR (Rs. 3,200,000/-), the WagonR is Rs. 1,000,000/- more expensive than Suzuki Alto. This difference can be expressed as given below:

$$3,200,000 - 2,200,000 = \text{Rs. } 1,000,000/-$$

This difference in the form of percentage of Suzuki Alto price is:

$$\frac{1,000,000}{2,200,000} = 0.4545 = 45.45\% \approx 45\%$$

In relative terms, the *WagonR* costs 45% more than Suzuki Alto.

In this example, we are comparing it to the price of the Suzuki Alto, therefore, the Suzuki Alto is the **reference value**. Note that while reporting the difference, the reference value comes after the word 'than' (please see the underlined text in the above statement.). The WagonR price is the **compared value**. The **absolute difference** is the actual difference between the compared value and the reference value:

$$\text{Absolute difference} = \text{compared value} - \text{reference value}$$

In the above example comparing the price of Suzuki Alto (Rs. 2,200,000/-) and WagonR (Rs. 3,200,000/-), the WagonR is Rs. 1,000,000/- more expensive than Suzuki Alto (it is called absolute difference). The **relative difference** describes the size of the absolute difference in comparison to the reference value and can be expressed as a percentage:

$$\text{Relative difference} = \frac{\text{compared value} - \text{reference value}}{\text{reference value}} \times 100$$

In the above example, the relative difference in the price of WagonR and Suzuki Alto is 69%.

If we take the WagonR price as the reference value and the Suzuki Alto price as the compared value, the absolute difference is given below:

$$2,200,000 - 3,200,000 = \text{Rs. } -1,000,000/-$$

The negative sign indicates that the Suzuki Alto price is Rs. 1,000,000/- less than WagonR.

The relative difference in this case with the Suzuki Alto price as compared value is:

$$\begin{aligned}\text{Relative difference} &= \frac{\text{Compared value} - \text{reference value}}{\text{reference value}} \times 100 \\ &= \frac{2,200,000 - 3,200,000}{3,200,000} \times 100 \\ &= \frac{-1,000,000}{3,200,000} = -.3125 \times 100 \\ &= -31.25\% \approx -31\%\end{aligned}$$

The negative sign indicates that the Suzuki Alto price is 31% less than the WagonR. In this way, we have two ways to express relative differences:

1. *WagonR* costs 45% more than Suzuki Alto.
2. *Suzuki Alto's* price is 31% less than the WagonR.

Both statements are correct. However, it is important to notice the compared and reference values in the given examples.

Example: Comparison of Salary

The average pay of teachers is different in the public and private sectors. Data show that the average pay of a government school teacher is 50,000/- per month and the monthly pay of a private school teacher is 45,000/-. Compare average pay in private schools to that in government schools in both absolute and relative terms.

Solution:

Based on the wording of the statement, we use the private school pay as the compared value and the government school pay as the reference value.

$$\begin{aligned}\text{Absolute difference} &= \text{compared value} - \text{reference value} \\ &= 45,000 - 50,000 \\ &= \text{Rs. } -5,000/-\end{aligned}$$

$$\begin{aligned}
 \text{Relative difference} &= \frac{\text{compared value} - \text{reference value}}{\text{reference value}} \times 100 \\
 &= \frac{45,000 - 50,000}{50,000} \times 100 \\
 &= \frac{\text{compared value} - \text{reference value}}{\text{reference value}} \\
 &= -.1 \times 100 \\
 &= -10\%
 \end{aligned}$$

In absolute terms, the average pay in private schools was Rs. 5,000/- less than the average pay in government schools. In relative terms, average pay in private schools was about 10% less than average pay in government schools. (You can verify that average pay in government schools was about 11% more than average pay in private schools.)

3.2.4 Percentage of Percentages

You may have observed the change in a quantity when the quantity itself is a percentage. For example, a bank increases its interest rate on a property loan from 4% to 5%. In absolute terms, the interest rate increased by 1%. The relative change in the interest rate is 25%, as calculated below:

$$\frac{\text{new value} - \text{reference value}}{\text{reference value}} \times 100\% = \frac{5\% - 4\%}{4\%} \times 100\% = 25\%$$

Example

The percentage of adults who report reading books fell from about 69% in 2001 to 22% in 2023. Describe this change in book readership.

Solution

The drop in readership from 69% to 22% represents a decline of $69 - 22 = 47$ percentage points. This is the absolute change in the percentage of adults who read the books. The relative change in readership is:

$$\frac{\text{new value} - \text{reference value}}{\text{reference value}} \times 100\% = \frac{22\% - 69\%}{69\%} \times 100\% \approx -68\%$$

The negative sign indicates a decrease in readership. We say that readership dropped by about 68% in relative terms or by 47 percentage points in absolute terms.

3.3 Ratio

In the previous section, we learnt that we use absolute and relative differences for comparing two quantities. Another way to compare is the ratio of the two quantities. The quantities with the same unit can be compared and the answer of the ratio is without units. For example, the ratio of two quantities Rs. 80,000/- and Rs. 20,000/- is:

$$\frac{80,000}{20,000} = \frac{4}{1} = 4$$

Here the unit of currency ‘Rupee’ is canceled at the top and bottom of the fraction. The ratio between the two quantities is 4 to 1. The ratio of price of two mobiles phones X1 (price of X1 is Rs. 50,000/-) and X2 (price of X2 is Rs. 25,000/-) is:

$$\frac{50,000}{25,000} = \frac{2}{1} = 2$$

We can say that the ratio of the price of X1 to X2 price is 2 to 1. We can also say that the price of the X1 mobile phone is twice the price of the X2 mobile phone. It can be stated that the ratio of X2 price to X1 price is 1 to 2 i.e., $\frac{1}{2}$ or 0.50 or 50%. Notice here that the unit of currency ‘Rupee’ is canceled leaving the ratio without units.

3.4 Abuses of Percentages

Percentages provide information for comparison and reference. However, there are certain points which need to be considered while calculating and using percentages.

3.4.1 Notice Reference Values

By shifting the reference value, we can get different results. Therefore, be aware of the reference values while calculating the percentage value. For example, because of some financial losses faced by the company, you agree to accept a temporary 10% cut in your monthly salary. Your employer promises to give you a 10% pay increase after six months. Will this pay raise restore your original salary?

We can answer this question by assuming some arbitrary number for your original weekly pay, such as Rs. 20,000/-. A 10% pay cut means that your pay will decrease by 10% of Rs. 20,000/-, or Rs. 2,000/-, so your weekly pay after the cut is

$$\text{Rs. } 20,000 - \text{Rs. } 2,000 = \text{Rs. } 18,000/-$$

The subsequent raise increases your pay by 10% of Rs. 18,000/-, or Rs.1,800/-, making your weekly pay

$$\text{Rs. } 18,000 + \text{Rs. } 1,800 = \text{Rs. } 19,800/-$$

Notice that the 10% pay cut followed by the 10% pay raise leaves you with less money than you started with. This result arises because the reference value for the calculations shifted during the problem: It was Rs. 20,000/- in the first calculation and Rs. 18,000/- in the second.

3.4.2 Less than Nothing

The word ‘percentage’ comes from two words: per and cent. It also means out of a total of 100 parts, the value or portion of a particular quantity. For example, a student “A” achieved 80% marks in the annual examination of class 9. It means that if the total marks are 100, the student has achieved 80 marks. We can say that it involves representing a percentage value out of 100. However, in real-world situations, we often see numbers that represent large “more than” percentages. For example, a price of Rs. 400/- is 300% more than a price of Rs. 100/-.

However, in most cases, it is not possible to have a “less than” percentage that is “greater than 100%”. To understand the reason, consider an advertisement claiming that replacing standard light bulbs with energy-efficient bulbs would use “200 percent less energy.” If you think about it, you will realize that such a savings is impossible. If the new light bulbs used 100% less energy, they would be using no energy at all. The only way they could use 200% less would be if they actually produced energy. Clearly, whoever wrote the advertisement made a mistake. So, in this example, the percentage value mentioned does not sound logical.

Let us consider another example. A store advertises that it will take “150% off” the price of all products. What should happen when you go to the counter to buy a Rs. 500/- item? If the price were 100% off, the item would be free. So if the price is 150% off, the store should pay you half the item’s cost i.e., Rs. 250/-. More likely, the store manager did not understand percentages.

Can a sportsman give 120% effort for a final match of Cricket? Can a glass of orange juice have a 120% of the daily requirement for Vitamins? Now think about these questions and write your answers.

3.4.3 Do not Average Percentages

As a general rule, you should never calculate the average of the percentages. Because it may provide a misleading answer. Suppose you got 70% of the questions correct on a midterm exam and 90% correct on the final exam. Can we conclude that you answered 80% of all the questions correctly? It might seem convincing apparently because, after all, 80% is the average (mean) of 70% and 90%. But it would be wrong unless both tests happened to have the same number of questions.

Let us look at it in detail. Suppose that the midterm had 20 questions and the final examination had 100 questions, making it 120 questions in total. Your 70% score on the midterm means that you answered 14 questions correctly (i.e., $70/100 \times 20 = 14$), while your 90% score on the final means you answered 90 questions correctly. Therefore, on the two exams combined, you answered 104 (i.e., $14 + 90 = 104$) out of 120 (i.e., $20 + 100 = 120$) questions correctly, which is 86.67%. This is much higher than the 80% “average” of the two individual exam percentages.

In baseball, a player’s batting average represents the percentage of at-bats in which he got a hit. For example, a batting average of .350 means the player got a hit 35% of the times he batted. Suppose a player had a batting average of .200 during the first half of the season and .400 during the second half of the season. Can we conclude that his batting average for the entire season was .300 (the average of .200 and .400)? Why or why not? Give an example that illustrates your reasoning.

The answer is no. For example, suppose he had 300 at-bats during the first half of the season and 200 at-bats during the second-half, for a total of 500 at-bats. His first-half batting average of .200 means he got hits on 20% of his 300 at-bats or $0.2 \times 300 = 60$ hits. His second-half batting average of .400 means he got hits on 40% of his 200 at-bats, or $0.4 \times 200 = 80$ hits. For the season, he got a total of $60 + 80 = 140$ hits in his 500 at-bats, so his season batting average was $140/500 = 28\%$, or .280—not the .300 found by averaging his first-half and second-half batting percentages. (In fact, the only case in which his season average would be .300 is if he had precisely the same number of at-bats in each half of the season.)

3.5 Large and Small Numbers

Large numbers like millions, billions, or trillions are common in our daily discourse, whether in discussions about government expenditures or in various technological contexts such as the storage capacity of phones and computers. However, the true comprehension of the magnitude of these numbers is not widespread. This section would explore several methods for providing a meaningful perspective on large or small numbers.

3.5.1 Scientific Notation

When dealing with large and small numbers, employing scientific notation proves to be a helpful technique. In this format, numbers are represented by a value between 1 and 10, multiplied by a power of 10. For instance, considering a billion as ten to the ninth power (10^9), we express 6 billion in scientific notation as 6×10^9 . Likewise, 420 can be written as 4.2×10^2 in scientific notation, and 0.67 as 6.7×10^{-1} . This approach facilitates a clearer understanding of the scale of numbers. The

power of 10 in these examples indicated the number of places the decimal place moved from right to left for a positive value of power (i.e., 10^9) and left to right for a negative value of power (10^{-1}).

Scientific notation is a format in which a number is expressed as a number between 1 and 10 multiplied by a power of 10

Scientific notation is a convenient way of representing numbers, especially large numbers, regardless of their magnitude. However, it is crucial to exercise caution and not let this ease of expression lead to misconceptions. As an illustration, the simplicity with which we can write the number 1080 might make it seem unremarkable, yet it surpasses the total count of atoms in the observable universe.

Examples

- 1) Total spending on education in the previous federal budget was 4,900,000,000,000. In scientific notation, it can be expressed as 4.9×10^{12} .
- 2) The diameter of a hydrogen nucleus is about 0.000000000000001 meter. It can be written as 1×10^{-15} .

(a) Procedure of Converting to Scientific Notation

To convert a number from ordinary notation to scientific notation:

Step–1: Move the decimal point to come after the first non-zero digit on the left side. (Note: For a quantity without a point, the point is present on the right side at the end. For example, 264 can be written as 264.00.)

Step–2: For the power of 10, use the number of places the decimal point moves. For example, the decimal points moved in the above mentioned quantity (2.64) is; the power is positive if the decimal point moves to the left and negative if it moves to the right.

Examples:

5034	Decimal moves 3 places to left ~~~~~	5.034×10^3
0.0024	Decimal moves places to the right ~~~~~	2.4×10^{-3}
169×10^2	Decimal moves 2 places to the left ~~~~~	$(1.69 \times 10^2) \times 10^2 = 1.69 \times 10^4$

In the last example, the two places decimal moves were added into 10^2 for expressing as one quantity.

(b) Procedure of Converting from Scientific Notation

To convert a number from scientific notation to ordinary notation:

Step–1: The power of 10 indicates how many places to move the decimal point; move it to the right if the power of 10 is positive and to the left if it is negative.

Step–2: If moving the decimal point creates any open places, fill them with zeros.

For example,

8.06×10^3	Move decimal 2 places to the right rrrrr	8060
2.8×10^5	Move decimal 5 places to the right rrrrr	280000
3.8×10^{-4}	Move decimal 4 places to the left rrrrr	0.00038

(c) Addition and Subtraction with Scientific Notation

In general, we must write numbers in ordinary notation before adding or subtracting.

Examples are given below:

$$\begin{aligned}
 (6 \times 10^3) + (4 \times 10^5) &= 6,000 + 400,000 \\
 &= 406,000 \\
 &= 4.06 \times 10^5
 \end{aligned}$$

$$\begin{aligned}
 (14.2 \times 10^3) + (51 \times 10^4) &= 14,200 + 510,000 \\
 &= 524,200 \\
 &= 5.24 \times 10^5
 \end{aligned}$$

When both numbers have the same power of 10, we can factor out the power of 10 first.

$$\begin{aligned}
 (4 \times 10^8) + (8 \times 10^8) &= (4 + 8) \times 10^8 \\
 &= 12 \times 10^8 \\
 &= 1.2 \times 10^9
 \end{aligned}$$

$$\begin{aligned}
 (5.6 \times 10^{-25}) + (3.8 \times 10^{-25}) &= (5.6 + 3.8) \times 10^{-25} \\
 &= 9.4 \times 10^{-25}
 \end{aligned}$$

(d) Multiplying or Dividing with Scientific Notation

Multiplying or dividing numbers in scientific notation simply requires two tasks:

- (i) operating on the powers of 10 and
- (ii) working with the other parts of the number, separately.

$$\begin{aligned}(2 \times 10^2) \times (6 \times 10^4) &= (2 \times 6) \times (10^2 \times 10^4) \\ &= 12 \times 10^6 \\ &= 1.2 \times 10^7\end{aligned}$$

$$\begin{aligned}\frac{3.6 \times 10^{-3}}{4.2 \times 10^{-5}} &= \frac{3.6}{4.2} \times \frac{10^{-3}}{10^{-5}} \\ &= 0.86 \times 10^{-2 - (-5)} \\ &= 0.86 \times 10^3 \\ &= 8.6 \times 10^2\end{aligned}$$

Self-Assessment Questions

1. Convert each of the following numbers from scientific to ordinary notation.
a. 2×10^6
 - i. 3×10^4
 - ii. 5.2×10^{-3}
 - iii. 2.8×10^{-2}
 - iv. 4.5×10^{-2}
 - v. 6.45×10^{-5}
2. Write the following numbers in scientific notation.
 - (i) 53829
 - (ii) 6784
 - (iii) 0.05674
 - (iv) 7893.56
 - (v) 452567
 - (vi) 568.00001
3. Solve the following problems and express the answer in scientific notation. You can round off your answer to one decimal place (e.g., 3.2×10^3).
 - (i) $(3 \times 10^3) \times (2 \times 10^2)$
 - (ii) $(4 \times 10^2) \times (3 \times 10^8)$
 - (iii) $(3 \times 10^3) + (2 \times 10^2)$
 - (iv) $(8 \times 10^3) + (2 \times 10^5)$
 - (v) $(7 \times 10^8) \div (3 \times 10^9)$
 - (vi) $(5 \times 10^6) + (14 \times 10^8)$
 - (vii) $(94 \times 10^5) \div (2 \times 10^3)$
 - (viii) $(8 \times 10^{12}) \div (4 \times 10^4)$

3.6 Significant Digits

The digits in a number that represent actual measurements are called significant digits. For example, 132 pounds has 3 significant digits and implies a measurement to the nearest pound, while 132.00 pounds has 5 significant digits and implies a measurement to the nearest hundredth of a pound.

Note that zeros are significant when they represent actual measurements, but not when they serve only to locate the decimal point. For example, when dealing with measurements, 147 pounds and 147.00 pounds do not have the same meaning. Writing weight as 147 pounds means to the nearest pound whereas 147.00 pounds means to the nearest hundredth of a pound, rather than to the nearest pound. We assume that the zeros in 147.00 pounds are significant until they represent an actual measurement. In contrast, we assume that the zeros in 600 centimeters are not significant, because they serve only to tell us that the decimal point comes to their right. Rewriting 600 *centimeters* as 6 *meters* makes it easier to see that only the 6 is a significant digit.

The only subtlety in counting significant digits arises when we cannot be sure whether zeros are truly significant. For example, suppose your professor states that there are 200 students in your class. Without further information, you have no way to know whether she means exactly 200 students or roughly 200. We can avoid this kind of ambiguity by writing numbers in scientific notation. In that case, zeros appear only when they are significant. For example, an enrollment of 2×10^2 implies a measurement to the nearest hundred students, while 2.00×10^2 implies exactly 200 students.

Summary: When Digits are significant?

Type of Digit	Significance
Nonzero digits	Always significant
Zeros that follow a nonzero digit and lie to the right of the decimal point (as in 5.80 or 5.00)	Always significant
Zeros between nonzero digits (as in 3006 or 4.008) or other significant zeros (such as the first zero in 60.0)	Always significant
Zeros to the left of the first nonzero digit (as in 0.003 or 0.00068)	Never significant
Zeros to the right of the last nonzero digit but before the decimal point (as in 20,000 or 410)	Not significant unless stated otherwise

3.6.1 Rounding

Rounding means making a number simpler but keeping its value close to what it was. For example, 73 rounded to the nearest ten is 70, because 73 is closer to 70. Similarly, 76 goes up to 80 because it is close to 80. The result is less accurate, but easier to use. The basic process of rounding numbers takes just two steps.

Step–1: Decide which decimal place (e.g., tens, ones, tenths, or hundredths) is the smallest that should be kept.

Step–2: Look at the number in the next place to the right (for example, if rounding to tenths, look at hundredths). If the value in the next place is less than 5, round down; if it is 5 or greater, round up.

For example, the number 382.2593 is given to the nearest ten-thousandth. It can be rounded in the following ways:

- 382.2593 rounded to the nearest thousandth is 382.259.
- 382.2593 rounded to the nearest hundredth is 382.26.
- 382.2593 rounded to the nearest tenth is 382.3.
- 382.2593 rounded to the nearest one is 382.
- 382.2593 rounded to the nearest ten is 380.
- 382.2593 rounded to the nearest hundred is 400.

3.6.2 Rounding with Significant Digits

For each of the following operations, give your answer with the specified number of significant digits.

- a. Solve the problem and give your answer with 2 significant digits.

$$7.7 \text{ mm} \times 9.92 \text{ mm}$$

$$7.7 \text{ mm} \times 9.92 \text{ mm} = 76.384 \text{ mm}^2 = 76 \text{ mm}^2$$

Because we are asked to give the answer with 2 significant digits, we round to 76 mm^2 .

- b. Solve the problem and give your answer with 4 significant digits.

$$240,000 \times 72,106$$

$$240,000 \times 72,106 = 1.730544 \times 10^{10} = 1.731 \times 10^{10}$$

Because we are asked to give the answer with 4 significant digits, we round to 1.731×10^{10} .

Self-Assessment Questions

State the number of significant digits and the implied meaning of the following numbers.

- a) 13.40 seconds (time)
- b) Length of 0.000067 meter (length)
- c) 0.0030 gram (Weight)
- d) 240,000 (population of a city)
- e) 3.60×10^6 (population of a city)

3.7 Measurement Errors

Measurement error is the amount of inaccuracy. Precision is a measure of how well a result can be determined (without reference to a theoretical or true value). It is the degree of consistency and agreement among independent measurements of the same quantity; also the reliability or reproducibility of the result. Measurement errors fall into two categories: random errors and systematic error.

3.7.1 Random and Systematic Errors

Random errors occur because of random and inherently unpredictable events in the measurement process. *Systematic errors* occur when there is a problem in the measurement system that affects all measurements in the same way, such as making them all too low or too high by the same amount. Suppose you are measuring the weight of a glass object, the weight of the object ranges between 3 pounds – 3.5 pounds when you put the glass object on the scale. We can say that the scale introduces a random error, because any particular measurement may be either too high or too low/falls within the above-mentioned range. If we notice that the minimum reading scale shows, is 0.5 pounds when there is nothing on it, this type of error is called a systematic error, because it is caused by an error in the measurement system—that is, an error that consistently (systematically) affects all measurements.

If we discover a systematic error, we can go back and adjust the affected measurements. In contrast, the unpredictable nature of random errors makes it impossible to correct them. However, you can minimize the effects of random errors by making many measurements and averaging them. For example, if you measure the baby's weight ten times, your measurements will probably be too high in some cases and too low in others. You can therefore get a better estimate of the baby's true weight by averaging the ten individual measurements.

3.7.2 Size of Errors: Absolute versus Relative Error

Often interested to know whether an error is big enough to be of concern or small enough to be unimportant, the concept of absolute versus relative error may be helpful. The absolute error describes how far a measured (or claimed) value lies from the true value.

$$\text{Absolute error} = \text{measured value} - \text{true value}$$

The relative error compares the size of the absolute error to the true value and is often expressed as a percentage:

$$\text{Relative Error} = \frac{\text{measured value} - \text{true value}}{\text{true value}} \times 100\%$$

The absolute and relative errors are positive when the measured or claimed value is greater than the true value and negative when the measured or claimed value is less than the true value. Note the similarity between the ideas of absolute and relative error and those of absolute and relative change or difference.

Suppose you go to the store and buy what you think is 6 pounds of sugar, but because the store's scale is poorly set, you actually get only 4 pounds. You would probably be upset by this 2-pound error. Now suppose you are buying food for a celebration in your village and you order 3,000 pounds of rice, but you actually receive only 2,998 pounds. You are short by the same 2 pounds as before, but in this case, the error probably doesn't seem very important.

In a more technical language, the 2-pound error in both cases is an absolute error; it describes how far the claimed or measured value lies from the true value. A relative error compares the size of an absolute error to the true value. The relative error for the first case is fairly large because the absolute error of 2 pounds is half the true weight of 4 pounds; we say that the relative error is $2/4$, or 50%. In contrast, the relative error for the second case is the absolute error of 2 pounds divided by the true hamburger weight of 2,998 pounds, which is only $2/2998 \approx 0.00067$, or 0.067%.

Examples

- 1) The true weight of a person is 125 pounds, but the scale, s/he uses to weigh, shows 130 pounds. Find the absolute and relative error in this case.

The measured value is the scale reading of 130 pounds, and the true value is 125 pounds. The absolute and relative errors are:

$$\begin{aligned} \text{Absolute Error} &= \text{measured value} - \text{true value} \\ &= 130 \text{ lb} - 125 \text{ lb} = 5 \text{ lb} \end{aligned}$$

$$\text{Relative Error} = \frac{\text{measured value} - \text{true value}}{\text{true value}} \times 100\%$$

$$= \frac{5 \text{ lb}}{125 \text{ lb}} \times 100\% = 4\%$$

The measured weight is too high by 5 pounds, or 4%.

- 2) The government claims that a developmental dam program costs 49.0 billion rupees, but an audit shows that the true cost is 50.0 billion rupees. Find the absolute and relative error in this case.

The claimed cost of 49.0 billion rupees can be treated as the measured value. The true value is the true cost of 50.0 billion rupees. The absolute and relative errors are:

Absolute Error = 49.0 billion rupees - 50.0 billion rupees

$$= -1.0 \text{ billion rupees}$$

$$\textbf{Relative Error} = \frac{49.0 \text{ billion} - 50.0 \text{ billion}}{50.0 \text{ billion}} \times 100\%$$

$$= -2\%$$

The claimed cost is too low by 1.0 billion rupees or 2%.

3.8 Accuracy and Precision in Reporting Results

Accuracy describes how closely a measurement approximates a true value. An accurate measurement has a small relative error. Precision describes the amount of detail in a measurement.

The goal of any measurement is to obtain a value that is as close as possible to the true value. Accuracy describes how close the measured value lies to the true value. Precision describes the amount of detail in the measurement. For example, suppose a census says that the population of your hometown is 72,453 but the true population is 96,000. The census value of 72,453 is quite precise because it seems to tell us the exact count, but it is not very accurate because it is nearly 25% smaller than the actual population of 96,000. Note that accuracy is usually defined by relative error rather than absolute error. For example, if a company projects sales of \$7.30 billion and true sales turn out to be \$7.32 billion, we say the projection was quite accurate because it had a relative error of less than 1%, even though the absolute error of \$0.02 billion represents \$20 million.

Example

Suppose the true weight of a person is 102.4 pounds. The scale at the doctor's office, which can be read only to the nearest quarter pound, says that that person weighs $102\frac{1}{4}$ pounds. The scale at the gym, which gives a digital readout to the nearest 0.1 pound, says that he weighs 100.7 pounds. Which scale is more precise? Which is more accurate?

Solution: The scale at the gym is more precise, because it gives his weight to the nearest tenth of a pound while the doctor's scale gives his weight only to the nearest quarter pound. However, the scale at the doctor's office is more accurate, because its value is closer to his true weight.

3.9 Combining Numbers

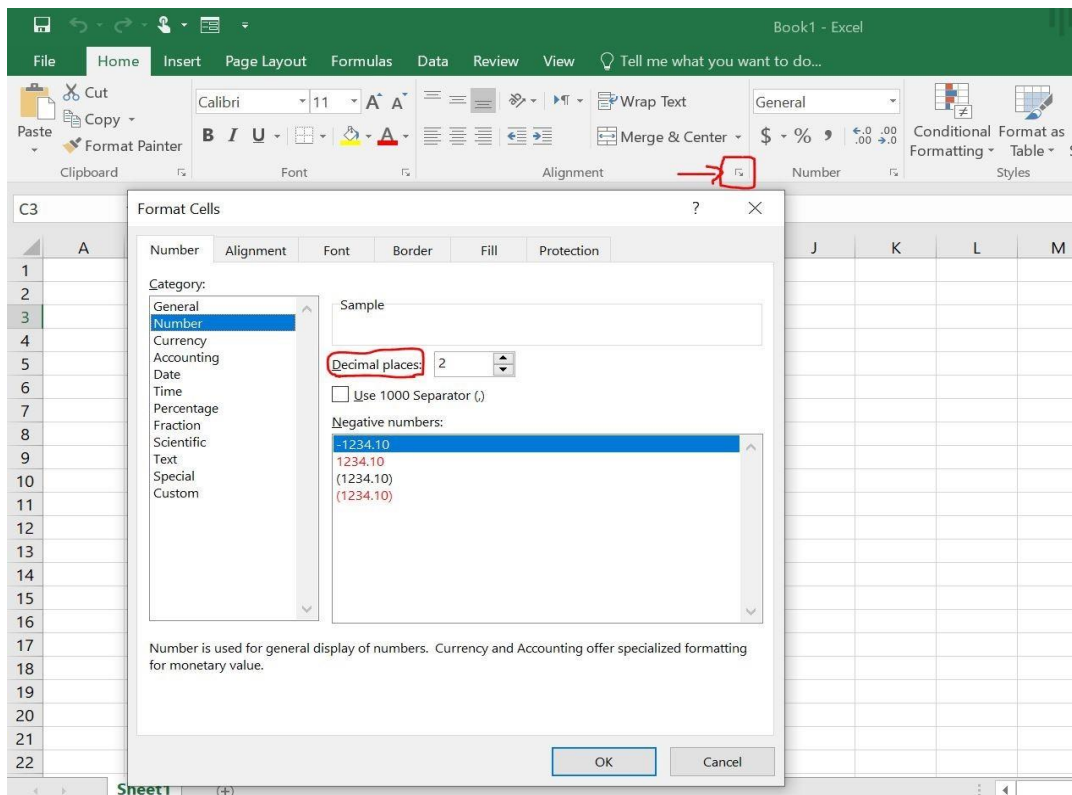
We must be very careful when we combine measured numbers. Otherwise, we may state answers with more certainty than they deserve. In scientific or statistical work, researchers conduct careful analyses to determine how to combine numbers properly. For most purposes, however, we can use two simple rounding rules given below.

- **Rounding rule for addition or subtraction:** Round your answer to the same precision as the least precise number in the problem.
- **Rounding rule for multiplication or division:** Round your answer to the same number of significant digits as the measurement with the fewest significant digits.

Note: To avoid errors, the rounding must only be done after completing all the operations, not during intermediate steps.

Suppose the population of your city is One day, your relative moves to your city to stay in your house. What is the population of your city now? You might be tempted to add your relative to the city's population, making the new population 300,001. However, the number 300,000 has only 1 significant digit, implying that the population is known only to the nearest 100,000 people. The number 300,001 has 6 significant digits, which implies that you know the population exactly. Clearly, your relative's move cannot change the fact that the population is known only to the nearest 100,000. The population is still 300,000, despite the addition of your relative.

In excel, the numbers are rounded to specify the precision. First, open the "Format Cells", dialog box, there is an option to format numbers in the cell that allows to choose how many decimal places are displayed.



Examples

1. A book written in 1985 states that the oldest Mayan ruins are 2000 years old. How old are they now?

A book written in 1985 is roughly 30 years old, so we might be tempted to add 30 years to 2000 years to get 2030 years for the age of the ruins. However, 2000 years is the less precise of the two numbers: It is precise only to the nearest 1000 years, while 30 years is precise to the nearest 10 years. Therefore, the answer also should be precise only to the nearest 1000 years:

2000 yr	+	30 yr =	2030 yr	≈	2000yr
Precise to nearest 1000		Precise to nearest 10	Must round to nearest 1000		Correct final answer

Given the precision of the age of the ruins, they are still 2000 years old, despite the 30-year age of the book.

2. The government in a town of 82,000 people plans to spend 41.5 million rupees this year. Assuming all this money must come from taxes, what average amount must the city collect from each resident?

We find the average tax by dividing the 41.5 million rupees, which has 3 significant digits, by the population of 82,000, which has 2 significant digits. The population has the fewest significant digits, so the answer should be rounded to match its 2 significant digits.

$$41,500,000 \text{ rupees} \div 82,000 \text{ persons} = \text{Rs. } 506.10/- \text{ per person} \approx \text{Rs. } 510/- \text{ per person}$$

3 significant
digits

2 significant
digits

Must round to 2
significant digits

Correct final
answer

The average resident must pay about Rs. 510/- in taxes.

3.10 Index Number

An index number provides a simple way to compare measurements made at different times or in different places. The value at one particular time (or place) is chosen as the reference value. The index number for any other time (or place) is:

$$\text{Index number} = \frac{\text{value}}{\text{reference value}} \times 100$$

The term index is commonly used for almost any kind of number that provides a useful comparison, even when the numbers are not standard index numbers. For example, body mass index (BMI) provides a way of comparing people by height and weight, but is defined without any reference value. Specifically, body mass index is defined as weight (in kilograms) divided by height (in meters) squared.

Example

Suppose the cost of gasoline today is 350/- per gallon. Using the 1980 price as the reference value, find the price index for gasoline today.

Solution: We use the 1980 price of 20/- per gallon as the reference value to find the price index for the 350/- gasoline price today.

$$\text{Index number} = \frac{\text{current price}}{\text{1980's price}} \times 100 = \frac{350}{20} \times 100 = 1750$$

This price index for the current price is 1750, which means the current gasoline price is 1750% of the 1980's price.

3.10.1 Consumer Price Index

Consumer Price Index (CPI) is based on an average of prices for a sample of more than 60,000 goods, services, and housing costs. It is computed and reported after regular intervals. The CPI allows us to compare overall prices at different times. For example, to find out how much higher typical prices were in 2015 than in 1995, we compute the ratio of the CPIs for the two years, using the shorthand CPI₂₀₁₅ to represent the CPI for 2015 (and similar notation for other years):

$$\frac{\text{CPI}_{2015}}{\text{CPI}_{1995}} = \frac{218.1}{152.4} = 1.43$$

Based on the CPI, typical prices in 2015 were 1.43 times those in 1995, or 43% higher. For example, a typical item that cost Rs.1000/- in 1995 would have cost Rs.1430/- in 2015.

Example

Suppose you needed Rs. 30,000/- to maintain a particular standard of living in 2000. How much would you have needed in 2020 to maintain the same living standard? Assume that the average price of your typical purchases has risen at the same rate as the Consumer Price Index (CPI).

Solution: We compare prices for two different years by comparing the CPIs for those years:

$$\frac{\text{CPI}_{2020}}{\text{CPI}_{2000}} = \frac{229.6}{172.2} = 1.333$$

That is, typical prices in 2020 were 1.333 times those in 2000. If you needed Rs.30,000/- for your standard of living in 2000, then in 2020 you would have needed $1.333 \times \text{Rs. } 30,000/- = \text{Rs. } 40,000/-$ for the same standard of living.

3.10.2 Rate of Inflation

The rate of inflation refers to the relative change in the CPI from one year to the next. For example, the inflation rate from 1978 to 1979 was the relative increase in the CPI between those two years:

$$\begin{aligned}\text{Inflation rate 1978 to 1979} &= \frac{\text{CPI}_{1979} - \text{CPI}_{1978}}{\text{CPI}_{1978}} \\ &= \frac{72.6 - 65.2}{65.2} \\ &= 0.113 = 11.3\%\end{aligned}$$

As measured by the CPI, the rate of inflation from 1978 to 1979 was 11.3%.

Main points:

1. Errors can occur in many ways, but generally can be classified as one of two basic types: random errors or systematic errors.
2. Whatever the source of an error, its size can be described in two different ways: as an absolute error or as a relative error.
3. Once a measurement is reported, we can evaluate it in terms of its accuracy and its precision.

SUMMARY

In this unit, we have observed the influence of numeracy in our everyday experiences. From simple tasks like counting and measuring to more complex applications such as financial management and data analysis, numbers serve as essential tools for understanding and interacting with the world around us.

Through our exploration of fundamental numerical concepts and practical applications, we have gained valuable insights into the importance of numerical literacy in facilitating informed decision-making, problem-solving, and effective communication. By honing our arithmetic skills and embracing numeracy as a foundational skill set, we have equipped ourselves with the tools necessary to navigate the challenges and opportunities of modern life. Moreover, our journey has emphasized the interconnectedness of numeracy with other domains of knowledge, from science and technology to economics and beyond. By recognizing the interdisciplinary nature of numerical literacy, we are prepared in a better way to engage with complex issues.

As we come to the end of this unit, let us reflect on the significance of numbers in shaping our daily experiences and recognize the importance of ongoing learning and development in this critical area. By continuing to cultivate our numerical skills and seeking out opportunities for growth, we can master the power of numbers to enrich our lives and make positive contributions to the world around us.

EXERCISES

Multiple-Choice Questions

- What is 25% expressed as a fraction?
a) $\frac{1}{2}$ b) $\frac{1}{4}$
c) $\frac{1}{5}$ d) $\frac{1}{8}$
Answer: b) $\frac{1}{4}$
- If the price of an item increases from 200 PKR to 250 PKR, what is the percentage increase?
a) 20% b) 25%
c) 30% d) 35%
Answer: b) 25%
- Which of the following is an abuse of percentages?
a) Averaging percentages directly
b) Using percentages to show change
c) Using percentages for comparison
d) Converting percentages into fractions
Answer: a) Averaging percentages directly
- Convert 0.00045 to scientific notation.
a) 4.5×10^{-4} b) 4.5×10^{-3}
c) 4.5×10^{-2} d) 4.5×10^{-5}
Answer: a) 4.5×10^{-4}
- What is the result of $(2.5 \times 10^3) + (3.5 \times 10^3)$ in scientific notation?
a) 6.0×10^3 b) 5.0×10^3
c) 7.5×10^3 d) 8.5×10^3
Answer: a) 6.0×10^3
- What is the absolute error if the measured value is 12.5 cm and the true value is 12.7 cm?
a) 0.2 cm b) 0.1 cm
c) 0.3 cm d) 0.4 cm
Answer: a) 0.2 cm
- How is precision different from accuracy?
a) Precision refers to correctness, accuracy to consistency.
b) Precision refers to consistency, accuracy to correctness.
c) Precision is about significant digits, and accuracy is about rounding.
d) Precision is for large numbers, accuracy for small numbers.
Answer: b) Precision refers to consistency, accuracy to correctness.

8. What does the Consumer Price Index (CPI) measure?
 - a) Inflation rate
 - b) Change in prices over time
 - c) Cost of living
 - d) All of the above

Answer: d) All of the above
9. What is the result of $(4.0 \times 10^2) \times (2.0 \times 10^3)$ in scientific notation?
 - a) 8.0×10^6
 - b) 8.0×10^5
 - c) 8.0×10^4
 - d) 8.0×10^3

Answer: b) 8.0×10^5
10. If a measurement is reported as 25.0 ± 0.5 , what is the relative error?
 - a) 2%
 - b) 4%
 - c) 6%
 - d) 8%

Answer: a) 2%

Short-Answer Questions

1. Convert 3.75 to a percentage.
2. Define "absolute error" and give one example.
3. Write down the procedure for converting a number to scientific notation.
4. How do you add two numbers in scientific notation? Provide an example.
5. Differentiate between accuracy and precision with a practical example.
6. What is the significance of rounding to significant digits in problem-solving?
7. Explain the abuses of percentages with an example.
8. Convert 7.5×10^2 back to standard form.
9. Calculate the percentage of 5% of 1000.
10. What does the Index Number represent in economic terms?

Subjective Questions

1.
 - a) If the price of an item decreases from 500 PKR to 400 PKR, calculate the percentage decrease.
 - b) Compare the income ratio of two individuals earning 20,000 PKR and 25,000 PKR. Express it in the simplest form.
2. Solve the following:
 - a) Convert 0.0075 into scientific notation.
 - b) Multiply $(2.0 \times 10^5) \times (4.0 \times 10^3)$ and express the result in scientific notation.
3. A scientist measures the length of a table as 120.5 cm. The actual length is 121.0 cm.
 - a) Calculate the absolute error.
 - b) Calculate the relative error as a percentage.

4. If the Consumer Price Index (CPI) for a base year is 100 and the Consumer Price Index (CPI) for the current year is 125, calculate the inflation rate. Explain how CPI affects daily life.
5.
 - a) Add $(3.0 \times 10^3) + (2.5 \times 10^2)$. Round the result to two significant digits.
 - b) Subtract $(6.0 \times 10^4) - (5.0 \times 10^3)$ and express the answer in scientific notation.
6. Calculate 30% of 50% of 800. Show all steps clearly and explain the concept of percentage of percentages.
7. Explain the differences between random and systematic errors in measurements with examples. Why is it important to understand these errors in order to report the results?

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Unit–4

MANAGING MONEY

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INTRODUCTION

Managing money is an essential skill that affects every aspect of our lives, from our day-today expenses to our long-term financial security. In today's fast-paced and ever-changing world, the ability to effectively manage finances has become increasingly important for individuals and families alike. Whether it is creating a budget, saving for emergencies, or planning for retirement, mastering the art of money management is crucial for achieving financial stability and success.

In this unit, we will explore the fundamentals of managing money, offering practical tips and strategies to help you take control of your finances and make informed decisions about your financial future. From understanding income and expenses to setting financial goals and making smart investments, we will cover a wide range of topics designed to empower you to achieve your financial aspirations.

Through real-life examples, interactive exercises, and actionable advice, we will delve into the key principles of money management and provide you with the tools and knowledge you need to navigate the complexities of personal finance with confidence and ease. Whether you are just starting your financial journey or looking to refine your existing financial habits, this guide will equip you with the skills and resources necessary to build a solid foundation for financial success.

LEARNING OUTCOMES

After completing the unit, you would be able to:

1. Understand the importance of effective money management in achieving financial stability and long-term goals.
2. Create and maintain a personal budget that reflects income, expenses, and financial priorities.
3. Identify and prioritize financial goals such as saving for emergencies, paying off debt, and investing for the future.
4. Demonstrate proficiency in basic financial calculations, including calculating interest, budgeting for expenses, and determining savings goals.
5. Develop critical thinking skills to evaluate financial products and services such as loans and investment opportunities.
6. Apply financial management principles to real-life scenarios, making informed decisions to achieve personal financial goals and enhance overall financial well-being.

4.1 Managing Money

Money is not everything, but it certainly has a great influence on our lives. Most people would like to have more money, and there is no doubt that more money allows you to do things that simply aren't possible with less. However, when it comes to personal happiness, studies show that the amount of money you have is less important than having your personal finances under control. People who lose control of their finances tend to suffer from financial stress, which in turn leads to higher divorce rates and other difficulties in personal relationships, higher rates of depression, and a variety of other ailments. In contrast, people who manage their money well are more likely to say they are happy, even when they are not particularly wealthy. So if you want to attain happiness—along with any financial goals you might have—the first step is to make sure you understand your personal finances enough to keep them well under control.

4.2 Basics of Budget

A budget is a tool that expresses activities in financial terms. A budget shows the money you need to spend to carry out your planned activities. The money you need to spend is called expenditures or costs. The money you need to make/receive is called income. The process of making the budget is called budgeting. It helps to control and monitor your finances.

Example:

Ali is not a very rich person. He is a student. He loves to sit down for tea at the college coffee shop. With tax and tips, he usually spends Rs. 25/- on his tea. He gets at least one a day (on average), and about every three days, he has a second cup of tea. He figures that it is not a big expense. Is it?

Solution: One cup of tea a day means 365 cups per year. A second cup of tea on every third day adds about $365 \div 3 = 121$ more cups (rounding down). That means $365 + 121 = 486$ lattes a year. At Rs. 25/- per cup, this comes to:

$$486 \times 25/- = \text{Rs. } 12,150/-$$

Ali's tea habit is costing him more than Rs. 12,150/- per year. That might not be much if he is financially well settled. But it is more than one month of rent for an average college student. He could easily build a savings balance of more than Rs. 12,150/- over the next ten years.

Making a budget means keeping track of how much money you have coming in and going out and then deciding what adjustments you need to make. The following box summarizes the four basic steps in making a budget.

A Four-Step Budget

1. List all your monthly income. Be sure to include an average monthly amount for any income you do not receive monthly (such as once-per-year payments).
2. List all your monthly expenses. Be sure to include an average amount for expenses that do not recur monthly, such as expenses for tuition, books, vacations, and holiday gifts.
3. Subtract your total expenses from your total income to determine your net monthly cash flow.
4. Make adjustments as needed.

The most difficult part of the budget process is making sure you include every expense in your list of monthly expenses. A good technique is to keep careful track of your expenses for a few months. For example, carry a small notepad with you and write down everything you spend, or use a personal budgeting app that will work with your phone or tablet. And do not forget your occasional expenses, or else you may severely underestimate your average monthly costs.

Once you have made your lists for steps 1 and 2, the third step is just arithmetic: Subtracting your monthly expenses from your monthly income gives you your overall monthly cash flow. If your cash flow is positive, you will have money left over at the end of each month, which you can use for savings. If your cash flow is negative, you have a problem: You will need to find a way to balance it out, either by earning more, spending less, using savings, or taking out a loan.

Example:

In addition to monthly expenses, Ali has the following college expenses that he pays twice a year: Rs. 35000/- for the tuition each semester, Rs. 7500/- in student fees each semester, and Rs. 800/- for textbooks each semester. How should you handle these expenses in computing your monthly budget?

Solution: Because these expenses are paid twice a year, the total amount paid over a whole year is:

$$2 \times (35000 + 7500 + 800) = 86,600/-$$

To average this total expense for the year on a monthly basis, we divide it by 12:

$$86,600 \div 12 \approx \text{Rs. } 7220/-$$

Your average monthly college expense for tuition, fees, and textbooks comes to a little less than Rs. 7220/-, so Ali should put Rs. 7220/- per month into his expense list.

4.3 Budget and Financial Goals

A careful analysis of the budget will be helpful. For example, many people find that they are spending a lot more in certain categories than they had imagined and that the items they thought were causing their biggest expenses are small compared to other items. Once you evaluate your current budget, you will almost certainly want to make adjustments to improve your cash flow for the future. However, there are no set rules for adjusting your budget, so you will need to use your critical thinking skills to come up with a plan that makes sense for you. You can also get help from your family members and teachers for making decisions about financial planning.

Figuring out your monthly budget is a crucial step in taking control of your personal finances, but it is only the beginning. Once you have understood your budget, you need to start looking at longer-term financial issues. The general principle is always the same: Before making any major expenditure or investment, be sure you figure out how it will affect your finances over the long term.

It is rare for a financial question to have a clear “best” answer for everyone. Instead, your decisions depend on your current circumstances, your goals for the future, and some unavoidable uncertainty. The key to financial success is to approach all your financial decisions with a clear understanding of the available choices. To achieve your financial goals, go through the following points:

1. Achieving your financial goals will almost certainly require that you build up savings over time. Although it may be difficult to save while you are in college, ultimately you must find a way to save. You will also need to understand how savings work and how to choose appropriate savings plans.
2. You will probably need to borrow money at various points in your life. You may already have credit cards, or you may be using student loans to help pay for college or university fees. In the future, you may need loans for large purchases, such as a car or a home. Because borrowing is very expensive, it is critical that you understand the basic mathematics of loans so you can make wise choices.
3. Many of our financial decisions have consequences on our taxes. Sometimes, these tax consequences can be large enough to influence our decisions. For example, the fact that interest on house payments is tax deductible while rent is not may influence your decision to rent or buy. While no one can expect to understand tax lawfully, it is important to have at least a basic understanding of how taxes are computed and how they can affect your financial decisions.
4. Finally, we do not live in isolation, and our personal finances are inevitably intertwined with those of the government. For example, when politicians allow the government to run deficits, it means that future politicians will have to collect more taxes from you or your children. So, understanding the federal budget is important for your future.

4.4 Simple versus Compound Interest Formula

The principal in financial formulas is the balance upon which interest is paid. Simple interest is interest paid only on the original investment and not on any interest added at later dates. Compound interest is interest paid both on the original investment and on all interest that has been added to the original investment.

Imagine that you deposit Rs. 1000/- in a bank Y, which promises to pay 5% interest each year. At the end of the first year, the bank sends you a check for:

$$5\% \times 1000 = 0.05 \times 1000 = \text{Rs. } 50/-$$

Because you receive a check for your interest, your balance with the bank remains Rs.1000/-, so you get the same Rs.50/- payment at the end of the second year, and the same again for the third year. Your total interest for the three years is $3 \times 50 = \text{Rs. } 150/-$. Therefore, your original investment of Rs. 1000/- grew in value to Rs. 1150/-. The Bank's method of payment represents simple interest, in which interest is paid only on your initial investment. The amount of money on which interest is paid is called the principal.

To understand compound interest, suppose that you place the Rs. 1000/- in a bank account in a bank Z that pays the same 5% interest once a year. Instead of paying you the interest directly, the bank adds the interest to your account. At the end of the first year, the bank deposits $5\% \times \text{Rs. } 1000/- = \text{Rs. } 50/-$ interest into your account, raising your balance to Rs.1050/. At the end of the second year, the bank again pays you 5% interest. This time, however, the 5% interest is paid on the balance of Rs.1050/-, so it amounts to:

$$5\% \times 1050 = 0.05 \times 1050 = \text{Rs. } 52.50/-$$

Adding this Rs. 52.50/- raises your balance to

$$1050 + 52.50 = \text{Rs. } 1102.50/-$$

This is the new balance on which your 5% interest is computed at the end of the third year, so your third interest payment is:

$$5\% \times 1102.50 = 0.05 \times 1102.50 = \text{Rs. } 55.13/-$$

Therefore, your balance at the end of the third year is

$$1102.50 + 55.13 = \text{Rs. } 1157.63/-$$

Despite identical interest rates, you end up with Rs. 7.63/- more if you use the bank Z instead of bank Y. The difference comes about because the bank Z pays you interest on the interest as well as on the original principal. This type of interest payment is called compound interest. This formula can be used to calculate profit on a certain amount of investment in a business.

4.2.1 Compound Interest Formula

Suppose that Ali made an investment of Rs. 500/- in a business for 30 years. Let us assume that the interest rate is 4% per year. For each year, we can calculate the interest and the new balance with interest. The first three columns of Table 2 show these calculations for 04 years.

Table 4.1: Calculating Compound Interest (starting principal P Rs. 500/-, annual interest rate APR 4%)

After 'n' Years	Interest	Balance	Or Equivalently
Year 1	$4\% \times 500 = 20$	$500 + 20 = 520$	$500 \times 1.04 = 520$
Year 2	$4\% \times 520 = 20.8$	$520 + 20.8 = 540.8$	$520 \times (1.04)^2 = 540.8$
Year 3	$4\% \times 540.8 = 21.63$	$540.8 + 21.63 = 562.43$	$520 \times (1.04)^3 = 562.43$
Year 4	$4\% \times 562.43 = 22.49$	$562.43 + 22.49 = 584.93$	$520 \times (1.04)^4 = 584.93$

As shown in the fourth column, there is a much easier way to calculate the profit rate. The 4% annual interest rate means that each end-of-year balance is 104% of, or 1.04 times, the previous year's balance. As shown in the last column of Table 4.1, we can get each balance as follows:

- The balance at the end of 1 year is the initial deposit of Rs. 500/- times 1.04:

$$500 \times 1.04 = 520/-$$

- The balance at the end of 2 years is the 1-year balance times 1.04, which is equal to the initial deposit times $(1.04)^2$:

$$500 \times 1.04 \times 1.04 = 500 \times (1.04)^2 = 540.8/-$$

- The balance at the end of 3 years is the 2-year balance times 1.04, which is equal to the initial deposit times $(1.04)^3$:

$$500 \times 1.04 \times 1.04 \times 1.04 = 500 \times (1.04)^3 = 562.43/-$$

- Continuing the pattern, we find that the balance after Y years is the initial deposit times 1.04 raised to the power of those number of years. For example, the balance after Y = 10 years is:

$$500(1.04)^{10} = 740.12/-$$

When interest is compounded just once a year, as it is in this case, the interest rate is called the **annual percentage rate, or APR**. The number of compounding periods is then simply the number of years Y over which the principal earns interest. We therefore obtain the following general formula for interest compounded once a year.

The Compound Interest Formula (for Interest Paid once a Year)

$$A = P \times (1 + \text{APR})^Y$$

A = accumulated balance after Y years

P = starting principal

APR = annual percentage rate (as a decimal) Y = number of years

Notes:

- (1) The starting principal, P, is often called the present value (PV), because we usually begin a calculation with the amount of money in an account at present.
- (2) The accumulated balance, A, is often called the future value (FV), because it is the amount that will be accumulated at some time in the future.
- (3) When using this formula, you must express the APR as a decimal rather than as a percentage.

In the example mentioned above, the annual interest rate is APR = 4% = 0.04, and interest is paid over a total of 25 years. The accumulated balance after Y = 25 years is:

$$\begin{aligned} A &= P \times (1 + \text{APR})^Y \\ &= 500 \times (1 + 0.04)^{25} \\ &= 500 \times 2.04^{25} \\ &= 1.02 \times 10^{28}/- \end{aligned}$$

As the administrator claimed, a 4% interest rate for 25 years would make the original 500/- amount grow to $1.02 \times 10^{28}/-$.

4.5 Saving Plans & Investment

Suppose you want to save money, perhaps for retirement or for your house expenses. You could deposit a lump sum of money today and let it grow through the power of compound interest. But if you do not have a large lump sum to start such an account, you can start depositing a small amount each month or after regular intervals to save it.

For most people, a more realistic way to save is by depositing smaller amounts on a regular basis. For example, you might put 500/- a month into savings. Such long-term savings plans are so popular that many have special names e.g., some get special tax treatment including National Saving Accounts (NSAs).

4.5.1 Savings Plan Formula

Suppose you deposited Rs. 400/- into a savings plan at the end of each month. To keep the numbers simple, suppose that your plan pays interest monthly at an annual rate of APR = 12%, or 1% per month.

- You begin with \$0 in the account. At the end of month 1, you make the first deposit of 400/-.
- At the end of month 2, you receive the monthly interest on the 400/- already in the account, which is $1\% \times 400/- = 4/-$. In addition, you make your monthly deposit of 400/-. Your balance at the end of the second month is:

$$400 + 4 + 400 = 804/-$$

- At the end of the third month, you receive 1% interest on the 400/-already in the account, or $1\% \times 800 = 8/-$. Adding your monthly deposit of 400/- , you have a balance at the end of the third month:

$$804 + 8 + 400 = 1212/-$$

Table 4.2 continues these calculations through 6 months. In principle, we could extend this table indefinitely—but it would take a lot of work. There is a formula “the savings plan formula” (next page) which would make these calculations easier and less time-consuming.

Table 4.2: Savings Plan Calculations (400/- monthly deposits; APR 12% or 1% per month)

Month	Prior Balance	Interest on a prior balance	End-of- month deposit	New balance
Month 1	0/-	—	400/-	400/-
Month 2	400/-	1%×400= 4	400/-	804/-
Month 3	804/-	1%×804= 8.04	400/-	1212.04/-
Month 4	1212.04	1%×1212.04= 2.12	400/-	1624.16/-
Month 5	1624.16	1%×1624.16= 16.24	400/-	2040.4/-
Month 6	2040.4	1%×2040.4= 20.40	400/-	2460.8/-

Savings Plan Formula (Regular Payments)

$$A = \text{PMT} \times \frac{\left[\left(1 + \frac{\text{APR}}{n} \right)^{(nY)} - 1 \right]}{\left(\frac{\text{APR}}{n} \right)}$$

A = accumulated savings plan balance

PMT = regular payment (deposit) amount

APR = annual percentage rate (as a decimal)

n = number of payment periods per year

Y = number of years

As with compound interest, the accumulated balance (A) is often called the future value (FV); the present value is the starting principal (P), which is 0 because we assume the account has no balance before the payments begin.

4.5.2 Total and Annual Return

In the examples, we assumed that you get a constant interest rate for a long period of time. In reality, interest rates usually vary over time. Consider a case in which you initially deposit Rs. 1000/- and it grows to Rs. 1500/- in 5 years. Although the

interest rate may have varied during the 5 years, we can still describe the change in both total and annual terms. Your total return is the percentage change in the investment value over the 5-year period:

$$\begin{aligned}\text{Total return} &= \frac{\text{new value} - \text{starting principal}}{\text{starting principal}} \times 100\% \\ &= \frac{1500 - 1000}{1000} \times 100\% = 50\%\end{aligned}$$

The total return on this investment is 50% over 5 years. Your annual return is the average annual rate at which your money grew over the 5 years. It is the constant annual percentage yield (APY) that would give the same result in 5 years. One way to determine this annual return is through trial and error. If you test $\text{APY} = 8.5\% = 0.085$ with a starting principal $P = 1000/-$ and a number of years $Y = 5$, you will find that the principal grows to approximately $A = 1500/-$:

$$A = P \times (1 + \text{APY})^Y = 1000 \times (1 + 0.085)^5 = 1503.66/-$$

You can find a more exact answer using the following formula for the annual return.

Total and Annual return

Consider an investment that grows from an original principal P to a later accumulated balance A . The total return is the percentage change in the investment value:

$$\text{Total return} = \frac{(A - P)}{P} \times 100\%$$

The annual return is the annual percentage yield (APY) that would give the same overall growth over Y years. The formula is:

$$\text{Annual return} = (A \div P)^{1/Y} - 1$$

This formula gives the annual return as a decimal; multiply by 100% to express it as a percentage.

Example 1: You invest 3000/- in the Clearwater mutual fund. Over 4 years, your investment grows in value to 8400/-. What are your total and annual returns for the 4-year period?

Solution: You have a starting principal $P = 3000/-$ and an accumulated value of $A = 8400/-$ after $Y = 4$ years. Your total and annual returns are:

$$\begin{aligned}\text{Total return} &= \frac{(A - P)}{P} \times 100\% \\ &= \frac{(8400 - 3000)}{3000} \times 100\% \\ &= 180\%\end{aligned}$$

$$\begin{aligned}\text{Annual return} &= (A \div P)^{1/Y} - 1 \\ &= (8400 \div 3000)^{1/4} - 1 \\ &= \sqrt[4]{2.8} - 1 \approx 0.294 = 29.4\%\end{aligned}$$

Your total return is 180%, meaning that the value of your investment after 4 years is 1.8 times its original value. Your annual return is approximately 0.294, or 29.4%, meaning that your investment has grown by an average of 29.4% each year.

Example 1: You purchased shares on a website for Rs. 2000/-. Three years later, you sold them for Rs. 1100/-. What were your total return and annual return on this investment?

Solution: You had a starting principal $P = 2000/-$ and an accumulated value of $A = 1100/-$ after $Y = 3$ years. Your total and annual returns were:

$$\begin{aligned}\text{Total return} &= \frac{(A - P)}{P} \times 100\% \\ &= \frac{(1100 - 2000)}{2000} \times 100\% \\ &= -45\%\end{aligned}$$

$$\begin{aligned}\text{Annual return} &= (A \div P)^{1/Y} - 1 \\ &= (1100 \div 2000)^{1/3} - 1 \\ &= \sqrt[3]{0.55} - 1 = -0.18\end{aligned}$$

Your total return was -45% meaning that your investment lost 45% of its original value. Your annual return was -0.18 or -18%, meaning that your investment lost an average of 18% of its value each year.

Note: In Excel, use the ^ symbol to raise to a power, with the fractional power in parentheses.

4.6 Investment: Types and Investment Considerations

4.6.1 Types of Investments

By combining what we have covered about savings plans with the ideas of total and annual return, we can now study investment options. Most investments fall into one of the three basic categories described as follows. **Three Basic Types of Investments.**

1. **Stock (or equity)** gives you a share of ownership in a company. You invest by purchasing shares of the stock, and the only way to get your money out is to sell the stock. Because stock prices change with time, the sale may give you either a gain or a loss on your original investment.
2. **A bond (or debt)** represents a promise of future cash. You usually buy bonds issued by either a government or a corporation. The issuer pays you simple interest (as opposed to compound interest) and promises to pay back your initial investment plus interest, later.
3. **Cash investments** include money you deposit into bank accounts, and certificates of deposit (CD). Cash investments generally earn interest.

There are two basic ways to invest in any of these categories: (1) You can invest directly, which means buying individual investments yourself (often through a broker). (2) You can invest indirectly by purchasing shares in a mutual fund, through which a professional fund manager invests your money along with the money of others participating in the fund.

4.6.2 Investment Considerations: Liquidity, Risk, and Return

No matter what type of investment you make, you should evaluate the investment in terms of three general considerations.

- **Liquidity:** How difficult is it to take out your money? An investment from which you can withdraw money easily, such as an ordinary bank account, is said to be liquid. The liquidity of an investment like real estate is much lower because real estate can be difficult to sell.
- **Risk:** Is your investment principal at risk? The safest investments are federally insured bank accounts: there is virtually no risk of losing the principal you've invested. Stocks and bonds are much riskier because they can drop in value, in which case you may lose part or all of your principal.
- **Return:** How much return (total or annual) can you expect on your investment? A higher return means you earn more money. In general, low-risk investments offer relatively low returns, while high-risk investments offer the prospects of higher returns—along with the possibility of losing your principal.

4.6.3 Historical Returns

One of the most difficult tasks of investing is trying to balance risk and return. Although there is no way to predict the future, historical trends offer at least some guidance. To study historical trends, financial analysts generally look at an index that describes the overall performance of some category of investment.

Example: Suppose your great-great-grandmother invested 100/- at the end of 1900 in each of three funds that tracked the averages of stocks, bonds, and cash, respectively. Assuming that her investments grew over the period of years approximately how much would each investment have been worth at the end of 2012?

Solution: We find the value of each investment with the compound interest formula (for interest compounded once a year), setting the interest rate (APR) to the average annual return for each category. In each case, the starting principal is $P = \text{Rs.}100/-$ and $Y = 112$ (the number of years from the end of 1900 to the end of 2012).

Stocks (annual return = 0.063) $A = P \times (1 + \text{APR})^Y$
 $= 100 \times (1 + 0.063)^{112} = 93,696.99/-$

Bonds (annual return = 0.020) $A = P \times (1 + \text{APR})^Y$
 $= 100 \times (1 + 0.020)^{112} = 918.80/-$

Cash (annual return = 0.009) $A = P \times (1 + \text{APR})^Y$
 $= 100 \times (1 + 0.009)^{112} = 272.78/-$

Notice the enormous difference in how much Rs.100/- grows with each type of investment. Unfortunately, the fact that stocks have clearly been the long-term investment of choice in the past is no guarantee that they will remain the best long-term investment for the future.

4.7 Loans, Credit Cards & Mortgages

4.7.1 Loans

For any loan, the principal is the amount of money owed at any particular time. Interest is charged on the loan principal. To pay off a loan, you must gradually pay down the principal. The loan term is the time you have to pay back the loan in full.

Suppose you borrow Rs. 1200/- at an annual interest rate of $\text{APR} = 12\%$, or 1% per month. At the end of the first month, you owe interest in the amount of $1\% \times 1200 = 12/-$. If you paid only this Rs. 12/- in interest, you would still owe Rs. 1200/-. That is, the total amount of the loan, called the loan principal, would

still be Rs. 1200/-. In that case, you would owe the same Rs. 12/- in interest the next month. In fact, if you paid only the interest each month, the loan would never be paid off and you would pay Rs. 12/- per month forever.

If you hope to make progress in paying off the loan, you need to pay part of the principal as well as interest. For example, suppose that you paid Rs. 200/- toward your loan principal each month, plus the current interest. At the end of the first month, you would pay Rs. 200/- toward principal plus Rs. 12/- for the 1% interest you owe, making a total payment of Rs. 212/-. Because you have paid Rs. 200/- toward the principal, your new loan principal would be $1200 - 200 = 1000/-$. At the end of the second month, you would again pay Rs. 200/- toward principal and 1% interest. But this time the interest is on the 1000/- that you still owe. Your interest payment therefore would be $1\% \times 1000 = 10/-$, making your total payment Rs. 210/-. Table 4.3 shows the calculations for the 6 months until the loan is paid off.

Table 4.3: Payments and principal for a Rs. 1200/- loan with principal paid off at Rs. 200/-

Month	Prior Principal	Interest on Prior Principal	Payment Toward Principal	Total Payment	New Principal
Month 1	1200/-	$1\% \times 1200 = 12/-$	200/-	212/-	1000/-
Month 2	1000/-	$1\% \times 1000 = 10/-$	200/-	210/-	800/-
Month 3	800/-	$1\% \times 800 = 8/-$	200/-	208/-	600/-
Month 4	600/-	$1\% \times 600 = 6/-$	200/-	206/-	400/-
Month 5	400/-	$1\% \times 400 = 4/-$	200/-	204/-	200/-
Month 6	200/-	$1\% \times 200 = 2/-$	200/-	202/-	0/-

With monthly payments as given in Table 4.3, your total payment decreases from month to month because of the declining amount of interest that you owe. It can easily be calculated. However, most people prefer to pay the same total amount each month because it makes planning a budget easier. A loan that you pay off with equal regular payments is called an installment loan (or amortized loan).

Suppose you wanted to pay off your Rs. 1200/- loan with 6 equal monthly payments. How much should you pay each month? Because the payments in Table 4 vary between Rs. 202/- and Rs. 212/-, it is clear that the equal monthly payments must lie somewhere in this range. The exact amount is not obvious, but we can calculate it with the loan payment formula.

Loan Payment Formula

$$PMT = \frac{P \times \left(\frac{APR}{n}\right)}{\left[1 - \left(1 + \frac{APR}{n}\right)^{(-nY)}\right]}$$

PMT = regular payment amount

P = starting loan principal (amount borrowed)

APR = annual percentage rate

n = number of payment periods per year

Y = loan term in years

In our current example, the starting loan principal is $P = 1200/-$, the annual interest rate is $APR = 12\%$, the loan term is $Y = \frac{1}{2}$ year (6 months), and monthly payments mean $n = 12$. The loan payment formula gives:

$$\begin{aligned} PMT &= \frac{P \times (APR \div n)}{[1 - (1 + APR \div n)^{(-nY)}]} \\ &= \frac{1200 \times (0.12 \div 12)}{[1 - (1 + 0.12 \div 12)^{(-12 \times 1/2)}]} \\ &= \frac{1200 \times (0.01)}{[1 - (1 + 0.01)^{(-6)}]} \\ &= \frac{12}{[1 - 0.942]} \\ &= \text{Rs. } 207.06/- \end{aligned}$$

The monthly payments would be Rs. 207.06/-. Please note that, as we expected, the payment is between Rs.202/- and Rs.212/-. And because installment loans gradually pay down the loan principal while the payments remain the same, the following two features apply to all installment loans:

- The interest due each month gradually decreases.
- The amount paid toward the principal each month gradually increases.

Early in the loan term, the portion going toward interest is relatively high and the portion going toward principal is relatively low. As the term proceeds, this pattern gradually reverses, and toward the end of the loan term, most of the payments go to the principal and relatively little to interest.

4.7.1 Credit Cards

Credit card loans differ from installment loans in that you are not required to pay off your balance in a particular set time period. Instead, you are required to make only a minimum monthly payment that generally covers all the interest but very little principal. As a result, it takes a very long time to pay off your credit card loan if you make only the minimum payments. If you wish to pay off your loan in a particular amount of time, you should use the loan payment formula to calculate the necessary payments.

Most credit cards have very high interest rates compared to other types of loans. As a result, it is easy to get into financial trouble if you get overextended with credit cards. The trouble is particularly bad if you miss your payments. In that case, you will probably be charged a late fee that is added to your principal, thereby increasing the amount of interest due the next month. With the interest charges operating like compound interest in reverse, failure to pay on time can put a person into an ever-deepening financial hole.

Example

Suppose you have a credit card balance of Rs. 2300/- with an annual interest rate of 21%. You decide to pay off your balance over 1 year. How much will you need to pay each month? Assume you make no further credit card purchases.

Solution: The amount borrowed is $P = \text{Rs. } 2300/-$, the interest rate is $\text{APR} = 0.21$, and you make $n = 12$ payments per year. Because you want to pay off the loan in 1 year, we set $Y = 1$. The required payments are:

$$\begin{aligned} \text{PMT} &= \frac{P \times (\text{APR} \div n)}{[1 - (1 + \text{APR} \div n)^{(-nY)}]} \\ &= \text{Rs. } 214.16/- \end{aligned}$$

You must pay Rs. 214.16/- per month to pay off the balance in 1 year.

4.7.2 Mortgages

A home mortgage is an installment loan designed specifically to finance a home. A down payment is the amount of money you must pay upfront in order to be given a mortgage or other loan. Closing costs are fees you must pay in order to be given the loan. They may include a variety of direct costs, or fees charged as points, where each point is 1% of the loan amount. In most cases, lenders are required to give you a clear assessment of closing costs before you sign for the loan.

One of the most popular types of installment loans is a home mortgage, which is designed specifically to help you buy a home. Mortgage interest rates generally are lower than interest rates on other types of loans because your home itself serves as a payment guarantee. If you fail to make your payments, the lender (usually a bank or mortgage company) can take possession of your home, through the process called foreclosure, and sell it to recover some or all of the amount loaned to you.

There are several considerations in getting a home mortgage. First, the lender may require a down payment, typically 10% to 20% of the purchase price. Then the lender will loan you the rest of the money needed to purchase the home. Most lenders also charge fees, or closing costs, at the time you take out a loan. Closing costs can be substantial and may vary significantly between lenders, so you should be sure that you understand them. In general, there are two types of closing costs:

- Direct fees, such as fees for getting the home appraised and checking your credit history, for which the lender charges a fixed dollar amount. These fees typically range from a few hundred dollars to a couple thousand dollars.
- Fees are charged as points, where each point is 1% of the loan amount. Many lenders divide points into two categories: an “origination fee” that is charged on all loans and “discount points” that vary for loans with different rates. For example, a lender might charge an origination fee of 1 point 11%² on all loans, then offer you a choice of interest rates depending on how many discount points you are willing to pay. Despite their different names, there is no essential difference between an origination fee and discount points.

You should watch out for any fine print that may affect the cost of your loan. For example, make sure that there are no prepayment penalties if you decide to pay off your loan early. Most people pay off mortgages early, either because they sell the home or because they decide to refinance the loan to get a better interest rate or to change their monthly payments.

Fixed Rate Mortgages: The simplest type of home loan is a fixed rate mortgage, in which you are guaranteed that the interest rate will not change over the life of the loan. Most fixed rate mortgages have a term of either 15 or 30 years, with lower interest rates on the shorter-term loans.

Example: You need a loan of \$100,000 to buy your new home. The bank offers a choice of a 30-year loan at an APR of 5% or a 15-year loan at 4.5% Compare your monthly payments and total loan cost under the two options. Assume that the closing costs are the same in both cases and therefore do not affect the choice.

Solution: Mortgages are installment loans, so we use the loan payment formula:

$$PMT = \frac{P \times (APR \div n)}{[1 - (1 + APR \div n)^{(-nY)}]}$$

Steps	Y= 30 years; APR = 0.05	Y= 30 years; APR = 0.045
1.	$PMT = \frac{P \times (APR \div n)}{[1 - (1 + APR \div n)^{(-nY)}]}$ $= \frac{100,000 \times (0.05 \div 12)}{[1 - (1 + 0.05 \div 12)^{(-12 \times 30)}]}$ $= \text{Rs.}536.82/-$	$PMT = \frac{P \times (APR \div n)}{[1 - (1 + APR \div n)^{(-nY)}]}$ $= \frac{100,000 \times (0.05 \div 12)}{[1 - (1 + 0.05 \div 12)^{(-12 \times 15)}]}$ $= \text{Rs.} 764.99/-$
2.	$30 \text{ yr} \times \frac{12 \text{ mo}}{\text{yr}} \times \frac{536.88 \text{ /-}}{\text{mo}}$ $= 193,255/-$	$15 \text{ yr} \times \frac{12 \text{ mo}}{\text{yr}} \times \frac{764.99 \text{ /-}}{\text{mo}}$ $= 137,698/-$

Note that the monthly payments on the 15-year loan are higher by about 765 - 537 = 228. However, the 15-year loan saves you about 193,255 - 137,698 = 55,557/- in total payments. That is, the 15-year loan saves you a lot in the long run, but it is a good plan only if you are confident that you can afford the higher monthly payments required for the next 15 years.

4.8 Income Tax

The basic ideas behind income taxes are relatively simple, and most of the arcane rules apply only to relatively small segments of the population. As a result, most people cannot only file their own taxes but also understand them well enough to make intelligent decisions about both personal finances and political tax questions. Figure 01 summarizes the steps in a basic tax calculation. We will follow the flow of the steps, defining terms as we go.

- The process begins with your **gross income**, which is all your income for the year, including wages, tips, profits from a business, interest or dividends from investments, and any other income you receive.
- Some gross income is not taxed (at least not in the year it is received), such as contributions to individual retirement accounts (IRAs) and other tax-deferred savings plans. These untaxed portions of gross income are called adjustments. Subtracting adjustments from your gross income gives your **adjusted gross income**.
- Most people are entitled to certain **exemptions** and **deductions**—amounts that you subtract from your adjusted gross income before calculating your taxes. Once you subtract the exemptions and deductions, you are left with your **taxable income**.
- A tax table or tax rate computation allows you to determine how much tax you owe on your taxable income. However, you may not actually have to pay this much tax if you are entitled to any **tax credits**, such as the child tax credit that parents can claim. Subtracting the amount of any tax credits gives your **total tax**.
- Finally, most people have already paid part or all of their tax bill during the year, either through withholding (by your employer) or by paying quarterly estimated taxes (if you are self-employed). You subtract the taxes that you've already paid to determine how much you still owe. If you have paid more than you owe, then you should receive a **tax refund**.
- Exemptions are a fixed amount per person (3900/- in 2013 for most teachers). You can claim the amount of an exemption for yourself and each of your dependents (for example, children whom you support).
- Deductions vary from one person to another. The most common deductions include interest on home mortgages, contributions to charity, and taxes you have paid to other agencies (such as state income taxes or local property taxes).

Example 1: Saira earned wages of Rs. 38,200/- , received Rs. 750/- in interest from a savings account, and contributed Rs. 1200/- to her benevolent fund. She was entitled to a personal exemption of Rs. 3900/- and to deductions totaling Rs. 6100/-. Find her gross income, adjusted gross income, and taxable income.

Solution: Saira's gross income is the sum of all her income, which means the sum of her wages and her interest:

$$\text{Gross income} = 38,200 + 750 = \text{Rs. } 38,950/-$$

Her Rs. 1200/- contribution to a benevolent fund counts as an adjustment to her gross income, so her adjusted gross income (AGI) is:

$$\begin{aligned}\text{AGI} &= \text{gross income} - \text{adjustments} \\ &= 38,950 - 1200 = \text{Rs. } 37,750/-\end{aligned}$$

To find her taxable income, we subtract her exemptions and deductions:

$$\begin{aligned}\text{Taxable income} &= \text{AGI} - \text{exemptions} - \text{deductions} \\ &= 37,750 - 3900 - 6100 = \text{Rs. } 27,750/-\end{aligned}$$

Her taxable income is Rs. 27,750/-.

Example 2: Suppose you have the following deductible expenditures: Rs. 2500/- for interest on a home mortgage, Rs. 900/- for contributions to charity, and Rs. 250/- for state income taxes. Your filing status entitles you to a standard deduction of Rs. 6100/-. How many itemized deductions, she can claim?

Solution: The total of your deductible expenditures is:

$$\text{Rs. } 2500/- + \text{Rs. } 900/- + \text{Rs. } 250/- = \text{Rs. } 3650/-$$

If you itemize your deductions, you can subtract Rs.3650/- when finding your taxable income.

SUMMARY

By the end of this unit, it is evident that financial literacy is a cornerstone of personal empowerment and long-term success. Throughout our exploration, we have delved into the fundamental principles of money management, gaining valuable insights and practical skills to navigate the complex landscape of personal finance.

From creating budgets and setting financial goals to understanding investment options and managing debt, we have equipped ourselves with the knowledge and tools necessary to make informed decisions about our finances. By prioritizing financial responsibility and adopting prudent financial habits, we can build a solid foundation for achieving our financial aspirations and securing our future financial well-being. Moreover, our journey has highlighted the importance of ongoing learning and adaptation in the ever-evolving world of finance. By staying informed about changes in the financial landscape and continuously refining our financial skills, we can adapt to new challenges and seize opportunities to grow our wealth and achieve our goals.

As we move forward, let us carry forward the lessons learned from this unit and apply them in our daily lives. Let us commit to practicing sound financial habits, seeking opportunities for growth, and empowering ourselves to take control of our financial futures. With dedication, diligence, and a commitment to lifelong learning, we can navigate the complexities of personal finance with confidence and achieve financial freedom and security.

EXERCISES

Multiple Choice Questions

1. What is the main purpose of a budget?
 - a) To plan income and expenses
 - b) To invest in risky assets
 - c) To reduce taxes
 - d) To spend on luxury items**Answer:** a) To plan income and expenses

2. Which of the following best describes financial goals?
 - a) Spending more on entertainment
 - b) Short-term and long-term targets for managing finances
 - c) Reducing income taxes entirely
 - d) Investing only in cash-based assets**Answer:** b) Short-term and long-term targets for managing finances

3. What is the key characteristic of a stock investment?
 - a) Low-risk and fixed return
 - b) Ownership in a company and potential high returns
 - c) Guaranteed returns with high liquidity
 - d) Fixed interest paid annually**Answer:** b) Ownership in a company and potential high returns

4. Why is liquidity important in investments?
 - a) It guarantees a higher return.
 - b) It allows quick access to money when needed.
 - c) It ensures low risk in investments.
 - d) It always results in tax savings.**Answer:** b) It allows quick access to money when needed.

5. Which of the following is an example of a cash investment?
 - a) Government bonds
 - b) Savings accounts
 - c) Mutual funds
 - d) Corporate stocks**Answer:** b) Savings accounts

6. What does "rate of return" mean in investment terms?
- a) The annual tax paid on an investment
 - b) The profit or loss earned on an investment over time
 - c) The amount of interest charged on loans
 - d) The guaranteed income from government bonds
- Answer:** b) The profit or loss earned on an investment over time
7. What is a common characteristic of credit cards?
- a) Fixed monthly repayment amounts
 - b) High interest rates on unpaid balances
 - c) Low initial costs with no additional fees
 - d) No credit checks required
- Answer:** b) High interest rates on unpaid balances
8. Which of these describes an absolute error?
- a) The total value of an error in measurement
 - b) The percentage difference from an expected value
 - c) The difference caused by rounding
 - d) A systematic mistake in calculations
- Answer:** a) The total value of an error in measurement
9. Why should percentages not be averaged directly?
- a) They represent relative values dependent on different bases.
 - b) They are not measurable values.
 - c) They always give inflated results.
 - d) Averaging them always reduces accuracy.
- Answer:** a) They represent relative values dependent on different bases.
10. What is the purpose of a Consumer Price Index (CPI)?
- a) To measure changes in investment risks
 - b) To track changes in the cost of living over time
 - c) To evaluate savings account returns
 - d) To calculate tax rates on income
- Answer:** b) To track changes in the cost of living over time

Short-Answer Questions

1. What is a budget, and why is it important?
2. How can financial goals influence spending and saving habits?
3. Name and explain three types of investments.
4. What are the primary differences between stocks, bonds, and cash investments?
5. Define liquidity and explain why it matters in financial planning.
6. What are the potential risks of using credit cards?
7. Explain the concept of “accuracy and precision” in financial reporting.
8. What is the significance of the Consumer Price Index (CPI)?
9. Describe the concept of "rate of return" in simple terms.
10. Explain the difference between random and systematic errors.

Subjective Questions

1.
 - a) Define budgeting and explain its components.
 - b) How can setting financial goals help individuals manage their finances better?
2.
 - a) Compare and contrast stocks, bonds, and cash investments based on liquidity, risk, and return.
 - b) What factors should an individual consider when selecting an investment?
3.
 - a) Explain the role of interest rates in managing credit cards and loans.
 - b) Discuss the advantages and disadvantages of using credit cards.
4.
 - a) Define and differentiate between random and systematic errors in financial calculations.
 - b) Why is precision important in financial reporting, and how can it affect decision-making?
5.
 - a) Explain the Consumer Price Index (CPI) and its role in measuring the rate of inflation.
 - b) How can changes in CPI affect an individual’s financial planning?

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Unit-5

FUNDAMENTALS OF GEOMETRY

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INTRODUCTION

We live in a three-dimensional world, and much of our understanding of that world is rooted in geometry. Surveying land for a new park, analyzing a satellite photograph of the Earth, navigating a ship at sea, and creating a medical image all rely on ideas that originated with the ancient Greeks. You can find the perimeter of a geometric shape like a circle or square with a simple formula. But what if you want to find the perimeter of a natural object with jagged edges such as a fern leaf?

Geometry is a branch of mathematics that explores the properties, shapes, sizes, and relationships of various figures and spaces. From simple lines and angles to complex three-dimensional shapes, geometry plays a fundamental role in understanding the structures and patterns that surround us in the physical world.

In this unit, we will explore the basic concepts and principles of geometry, laying the groundwork for further exploration and discovery. Through engaging activities, interactive lessons, and real-world examples, we will delve into the fascinating realm of geometric shapes, measurements, and transformations.

Whether you are new to geometry or looking to refresh your understanding of geometric concepts, this unit will provide you with the tools and knowledge to navigate the rich terrain of geometry with confidence and enthusiasm. By the end of this unit, you will have developed a solid foundation in geometry, enabling you to analyze shapes, solve geometric problems, and appreciate the beauty and elegance of geometric relationships.

LEARNING OUTCOMES

After completing the unit, you will be able to:

1. Develop a fundamental understanding of geometric concepts such as points, lines, angles, and shapes.
2. Identify and classify different types of geometric figures, including polygons, circles, and solids.
3. Demonstrate proficiency in measuring and calculating geometric properties, such as length, area, perimeter, volume, and surface area.
4. Recognize the practical applications of geometry in various fields, such as architecture, engineering, art, and design.
5. Develop spatial awareness and visualization skills to analyze and manipulate geometric figures in two-dimensional and three-dimensional space.
6. Utilize geometric reasoning and problem-solving strategies to tackle real-world problems and mathematical challenges.

5.1 Introduction to Geometry

The word geometry literally means “earth measure.” Many ancient cultures developed geometrical methods to survey flood basins around agricultural fields and to establish patterns of planetary and star motion. However, geometry was always more than just a practical science, as we can see from the use of geometric shapes and patterns in ancient art.

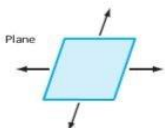
The Greek mathematician Euclid (c. 325–270 B.C.E.) summarized Greek knowledge of geometry in a 13-volume textbook called *Elements*. The geometry described in Euclid’s work, now called Euclidean geometry, is the familiar geometry of lines, angles, and planes.

5.2 Points, Lines and Planes

Geometric objects, such as points, lines, and planes, represent idealizations that do not exist in the real world. A geometric point is imagined to have zero size. No real object has zero size, but many real phenomena approximate geometric points. Stars, for example, appear as points of light in the night sky.

A geometric line is formed by connecting two points along the shortest possible path. It has infinite length and no thickness. Because no physical object is infinite in length, we usually work with line segments or pieces of a line. A long taut wire is a good approximation to a line segment. A geometric plane is a perfectly flat surface that has infinite length and width, but no thickness. A smooth tabletop is a good approximation to a segment of a plane.

Table 5.1

Point	
Line	
Plane	

5.3 Plane Geometry

Plane geometry is the geometry of two-dimensional objects. Here we examine problems involving circles and polygons, which are the most common two-dimensional objects.

All points on a circle are located at the same distance—the radius—from the circle's center. The diameter of a circle is twice its radius, which means it is the distance across the circle on a line passing through its center.

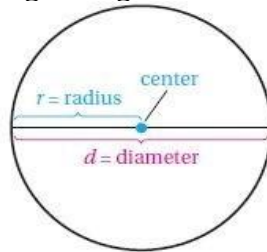


Figure 5.1. Circle

A polygon is any closed shape in the plane made from straight-line segments. The root poly comes from a Greek word for “many,” so a polygon is a many-sided figure. A regular polygon is a polygon in which all the sides have the same length and all interior angles are equal. Table 10.1 shows several common regular polygons and their names.

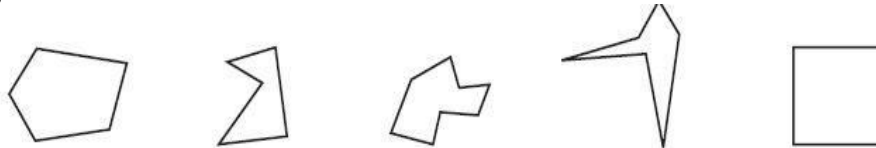


Figure 5.2. Shapes of Polygons

Table 5.2: Shapes and Sides of Polygons

Sides	Name	Picture	Sides	Name	Picture
3	Equilateral Triangle		6	Regular Hexagon	
4	Square		8	Regular Octagon	
5	Regular Pentagon		10	Regular Decagon	

Triangles are among the most important of all polygons, and they take many different forms. All three sides of an equilateral triangle have equal length, making it a regular polygon. An isosceles triangle has exactly two sides of equal length. A right triangle contains one right (90°) angle. By sketching some triangles—

particularly triangles with one very large angle and two very small angles - you can observe that the sum of the three angle measures of a triangle is always 180° .

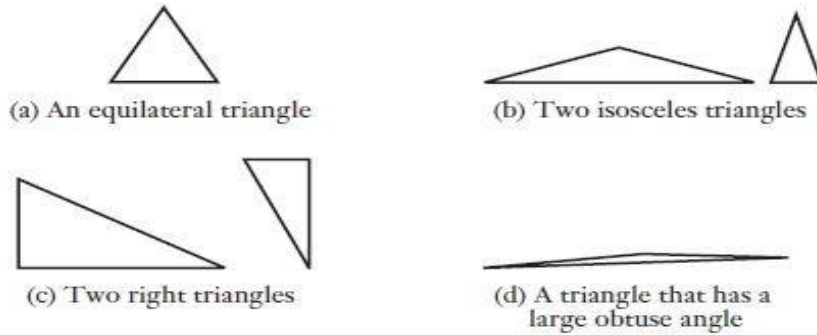


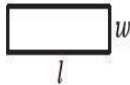
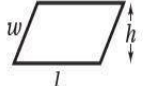
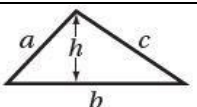
Figure 5.3. Different types of triangles

5.3.1 Perimeter

The perimeter of a plane object is simply the length of its boundary. We can find the perimeter of a polygon by adding the lengths of the individual edges. The perimeter of a circle, called the *circumference*, is related to its diameter or radius by the universal constant ‘ π ’ (pronounced “pie”), which has a value of approximately 3.14:

$$\begin{aligned}\text{Circumference of circle} &= \pi \times \text{diameter} = \pi \times d \\ &= 2 \times \pi \times \text{radius} = 2 \times \pi \times r\end{aligned}$$

Table 5.3: Formula for Perimeter and Area of Two-Dimensional Shapes

Shape of object	Picture	Formula for Perimeter	Area
Circle		$2\pi r = \pi d$	πr^2
Square		$4l$	l^2
Rectangle		$2l + 2w$	lw
Parallelogram		$2l + 2w$	lh
Triangle		$a + b + c$	$\frac{1}{2}bh$

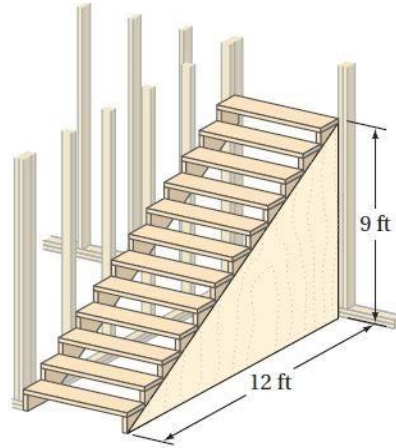
Example

You have built a stairway in a new house and want to cover the triangular space beneath the stairs with plywood. The figure shows the region to be covered. What is the area of this region?

Solution: The region to be covered is triangular, with a base of 12 feet and height of 9 feet. The area of this triangle is:

$$\begin{aligned}\text{area} &= 1/2 \times b \times h \\ &= 1/2 \times 12 \text{ ft} \times 9 \text{ ft} \\ &= 54 \text{ ft}^2\end{aligned}$$

The area of the region to be covered is 54 square feet.



5.4 Three-Dimensional Geometry

Two of the most important properties of a three-dimensional object, such as a box or a sphere, are its volume and its surface area. Figure 5.4 given below gives the names of several familiar three-dimensional objects, along with their volume and surface area formulas. Some of the formulas in Figure 5.4 given below are easily understandable. For example, the volume formula for a box or a cube is just the familiar “length $\times$ width $\times$ height” formula. The surface area formula for a box or a cube is the result of adding the areas of the six (rectangular) faces of the object.

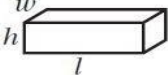
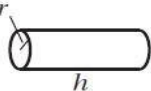
Object	Picture	Surface Area	Volume
Sphere		$4\pi r^2$	$\frac{4}{3}\pi r^3$
Cube		$6l^2$	l^3
Rectangular prism (box)		$2(lw + lh + wh)$	lwh
Right circular cylinder		$2\pi r^2 + 2\pi rh$	$\pi r^2 h$

Figure 5.4. Formula of Surface Area and Volume of three-dimensional objects

Example

A water reservoir has a rectangular base that measures 30 meters by 40 meters and vertical walls 15 meters high. At the beginning of the summer, the reservoir was filled to capacity. At the end of the summer, the water depth was 4 meters. How much water was used?

Solution: The reservoir has the shape of a rectangular prism, so the volume of water in the reservoir is its length times its width times its depth. When the reservoir was filled at the beginning of the summer, the volume of water was:

$$30 \text{ m} \times 40 \text{ m} \times 15 \text{ m} = 18,000 \text{ m}^3$$

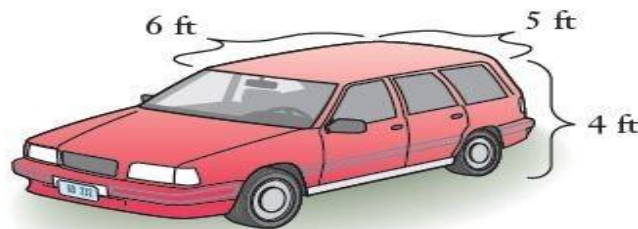
At the end of the summer, the amount of water remaining was:

$$30 \text{ m} \times 40 \text{ m} \times 4 \text{ m} = 4800 \text{ m}^3$$

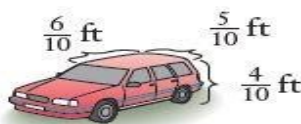
Therefore, the amount of water used was $18,000 \text{ m}^3 - 4800 \text{ m}^3 = 13,200 \text{ m}^3$.

5.5 Scaling Laws

Scaling is a process by which a real object is modeled by a similar object whose dimensions are proportionally larger or smaller. For example, an architect might make a scale model of an auditorium in which all lengths are smaller by a factor of 100 or a biologist might make a scale model of a cell in which all lengths are larger by a factor of 10,000.



Actual car



Model car

Suppose that we make an engineering model of a car with a scale factor of 10. The actual car is 10 times as long, 10 times as wide, and 10 times as tall as the model car. How are the surface area and volume of the actual car related to the surface area and volume of the model? Consider the area of the car roof, which is its length times its width. Because the length and width of the actual car roof are each 10 times their size in the model, the area of the actual car roof is:

$$\begin{aligned}\textbf{Actual Roof of Area} &= \text{Actual roof length} \times \text{Actual roof width} \\ &= (10 \times \text{model roof length}) \times (10 \times \text{model roof width}) \\ &= 10^2 \times \text{model roof length} \times \text{model roof width} \\ &= 10^2 \times \text{model roof area}\end{aligned}$$

That is, the actual roof area is greater than the model roof area by the square of the scale factor ($10^2 = 100$, in this case).

We can do a similar calculation for the volume by considering, for example, the box-shaped (rectangular prism) passenger compartment.

$$\begin{aligned}\textbf{Actual Volume} &= \text{Actual length} \times \text{Actual width} \times \text{Actual height} \\ &= (10 \times \text{model length}) \times (10 \times \text{model width}) \times (10 \times \text{model height}) \\ &= 10^3 \times \text{model length} \times \text{model width} \times \text{model height} \\ &= 10^3 \times \text{model volume}\end{aligned}$$

The actual car volume is greater than the model car volume by the cube of the scale factor ($10^3 = 1000$, in this case).

5.6 Surface Area to Volume Ratio

Another important concept in scaling is the relative scaling of areas and volumes. We define the surface-area-to-volume ratio for any object as its surface area divided by its volume:

$$\text{Surface-area-to-volume ratio} = \frac{\text{surface area}}{\text{volume}}$$

Because surface area scales with the square of the scale factor and volume scales with the cube of the scale factor, the surface-area-to-volume ratio must scale with the reciprocal of the scale factor:

$$\text{scaling of surface-area-to-volume ratio} = \frac{\text{scale factor}^2}{(\text{scale factor})^3} = \frac{1}{\text{scale factor}}$$

Therefore, when an object is “scaled up,” its surface-area-to-volume ratio decreases. When an object is “scaled down,” its surface-area-to-volume ratio

increases. Larger objects have smaller surface-area-to-volume ratios than similarly proportioned small objects. Smaller objects have larger surface-area-to-volume ratios than similarly proportioned large objects.

SUMMARY

In this unit, we have explored the fundamental concepts and principles that underpin the study of geometry from the simplicity of points and lines to the complexity of three-dimensional solids. Throughout this unit, we have developed a deeper understanding of geometric figures, their properties, and their applications in various fields. We have honed our skills in measuring and calculating geometric quantities from lengths and areas to volumes and surface areas, enabling us to analyze and solve problems with confidence and precision.

As we move forward, let us carry forward the knowledge and skills gained from this unit, continuing to explore and appreciate the wonders of geometry in our daily lives. Whether in architecture, engineering, art, or design, the principles of geometry will continue to shape and inspire our understanding of the world and our place within it.

EXERCISES

Multiple Choice Questions

1. Which of the following is true about a plane in geometry?
 - a) It has only one dimension.
 - b) It has two dimensions and extends infinitely.
 - c) It has three dimensions and a fixed size.
 - d) It is the line where two surfaces meet.**Answer:** b) It has two dimensions and extends infinitely.

2. What is the perimeter of a rectangle with a length of 8 units and a width of 5 units?
 - a) 13 units
 - b) 26 units
 - c) 40 units
 - d) 48 units**Answer:** b) 26 units

3. In geometry, what is a point?
 - a) A line that extends infinitely in one direction
 - b) A flat surface with no thickness
 - c) An exact location with no size or dimension
 - d) A measurable length between two lines**Answer:** c) An exact location with no size or dimension

4. What is the formula for the perimeter of a circle?
 - a) $\pi \times r^2$
 - b) $2\pi r$
 - c) πr
 - d) $4r$**Answer:** b) $2\pi r$

5. Which geometric shape does not have a defined perimeter?
 - a) Circle
 - b) Square
 - c) Point
 - d) Rectangle**Answer:** c) Point

6. A triangle has sides of 3 cm, 4 cm, and 5 cm. What is its perimeter?
 - a) 9 cm
 - b) 12 cm
 - c) 15 cm
 - d) 18 cm**Answer:** b) 12 cm

7. What is the relationship between a line and a plane?

- a) A plane contains at least three lines.
- b) A line is the intersection of two planes.
- c) A plane is formed by one line.
- d) A line cannot lie on a plane.

Answer: b) A line is the intersection of two planes.

8. What is scaling in geometry?

- a) Increasing the length of one side of a shape
- b) Enlarging or reducing the size of a shape proportionally
- c) Drawing multiple shapes on the same plane
- d) Measuring the perimeter of a shape

Answer: b) Enlarging or reducing the size of a shape proportionally

9. How many edges does a triangle have?

- a) 2
- b) 3
- c) 4
- d) 6

Answer: b) 3

10. What is the perimeter of a square with a side length 6 units?

- a) 12 units
- b) 24 units
- c) 18 units
- d) 36 units

Answer: b) 24 units

Short-Answer Questions

1. Define a point, line, and plane in geometry.
2. Write the formula for the perimeter of a circle, rectangle, and triangle.
3. How does scaling affect the dimensions of a geometric figure?
4. What is the perimeter of a parallelogram if its sides are 6 cm and 8 cm?
5. Explain the difference between a line segment and a line.
6. Give an example of a situation where scaling is used in real life.
7. How do you find the perimeter of a square?
8. State any two characteristics of a plane in geometry.
9. What is the significance of the perimeter in geometry?
10. Write the formula for the perimeter of a parallelogram and explain it briefly.

Subjective Questions

1.
 - a) Define and differentiate between a point, line, and plane.
 - b) Provide real-life examples of each.
2.
 - a) Explain the concept of perimeter.
 - b) Calculate the perimeter of a rectangle with a length 10 cm and width 7 cm, a circle with a radius of 4 cm, and a triangle with sides of 5 cm, 6 cm, and 7 cm.
3.
 - a) Explain the concept of scaling in geometry.
 - b) Provide examples of how scaling is used in architecture and map-making.
4.
 - a) Why is calculating the perimeter important in construction and landscaping?
 - b) A rectangular garden has a length of 20 m and a width of 10 m. A fence is to be built around it. What is the total length of the fence needed?
5.
 - a) Discuss the properties of a square and a parallelogram in detail.
 - b) How are these shapes used in design and engineering?
6. Explain scaling laws and its implications in real-life.

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Unit–6

**SETS AND ALGEBRAIC
EXPRESSIONS**

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INTRODUCTION

In mathematics, sets and algebraic expressions are foundational concepts that serve as building blocks for a wide range of mathematical topics and applications. Sets provide a framework for organizing and categorizing elements, while algebraic expressions offer a language for representing mathematical relationships and patterns.

In this unit, we will embark on an exploration of sets and algebraic expressions, uncovering their significance and relevance in various mathematical contexts. We will begin by delving into the basic concepts of sets. Through interactive examples and exercises, we will learn how to represent, manipulate, and analyze sets to solve problems and draw conclusions.

Building upon our understanding of sets, we will then transition to the realm of algebraic expressions. We will explore the fundamentals of algebra, including variables, constants, coefficients, and terms. By mastering algebraic expressions, we will gain the ability to represent mathematical relationships symbolically and manipulate expressions to simplify and solve equations. Throughout this unit, we will emphasize the connections between sets and algebraic expressions, highlighting how these concepts complement and inform each other. We will also explore practical applications of sets and algebraic expressions in various fields from mathematics and science to economics and engineering.

LEARNING OUTCOMES

After completing the unit, you would be able to:

1. Understand the fundamental concepts of sets, including elements, subsets, unions, intersections, and complements.
2. Identify and analyze patterns in algebraic expressions, including variables, constants, coefficients, and terms.
3. Solve equations and inequalities involving algebraic expressions, and interpret the solutions in the context of the problem.
4. Explore real-world applications of sets and algebraic expressions in various fields, such as statistics, economics, and computer science.
5. Develop critical thinking and problem-solving skills through the application of sets and algebraic expressions to solve complex problems.

6.1 Sets and their Operations

A set is a collection of distinct objects, considered as an entity. These objects are called elements or members of the set. Sets are denoted by curly braces {}, and the elements are listed within the braces, separated by commas. Sets can be finite or infinite. For example:

- A = {1, 2, 3, 4, 5} is a set containing the numbers 1, 2, 3, 4, and 5.
- B = {apple, orange, banana} is a set containing the fruits apple, orange, and banana.
- C = {x | x is a prime number less than 10} is a set defined by a property; in this case, containing prime numbers less than 10.

There are different types of sets as given below:

- i. **The universal set** contains all the elements for the sample. For example, the universal set of digits is:

$$D = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

The set of all students in a school (Universal set) contains students from various classes (subsets). The set of all fruits available in a market (Universal set) includes subsets like apples, bananas, and oranges.

- ii. **A subset** is a set whose elements all belong to another set. For example, set Q is, the set of odd digits $Q = \{1, 3, 5, 7, 9\}$ is a subset of set D.

$$Q \subset D \text{ (Q is a subset of D)}$$

The set of all fruits available in a market (Universal set) includes subsets like apples, bananas, and oranges. So, the set of apples is a subset of the universal set of all the fruits available in a market. The set of all teachers in a school is a subset of the universal set of all staff members in the school.

- iii. **Complement** includes all the elements of the universal set that do not belong to a subset of it. For example, $Q' = \{0, 2, 4, 6, 8\}$ is the complement of Q. Notation used is the prime symbol, Q' or not Q.

If the universal set is the set of all students in a class and A is the set of students who play cricket, the complement of A is the set of students who do not play cricket.

Example 2: If the universal set is all days of the week and B is the set of weekdays, the complement of B is the set of weekend days.

- iv. **The empty set** is a set with no elements. For example, the set of odd numbers that are divisible by 2 is the empty set. Notation used: $\{ \}$ or $\emptyset$. For example, the set of people taller than 20 feet,
- v. **Disjoint sets** are two or more sets having no elements in common. For example, the set of fruits and set of vegetables are disjoint. For example, if set A is the set of cats and B is the set of laptops, both sets are disjoint sets because they have no common element.
- vi. **Finite set** is a set with a countable number of elements. For example, the set $E = \{2, 4, 6, 8\}$, the set of all books in a library (countable), and the set of all players in a cricket team (11 players).
- vii. **Infinite set** is a set with an infinite number of elements. For example, the set of natural numbers, $N = \{1, 2, 3, \dots\}$, the set of all stars in the universe, and the set of all people living on Earth over time (uncountable).

Operations on sets include union, intersection and difference.

S#	Set Notation	Meaning	Example
1.	$A \cup B$ A union B A OR B	- Any element that is in either of the sets - Any element that is in at least one of the sets	1) Set A: People who travel by car. Set B: People who travel by bike. Union ($A \cup B$): People who travel by car, bike, or both. 2) Set A: Students who play football. Set B: Students who play basketball. Union ($A \cup B$): Students who play football, basketball, or both.
2.	$A \cap B$ An intersection B A AND B	Only elements that are in both A and B	1) Set A: People who travel by car. Set B: People who travel by train. Intersection ($A \cap B$): People who travel by both car and train. 2) Set A: Students who play football. Set B: Students who play basketball. Intersection ($A \cap B$): Students who play both football and basketball.
3.	$A \setminus B$ A minus B	Elements found in set A but excluding the ones that are also in set B	1) Set A: Students who participate in sports. Set B: Students who participate in a speech competition. $A \setminus B$: Students who participate in sports but not in the speech competition. 2) Set A: People who use public transportation. Set B: People who use private vehicles. $A \setminus B$: People who use public transportation but do not own private vehicles.

4.	A' A compliment Not A	All elements in the universal set outside of A	1) Universal Set: All students in the school. Set A: Students who participate in sports. Complement (A'): Students who do not participate in sports. 2) Universal Set: All internet users. Set A: People who use Facebook. Complement (A'): People who do not use Facebook.
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Self-Assessment Questions

- Universal Set U: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
Set A: {2, 4, 6, 8}, Set B: {4, 5, 6, 7}

 - $A \cup B$ (union).
 - $A \cap B$ (intersection).
 - $A \setminus B$ (elements in A but not in B).
 - A' (complement of A).
- Set A: Students who play cricket.
Set B: Students who play football. Find

 - $A \cup B$.
 - $A \cap B$.
 - $A \setminus B$.
 - $B \setminus A$ represent?
- Universal Set (U): All employees in a company.
A: Employees who have completed training in data analysis.
B: Employees who have completed training in marketing. Find

 - $A \cup B$
 - $A \cap B$
 - $A \setminus B$
 - A'
- A: Students who participate in art club.
B: Students who participate in science club.
If A: {Ali, Sara, Hina, Ahmed} and B: {Sara, Ahmed, Zain}, find:

 - $A \cup B$
 - $A \cap B$
 - $A \setminus B$

6.2 Algebra

Algebra is a branch of mathematics that deals with symbols and the rules for manipulating those symbols. It involves solving equations, working with variables, and studying mathematical expressions. Algebra is fundamental to many areas of mathematics and is used extensively in various fields such as physics, engineering, computer science, economics, and more. A major goal of algebra is to solve equations for some variables. Here we review four basic rules that are often useful in this process.

6.3 Algebraic Expressions

Algebraic expressions are mathematical expressions that involve variables, constants, and mathematical operations (such as addition, subtraction, multiplication, division, exponentiation, and roots). These expressions can represent quantities or relationships between quantities. Variables in algebraic expressions can take on different values, and the expression's value may change accordingly. For example:

- (i) $2x + 3$ is an algebraic expression where x is a variable.
- (ii) $3x^2 - 5x + 2$ is another algebraic expression.
- (iii) $a + b$ is an algebraic expression where a and b are variables.
- (iv) $y = mx + b$ represents a linear equation, where m and b are constants representing the slope and y -intercept, respectively.

Terms related to basic algebra expressions are mentioned in the image given below.

- a) Exponent (1)
- b) Expression (2)
- c) Polynomial (Monomial, binomial and trinomial) (3)
- d) Like terms and Unlike terms (4)
- e) Constants (5)

$$3x^2 - 2xy + c$$

Figure 6.1: Algebraic Expression

Algebraic expressions are often used to model real-world situations, solve equations, and perform calculations. While both sets and algebraic expressions are foundational in mathematics, they serve different purposes: sets are used to represent collections of objects, while algebraic expressions are used to represent mathematical relationships and operations involving variables and constants.

6.4 Basic Rules of Algebra

Four basic rules of algebra are given below:

1. You can interchange the left and right sides of an equation. That is, if $x = y$, it is also true that $y = x$.
2. You can add or subtract the same quantity on both sides of an equation.
3. You can multiply or divide both sides of an equation by the same quantity, as long as you do not multiply or divide by zero.
4. You can raise both sides of an equation to the same power or take the same root on both sides (which is equivalent to raising both sides to the same fractional power).

Adding and Subtracting

Example 1: Solve the equation $x - 9 = 3$ for 'x'.

Solution: We isolate 'x' by adding 9 to both sides:

$$x - 9 + 9 = 3 + 9$$

Hence, $x = 12$

Example 2: Solve the equation $y + 6 = 2y$ for 'y'.

Solution: We put all terms with 'y' on the right side by subtracting 'y' from both sides:

$$y + 6 - y = 2y - y$$

Interchanging the two sides, the answer is $y = 6$. Hence, $6 = y$

Example 3: Solve the equation $8q - 17 = p + 4q - 2$ for p.

Solution: We isolate p by subtracting $4q$ from both sides while also adding 2 to both sides:

$$\begin{aligned} 8q - 17 - 4q + 2 &= p + 4q - 2 - 4q + 2 \\ 4q - 15 &= p \end{aligned}$$

Interchanging the two sides, the answer is $p = 4q - 15$.

Multiplying and Dividing

Example 1: Solve the equation $4x = 24$ for 'x'.

Solution: We isolate by dividing both sides by 4:

$$\frac{4x}{4} = \frac{24}{4}$$

Hence, $x = 6$.

Example 2: Solve the equation $\frac{3z}{4} - 2 = 10$ for 'z'.

Solution: First, we isolate the term containing z by adding 2 to both sides:

$$\begin{aligned}\frac{3z}{4} - 2 &= +2 = 10 + 2 \\ \frac{3z}{4} &= 12\end{aligned}$$

Now we multiply both sides by $\frac{4}{3}$ on both sides:

$$\begin{aligned}\frac{3z}{4} \times \frac{4}{3} &= 12 \times \frac{4}{3} \\ z &= 16\end{aligned}$$

Example 3: Solve the equation $7w = 3s + 5$ for 's'.

Solution: We isolate the term containing 's' by subtracting 5 from both sides:

$$\begin{aligned}7w - 5 &= 3s + 5 - 5 \\ 7w - 5 &= 3s\end{aligned}$$

Next, we divide both sides by 3 to isolate 's', then interchange the two sides to write the final answer.

$$\begin{aligned}\frac{7w - 5}{3} &= \frac{3s}{3} \\ s &= \frac{7w - 5}{3}\end{aligned}$$

Powers and Roots

Example: Find the positive solution of the equation $x^4 = 16$.

Solution: We solve for x by raising both sides to the 4th power:

$$(x^4)^{1/4} = 16^{1/4}$$

This leaves x on the left side (from the rule $(x^n)^m = x^{n \times m}$), and on the right, the $\frac{1}{4}$ power is the same as the fourth root:

$$x^4 \times \frac{1}{4} = 16^{\frac{1}{4}}$$

$$x = \sqrt[4]{16} = 2$$

The positive solution to the equation is $x = 2$.

6.5 Quadratic Equation

A quadratic equation is a type of equation that can be written in the form:

$$ax^2 + bx + c = 0$$

Where:

- x is the variable.
- a , b , and c are constants, with $a \neq 0$.

The equation involves the square of the unknown variable (x^2).

Example

Consider the quadratic equation:

$$x^2 + 5x + 6 = 0$$

Here, $a=1$, $b=5$, and $c=6$.

Solution:

To solve for x , we can factor the equation. We look for two numbers that multiply to give c (which is 6) and add up to give b (which is 5).

The numbers are 2 and 3, because:

$$2 \times 3 = 6 \quad \text{and} \quad 2 + 3 = 5$$

Now, we can factor the quadratic equation as:

$$(x+2)(x+3)=0$$

To find the values of x , set each factor equal to zero:

$$x+2=0 \quad \text{or} \quad x+3=0$$

Solving these:

$$x = -2 \quad \text{or} \quad x = -3$$

Answer: The solutions to the equation $x^2 + 5x + 6 = 0$ are $x = -2$ and $x = -3$.

6.5.1 Applications of Quadratic Equation

Quadratic equations are useful because they help solve many real-world problems, especially when something is changing in a curved or non-linear way. Here are a few examples of how quadratic equations are used in everyday life:

- i. **Throwing a Ball or Object (Projectile Motion):** When you throw a ball into the air, it follows a curved path. Quadratic equations are used to calculate how high the ball will go, how long it will stay in the air, and where it will land. For example, if you throw a ball, a quadratic equation can help you determine the time it takes to hit the ground.
- ii. **Business and Finance:** In business, quadratic equations can be used to calculate profits, especially when profits increase and then decrease as a result of factors like production costs or demand. For example, if a company makes a certain number of sales and then finds that making more sales will lead to higher costs, quadratic equations can help figure out the best number of sales to maximize profit.
- iii. **Designing and Building (Architecture):** In construction and design, quadratic equations can help determine the best angles or sizes for structures. For instance, if you're designing a ramp or arch, a quadratic equation can be used to figure out the right curve for strength and safety.
- iv. **Planning and Time Management:** Sometimes, problems in scheduling or planning (like how long it will take to finish a project) can be modeled by quadratic equations. It helps in finding out when certain tasks are best to complete, especially when you're dealing with increasing and decreasing time limits.

Self-Assessment Questions

1. The area of a rectangular garden is 120 square meters. The length is 5 meters more than the width. Find the dimensions (length and width) of the garden.

Let the width be x . Then the length will be $x+5$. Use the equation for the area of a rectangle:

$$x(x+5)=120$$

Solution:

The area of the garden is given by:

$$x(x+5)=120$$

Simplify:

$$x^2+5x-120=0$$

Factor the quadratic equation:

$$(x-10)(x+12)=0$$

So, $x=10$ or $x=-12$. Since a negative width is not possible, the width is 10 meters. The length is $x+5=10+5=15$ meters. So, the dimensions are 10 meters (width) and 15 meters (length).

2. A ramp is being constructed for wheelchair access. The equation that describes the ramp's slope is:

$$Y = -0.5x^2 + 4x + 2$$

where y is the height and x is the horizontal distance. Find the height when the ramp reaches 6 meters horizontally.

Solution:

The equation for the ramp's slope is:

$$y = -0.5x^2 + 4x + 2$$

We need to find the height when the horizontal distance is $x=6$ meters:

$$y = -0.5(6)^2 + 4(6) + 2$$

$$y = -0.5(36) + 24 + 2$$

$$y = -18 + 24 + 2 = 8$$

Answer: The height of the ramp at 6 meters is 8 meters.

3. An investment grows according to the equation:

$$A(t) = -0.5t^2 + 5t + 1000$$

where $A(t)$ is the amount of money in the account after t years. After how many years will the amount in the account drop back to the initial investment of Rs. 1000/-?

Solution:

The amount of money in the account is given by:

$$A(t) = -0.5t^2 + 5t + 1000$$

We need to find when the amount drops back to the initial investment, i.e., $A(t)=1000$:

$$-0.5t^2 + 5t + 1000 = 1000$$

Simplify:

$$-0.5t^2 + 5t = 0$$

Factor:

$$t(-0.5t+5)=0$$

So, $t=0$ or $t=10$.

Answer: Since $t=0$ represents the initial time, the amount will drop back to Rs. 1000/- after 10 years.

4. A farmer has 100 meters of fencing to create a rectangular enclosure. The length is 10 meters longer than the width. What are the dimensions of the enclosure that will maximize the area?

Solution:

The farmer has 100 meters of fencing. The length is 10 meters more than the width.

Let the width be x , and the length will be $x+10$.

The perimeter is given by:

$$2(x+(x+10))=100$$

Simplify:

$$2(2x+10)=100$$

$$4x+20=100$$

$$4x=80$$

$$x=20$$

So, the width is 20 meters and the length is $20+10=30$ meters.

The area will be:

$$A = 20 \times 30 = 600 \text{ square meters}$$

5. A tower is being built and its height after two years is given by:

$$h(t) = 10t^2 + 50t + 100$$

Find the height of the tower after 3 years.

Solution:

The height of the tower after two years is given by:

$$h(t) = 10t^2 + 50t + 100$$

Find the height after 3 years:

$$h(3) = 10(3)^2 + 50(3) + 100$$

$$h(3) = 10(9) + 150 + 100 = 90 + 150 + 100 = 340\text{meters}$$

So, the height of the tower after 3 years is 340 meters.

SUMMARY

In this unit, we have covered the foundational principles of sets and algebraic expressions, uncovering their roles as fundamental tools for mathematical reasoning and problem-solving. We began by delving into the world of sets, where we learned to define sets, and represent them using set notation. Transitioning to algebraic expressions, we explored the language of algebra, where variables, constants, and terms serve as the building blocks for mathematical expressions and equations. By learning techniques for simplifying expressions and solving equations, we developed the ability to analyze and manipulate algebraic relationships with confidence and proficiency.

EXERCISES

Multiple Choice Questions

1. Which of the following represents the union of two sets A and B?
 - a) $A \cap B$
 - b) $A - B$
 - c) $A \cup B$
 - d) $A \subset B$**Answer:** c) $A \cup B$

2. What is the result of the intersection ($A \cap B$) of sets $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$?
 - a) $\{1, 4\}$
 - b) $\{2, 3\}$
 - c) $\{1, 2, 3, 4\}$
 - d) $\{2, 3, 4\}$**Answer:** b) $\{2, 3\}$

3. Which of the following is the complement of the set $A = \{1, 2, 3\}$ if the universal set is $U = \{1, 2, 3, 4, 5\}$?
 - a) $\{1, 2\}$
 - b) $\{4, 5\}$
 - c) $\{1, 2, 3\}$
 - d) $\{4\}$**Answer:** b) $\{4, 5\}$

4. Simplify: $3x + 5 - 2x + 7$.
 - a) $5x + 12$
 - b) $x + 12$
 - c) $5x + 7$
 - d) $x + 5$**Answer:** b)

5. What is the solution of the quadratic equation $x^2 - 5x + 6 = 0$?
 - a) $x = 1, 6$
 - b) $x = -1, -6$
 - c) $x = 2, 3$
 - d) $x = -2, -3$**Answer:** c) $x = 2, 3$

6. If the set $A=\{a,b,c\}$ and $B=\{b,c,d\}$, what is $A-B$?
- a) $\{a, b\}$
 - b) $\{a\}$
 - c) $\{b, c, d\}$
 - d) $\{d\}$
- Answer:** b) $\{a\}$
7. Simplify: $2x - 3(x - 2) = 5$.
- a) $x = 3$
 - b) $x = 5$
 - c) $x = -5$
 - d) $x = 2$
- Answer:** a) $x = 3$
8. What is the result of $(A \cup B) \cap C$ if $A=\{1,2\}$, $B=\{2,3\}$, and $C=\{3,4\}$?
- a) $\{1, 2, 3\}$
 - b) $\{2, 3\}$
 - c) $\{3\}$
 - d) $\{1, 2, 3, 4\}$
- Answer:** c) $\{3\}$
9. Which of the following is the quadratic form of the equation $x^2 - 4x + 4 = 0$?
- a) $x = 4$
 - b) $x = 2$
 - c) $x = 1$
 - d) $x = 0$
- Answer:** b) $x = 2$
10. Which of the following is an example of an algebraic expression?
- a) $x + 2 = 5$
 - b) $3x^2 + 4x - 7$
 - c) $x = 3$
 - d) $x - 5$
- Answer:** b) $3x^2 + 4x - 7$

Short-Answer Questions

1. What is the union of two sets? Give an example.
2. Explain the difference between the complement and the intersection of sets.
3. Identifying and naming the patterns in algebraic expressions including variables, constants, coefficients, and terms.
4. Define a quadratic equation and give an example.

Subjective Questions

1. Define the set and explain its components, including elements, subsets, unions, intersections, and complements.
2. Discuss the applications of sets in our daily life.
3. Elaborate on the uses of quadratic equations in our daily life.
4. Solve the quadratic equation $x^2 + 6x + 9 = 0$ by factoring.
5. Given two sets $A = \{2, 4, 6\}$ and $B = \{1, 4, 5\}$, find $A \cup B$, $A \cap B$, and $A - B$.

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Unit–7

INTRODUCTION TO STATISTICAL STUDIES

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INTRODUCTION

Statistics is a powerful tool for understanding the world around us by analyzing data and drawing meaningful conclusions. In this unit, we will explore the essential characteristics and various types of statistical studies, providing a foundation for understanding how data are collected, analyzed, and interpreted in different contexts. Statistical studies are essential for making informed decisions and drawing accurate conclusions about populations. By understanding the characteristics of statistical studies and the different types of study designs, we can better interpret the results and evaluate the validity of statistical findings.

In this unit, we will begin by examining the fundamental characteristics of statistical studies, including the population, sample, variables, and data collection methods. We will explore the importance of randomization, bias, and reliability in ensuring the validity of study results. Building upon this foundation, we will then explore different types of statistical studies, including observational studies, experiments, surveys, and retrospective studies. Each type of study design has its strengths and limitations, and understanding these differences is crucial for selecting the appropriate approach for a given research question. Throughout this unit, we will emphasize the importance of critical thinking and skepticism in evaluating statistical studies. By learning to recognize potential sources of bias and error, we can become more discerning consumers of statistical information and make more informed decisions in our personal and professional lives.

LEARNING OUTCOMES

After completing this unit, you will be able to:

1. Understand the fundamental characteristics of statistical studies, including populations, samples, variables, and data collection methods.
2. Recognize the importance of randomization, bias, and reliability in ensuring the validity of study results.
3. Differentiate between the various types of statistical studies, including observational studies, experiments, surveys, and retrospective studies.
4. Interpret the strengths and limitations of each type of study design and understand when it is appropriate to use them.
5. Develop critical thinking skills to evaluate the quality and validity of statistical studies, including assessing potential sources of bias.

6. Understand the techniques for selecting representative samples, minimizing bias, and maximizing the reliability of study results.
7. Explore real-world examples of statistical studies and analyze how different study designs are used to address research questions and inform decision-making.
8. Apply the knowledge and skills gained from this unit to critically evaluate research findings, make informed decisions, and communicate effectively about statistical studies in various contexts.

7.1 Introduction to Statistics

Statistics is defined as the science of collecting, organizing, presenting, analyzing and interpreting numerical data for making a more effective decision. Statistics is concerned with scientific methods for collecting, organizing, summarizing, presenting and analyzing data as well as deriving valid conclusions and making reasonable decisions on the basis of this analysis. Statistics is concerned with the systematic collection of numerical data and its interpretation. The word ‘statistic’ can be referred to:

- Numerical facts, such as the number of people living in a particular area.
- The study of ways of collecting, analyzing and interpreting the facts or data.

The subject of statistics plays a major role in modern society. It is used to determine whether a new drug is effective in treating a particular disease e.g., fever, pain or Hepatitis, cancer, etc. It is involved when agricultural inspectors check the safety of the food supply. It is used in every opinion poll and survey. In business, it is used for market research. Sports statistics are part of daily conversation for millions of people. Statistics can be divided into two main areas or branches.

- (i) **Descriptive Statistics:** Descriptive statistics involve the procedures used to describe a given collection of data. It covers the measures of central tendency, variability frequency and percentages.
- (ii) **Inferential Statistics:** Inferential statistics involve the procedures that let us infer our findings beyond the particular sample at hand to the larger population represented by that sample. It covers correlation, chi-square, t-test, and ANOVA.

7.2 Steps of a Statistical Study

The basic steps of a study are given below:

- (i) State the goal of your study. It involves determining the population you want to study and exactly what you want to learn about it.
- (ii) Choose a representative sample from the population.
- (iii) Collect raw data from the sample and summarize these data by finding sample statistics of interest.
- (iv) Use the sample statistics to infer the population parameters.
- (v) Draw conclusions: Determine what you learned and whether you achieved your goal. The steps of the statistical study are evident in the image given below.

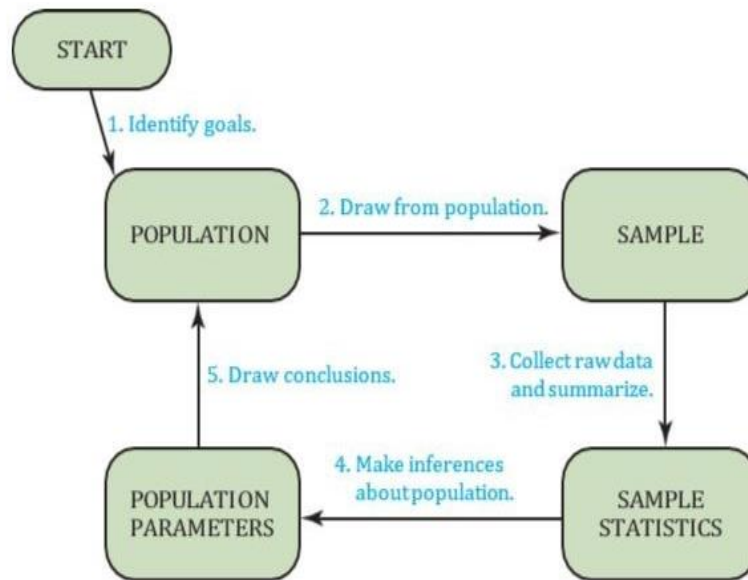


Figure 7.1. Elements of Statistical Study

There are different ways to conduct a statistical study; we will discuss it in the upcoming topic ‘types of statistical study’. Let us see an example of a statistical study. Ali conducted a study to find out the favorite sports of the people of Pakistan. He developed a survey form to collect the data from the people. For this study, Ali approached 7000 people from various parts of the country (in order to get representation from various provinces and parts of the country) and get their responses on their favorite sports. Ali entered all the ratings in MS Excel, analyzed the data by applying statistical formulas, and shared his results in the form of tables and graphs. Notice here that Ali collected data only from 7000 people and shared the results about the favorite sports of the people of Pakistan (imagine the total population of the country as compared to 7000 people approached for collecting data for this study). In Statistical language, we can say that the purpose of this study was to find out the favorite sports of the people of Pakistan i.e., the population of this study was all the people of Pakistan. The data about the favorite sports of the people of Pakistan are called **population parameters**. Ali studied a smaller group (i.e., a **sample of the study**) to learn about the **population** of the study (i.e., people of Pakistan are a very large group of people). In this study, the sample consisted of 7000 people from various parts of the country. The individual forms, Ali received from the sample with their responses about their favorite sports, represent the **raw data**. Ali entered these responses into MS Excel or SPSS (MS Excel and SPSS can be conveniently used for statistical data analysis), and presented the results in the form of percentages in tables and graphs. These percentages or numbers are called **sample statistics**.

- **Data** is raw, unorganized facts that need to be processed. Data can be something simple and seemingly random and useless until it is organized.
- The **population** in a statistical study is the complete set of people or things being studied. The **sample** is the representative subset of the population from which the raw data are actually obtained.
- We usually think of a **population** as a group of people. However, in statistics, a population can be any kind of group—people, animals, or things. For example, in a study of the tuition cost of college education, the population might be all colleges and universities, and the population parameters might include prices for tuition, fees, and housing.
- Experts have provided formulas and tables to decide about the **size (e.g., number of subjects) of the sample** for a given size of population. You must select the appropriate option for the population (i.e., what is the exact size or number of subjects of the population or if you do not know about the size of the population, select the ‘unknown’ or ‘infinite’ size of the population), you would get the appropriate size of sample for your study.
- **Population parameters** are specific numbers describing the characteristics of the population. **Sample statistics** are numbers describing characteristics of the sample, found by consolidating or summarizing the raw data collected from the sample.

7.3 Sampling Techniques

A representative sample is a sample in which the relevant characteristics of the sample members are generally the same as those of the population. The sampling methods, commonly used, are given below:

- i. **Simple Random Sampling:** In a simple random sampling technique, a sample of items is chosen in such a way that every sample of the same size has an equal chance of being selected.
- ii. **Systematic Sampling:** In systematic sampling, the sample is chosen by selecting every k th individual from the list, such as selecting every 10th or every 50th member of the population.
- iii. **Convenience Sampling:** Convenience sampling allows one to choose a sample that is convenient to select, such as people who happen to be in the same classroom.
- iv. **Stratified Sampling:** We use this method when we are concerned about differences among subgroups, or strata, within a population. We first identify the subgroups and then draw a simple random sample within each subgroup. The total sample consists of all the samples from the individual subgroups.

1.4 Data and Its Types

Qualitative data describe qualities or categories. Quantitative data represent counts or measurements. Essay grades are generally subjective because different teachers might score the same essay differently. We say that the grade categories are qualitative, because they represent qualities such as bad or good. In contrast, scores on a multiple-choice exam are quantitative, because they represent an actual count (or measurement) of the number of correct answers.

Data are collected for a variable. In statistics, a variable is any characteristic, number, or quantity that can be measured or counted. Variables can take on different values depending upon its characteristics.

Examples: Classify each of the following types of data as either qualitative or quantitative.

- a) Brand names of shoes in a consumer survey
- b) Heights of students
- c) Audience ratings of a film on a scale of 1 to 5, where 5 means excellent.

Solution

- a) Brand names are non-numerical categories, so they are qualitative data.
- b) Heights are measurements, so they are quantitative data.
- c) Although the film rating categories involve numbers, the numbers represent subjective opinions about a film, not counts or measurements. These data are therefore qualitative, despite being stated as numbers.

The variables in your data can be classified into four types on the basis of the measurement scale used to measure it.

Table 7.1: Types of data variables and the measurement scales

Type of Scale	Data Characteristics	Numerical Description	Examples	Descriptive Statistics
Nominal	Classification but no order, distance, or origin	Counting	Gender (1=Male, 2=Female)	Frequency in each category Percent in each category Mode

Ordinal	Classification and order but no distance or unique origin	Rank ordering	Academic status (1=Freshman, 2=Sophomore, 3=Junior, 4=Senior), Age group, Income level, Employment Status	Median Range Percentile ranking
Interval	Classification, order, and distance but no unique origin	Arithmetic operations that preserve order and magnitude	Temperature in degrees Satisfaction on semantic differential scale	Mean Standard deviation Variance
Ratio	Classification, order, distance and unique origin	Arithmetic operations on actual quantities	Age in years Income in Saudi riyals	Geometric mean Coefficient of variation

The data at nominal and ordinal scales are represented with respect to their category names. However, while using software like SPSS for quantitative/statistical analysis, these categories may be assigned some numerical code for easy calculations. This numerical code does not represent the value of that category (it is only applicable to nominal and ordinal scales) but it only serves the identity or differentiation of one category from the other categories of that variable (see the example in the table given above).

7.5 Binning Data

When we deal with quantitative data categories, it is often useful to group, or bin, the data into categories that cover a range of possible values. For example, in a table of income levels, it might be useful to create bins of Rs. 0/- to Rs. 20,000/-, Rs. 20,001/- to Rs. 40,000/-, and so on. In this case, the frequency of each bin is simply the number of people with incomes in that bin.

Example: Consider the following set of 20 scores from a 100-point exam:

76 80 78 76 94 75 98 77 84 88 81 72 91 72 74 86 79 88 72 75

Determine appropriate bins, and make a frequency table.

Solution: The scores range from 72 to 98. One way to group the data is with 5-point bins.

The first bin represents scores from 95 to 99, the second bin represents scores from 90 to 94, and so on. Note that there is no overlap between bins. We then count the frequency (the number of scores) in each bin. For example, only 1 score is in bin 95 to 99 (the high score of 98) and 2 scores are in bin 90 to 94 (the scores of 91 and 94). Table given below shows the complete frequency table. In this case, we interpret the cumulative frequency of any bin to be the total number of scores in or above that bin. For example, the cumulative frequency of 6 for the bin 85 to 89 means that 6 scores are either between 85 and 89 or higher than 89.

Table 7.2: Frequency table for binned data

Scores	Frequency
95-99	1
90-94	2
85-89	3
80-84	3
75-79	7
70-74	4
Total	20

7.6 Types of a Statistical Study

Broadly speaking, most statistical studies fall into one of three categories: surveys, observational studies and experiments. In an observational study and surveys, researchers observe or measure the characteristics of the sample members but do not attempt to influence or modify these characteristics. In an experiment, researchers apply a treatment to some or all of the sample members and then observe the effects of the treatment. The study conducted by Ali, is an observational study. The study conducted to check the after-effects of a medicine for vitamin C is an example of an experiment.

It is difficult to determine whether an experimental treatment works unless you compare groups that receive the treatment to groups that do not. In the vitamin C study, for example, researchers might create two groups of people: a treatment group that takes large doses of vitamin C and a control group that does not take vitamin C. The treatment group in an experiment is the group of sample members who receive the treatment being tested. The control group in an experiment is the group of sample members who do not receive the treatment being tested. An experiment is single-blind if the participants do not know whether they are members of the treatment group or members of the control group, but the

experimenters do know. An experiment is double-blind if neither the participants nor the experimenters (people administering the treatment) know who belongs to the treatment group and who belongs to the control group. It is important for the treatment and control groups to be selected randomly and to be alike in all respects except for the treatment.

Having a control group is usually crucial to interpreting the results of experiments. In an experiment, it is important for the treatment and control groups to be alike in all respects except for the treatment. For example, if the treatment group consisted of active people with good diets and the control group consisted of sedentary people with poor diets, we could not attribute any differences in colds to vitamin C alone. To avoid this type of problem, assignments to the control and treatment groups must be done randomly.

A case-control study (or retrospective study) is an observational study that resembles an experiment because the sample naturally divides into two (or more) groups. The participants who engage in the behavior under study form the cases, which makes them like a treatment group in an experiment. The participants who do not engage in the behavior are the controls, making them like a control group in an experiment.

The margin of error in a statistical study is used to describe a confidence interval that is likely to contain the true population parameter. We find this interval by subtracting and adding the margin of error from the sample statistic obtained in the study. That is, the confidence interval is

from (sample statistic - margin of error)
to (sample statistic + margin of error)

In social sciences, we assume the confidence interval of 95% with a margin of error of 5% whereas in physical and medical sciences, a valid and effective study must assume the confidence interval of 99% with a margin of error of 1%. Please read about the importance and reason for this limit of confidence interval in various fields. What impact does the margin of error have on the study results and its implications?

Examples:

For each of the following questions, which type of statistical study is most appropriate to lead to an answer? Give a reason for your answer.

- a) What is the average income of a computer engineer?
- b) Do seat belts save lives?
- c) Can lifting weights improve runners' times in a 10-kilometer race?
- d) Can a new herbal remedy reduce the severity of colds?

7.7 Evaluating a Statistical Study

There are eight guidelines for evaluating a statistical study, as given below.

1. **Get a big picture view of the study:** You should understand the goal of the study, the population that was under study, and whether the study was observational or an experiment.
2. **Consider the source:** Statistical studies are supposed to be objective, but the people who carry them out and fund them may be biased. Look for any potential sources of bias on the part of the researchers.
3. **Look for bias in the sample:** Decide whether the sample is the true representative of the population. Selection bias occurs whenever researchers select their sample in a way that tends to make it unrepresentative of the population. For example, a pre-election poll that surveys only registered voters who are workers of one particular political party, has selection bias because it is unlikely to reflect the opinions of all voters. Participation bias occurs whenever people choose whether to participate. For example, if participation in a survey is voluntary, people who feel strongly about the survey issue are more likely to participate, so their opinions may not represent the larger population that is less emotionally attached to the issue. (Surveys or polls in which people choose whether to participate are often called self-selected or voluntary response surveys.)
4. **Look for problems in defining or measuring the variables of interest:** A variable is any item or quantity that can vary or take on different values. The variables of interest in a statistical study are the items or quantities that the study seeks to measure. For example, in the previous example at the start of this unit, Ali studied about favorite sports and the number of playing those sports. Ambiguity in the variables can make it difficult to interpret reported results.
5. **Beware of confounding variables:** Variables that are not intended to be part of the study can sometimes make it difficult to interpret results properly. Such variables are often called confounding variables, because they confound (confuse) a study's results. Confounding variables can be discovered simply by thinking hard about factors that may have influenced a study's results. If the study neglected potential confounding variables, the results of the study may not be valid.
6. **Consider the setting and wording in surveys:** Look for anything that might tend to produce inaccurate or dishonest responses. Dishonest responses are particularly likely when the survey concerns sensitive subjects, such as personal habits or income. For example, the question "Do you cheat on your income taxes?" may/may not elicit honest answers from those who cheat, especially if the setting of the study does not guarantee complete confidentiality.

Sometimes just the order of the words in a question can affect the outcome. A poll conducted in Germany asked the following two questions:

- (i) Would you say that traffic contributes more or less to air pollution than industry?
- (ii) Would you say that industry contributes more or less to air pollution than traffic?

The only difference is the order of the words traffic and industry, but this difference dramatically changed the results in the study. With the first question, 45% answered traffic and 32% answered industry. With the second question, only 24% answered traffic while 57% answered industry.

- 7. **Check that results are presented fairly:** Even when a statistical study is done well, it may be misrepresented in graphs or concluding statements. Check whether the study really supports the conclusions that are presented in the media.
- 8. **Stand back and consider the conclusions:** Evaluate whether the study achieved its goals. If so, do the conclusions make sense and have practical significance? Are there any alternative explanations for the results?

Example 1

Radon is a radioactive gas produced by natural processes (the decay of uranium) in the ground. The gas can leach into buildings through the foundation and can accumulate in relatively high concentrations if doors and windows are closed. Imagine a study that seeks to determine whether radon gas causes lung cancer by comparing the lung cancer rate in Colorado, where radon gas is fairly common, with the lung cancer rate in Hong Kong, where radon gas is less common. Suppose the study finds that the lung cancer rates are nearly the same. Is it fair to conclude that radon is not a significant cause of lung cancer?

Solution: The variables under study are the amount of radon and lung cancer rate. However, because smoking can also cause lung cancer, the smoking rate may be a confounding variable in this study. In particular, the smoking rate in Hong Kong is much higher than the smoking rate in Colorado, so any conclusions about radon and lung cancer must take the smoking rate into account. In fact, careful studies have shown that radon gas can cause lung cancer, and the U.S. Environmental Protection Agency (EPA) recommends taking steps to prevent radon from building up indoors.

Example 2

An experiment is conducted in which the weight losses of people who try a new “Fast Diet Supplement” are compared to the weight losses of a control group of people who try to lose weight in other ways. After eight weeks, the results show that the treatment group lost an average of $\frac{1}{2}$ pounds more than the control group. Assuming that it has no dangerous side effects, does this study suggest that the Fast Diet Supplement is a good treatment for people wanting to lose weight?

Solution: Compared to the average person’s body weight, the difference of 1 pound hardly matters at all. So even if the study is flawless, the results do not seem to have much practical significance.

SUMMARY

Throughout this unit, we have journeyed through the essential characteristics of statistical studies and explored the various types of study designs used to collect and analyze data. We began by understanding the fundamental components of statistical studies, including populations, samples, variables, and data collection methods. We learned about the importance of randomization and bias in ensuring the validity of study results, and we developed critical thinking skills to evaluate the quality and validity of statistical studies.

Building upon this foundation, we explored different types of statistical studies, including observational studies, experiments, surveys, and retrospective studies. We recognized the strengths and limitations of each type of study design and learned when it is appropriate to use them based on the research question and context. With a solid understanding of the characteristics and types of statistical studies, we are better equipped to make informed decisions, communicate effectively about statistical findings, and contribute to evidence-based decision-making in our personal and professional lives.

EXERCISES

Multiple Choice Questions

1. Which of the following is an example of descriptive statistics?
 - a) Predicting election results
 - b) Calculating the average score of a class
 - c) Testing the effectiveness of a new drug
 - d) Estimating population size**Answer:** b) Calculating the average score of a class
2. Which step is NOT part of a statistical study?
 - a) Collecting data
 - b) Analyzing data
 - c) Changing results
 - d) Drawing conclusions**Answer:** c) Changing results
3. What is the primary goal of inferential statistics?
 - a) To summarize data
 - b) To make predictions about a population based on a sample
 - c) To calculate averages
 - d) To present data visually**Answer:** b) To make predictions about a population based on a sample
4. Which type of probability sampling involves selecting every nth individual from a population?
 - a) Simple random sampling
 - b) Systematic sampling
 - c) Stratified sampling
 - d) Cluster sampling**Answer:** b) Systematic sampling
5. In which type of probability sampling is the population divided into subgroups, and samples are taken from each subgroup?
 - a) Simple random sampling
 - b) Systematic sampling
 - c) Stratified sampling
 - d) Cluster sampling**Answer:** c) Stratified sampling
6. What is meant by binning data?
 - a) Sorting data in ascending order
 - b) Grouping data into intervals
 - c) Ignoring missing data
 - d) Calculating statistical measures**Answer:** b) Grouping data into intervals

7. What type of data is “eye color”?
- a) Quantitative
 - b) Qualitative
 - c) Continuous
 - d) Discrete
- Answer:** b) Qualitative
8. Which of the following is a continuous variable?
- a) Number of students in a class
 - b) Weight of a person
 - c) Number of cars in a parking lot
 - d) Type of vehicles owned
- Answer:** b) Weight of a person
9. Which sampling technique ensures each member of the population has an equal chance of being selected?
- a) Cluster sampling
 - b) Stratified sampling
 - c) Simple random sampling
 - d) Convenience sampling
- Answer:** c) Simple random sampling
10. What is the first step of a statistical study?
- a) Collecting data
 - b) Formulating a hypothesis
 - c) Identifying the problem or question
 - d) Analyzing results
- Answer:** c) Identifying the problem or question
11. In evaluating a statistical study, what is crucial to assess?
- a) The clarity of the conclusion
 - b) The relevance of the hypothesis
 - c) The design and methods used
 - d) All of the above
- Answer:** d) All of the above

Short-Answer Questions

1. Define descriptive and inferential statistics with one example each.
2. What are the key steps in conducting a statistical study?
3. Differentiate between random and non-random sampling techniques.
4. Explain the concept of binning data with an example.
5. What are the two main types of data? Provide an example for each.

Subjective Questions

1. Discuss the importance of descriptive and inferential statistics in decision-making. Include examples where each type is used effectively in real life.
2. Describe the different sampling techniques. Discuss their advantages and disadvantages with examples.
3. Explain the types of statistical studies. Highlight their differences and give examples of scenarios where each type is appropriate.
4. Evaluate the importance of properly designed statistical studies. Discuss the common errors that might occur during data collection and analysis.
5. Analyze a scenario of your choice using the steps of a statistical study. For example, conduct a basic survey to analyze students' preferences for different subjects.
6. Based on the information provided for each study, what suggestions or comments will you give to make it an effective study with respect to the design of the study, sampling, and other aspects?
 - (i) Researchers planned a survey study to determine whether people living in rural areas are happier than those living in urban areas.
 - (ii) A government study is designed to determine the percentage of taxpayers who understate their income, based on people who had their tax returns audited.
 - (iii) In a study designed to determine whether bicyclists who wear helmets have fewer accidents, researchers tracked 500 riders with helmets for one month.
 - (iv) In a study of obesity among children, researchers monitor the eating and exercise habits of the participating children, carefully recording everything they eat and all their activities.

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Unit–8

DESCRIPTIVE STATISTICS

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INTRODUCTION

Descriptive statistics serves as the bedrock upon which much of statistical analysis is built. It provides the tools and techniques necessary to understand, summarize, and visualize data in a meaningful and interpretable way. Descriptive statistics allows us to explore the characteristics of a dataset providing insights into its central tendencies, variability, and distribution. By examining key summary measures and visual representations, we can uncover patterns, trends, and relationships within the data, enabling us to make informed decisions and draw meaningful conclusions.

In this unit, we will explore the fundamental concepts and methods of descriptive statistics. We will begin by learning about measures of central tendency such as the mean, median, and mode, which provide a summary of the typical or central value of a dataset. Building upon this foundation, we will delve into measures of variability, including the range, variance, and standard deviation, which quantify the spread or dispersion of data points around the central tendency. Understanding these measures is essential for assessing the variability and reliability of data.

This unit will emphasize the importance of critical thinking and interpretation in descriptive statistics. By developing a solid understanding of descriptive statistics, you will be equipped with the skills and knowledge necessary to analyze and interpret data in a wide range of contexts, from scientific research to business analytics.

LEARNING OUTCOMES

After completing the unit, you would be able to:

1. Understand the fundamental concepts of descriptive statistics, including measures of central tendency, variability, and distribution.
2. Calculate and interpret measures of central tendency, such as the mean, median, and mode, to summarize the typical value of a dataset.
3. Calculate and interpret the measures of variability, including the range, variance, and standard deviation, to assess the spread or dispersion of data points.
4. Apply descriptive statistics techniques to analyze and interpret data in various contexts, including scientific research, business analytics, and social sciences.
5. Communicate findings effectively using descriptive statistics, presenting summary measures and visual representations in a clear and concise manner to stakeholders or audiences.
6. Develop a deeper appreciation for the role of descriptive statistics in understanding and interpreting data, and recognize its importance as a foundational skill in statistical analysis and decision-making.

8.1 Descriptive Statistics

Descriptive statistics refers to the branch of statistics that focuses on summarizing and describing the main features of a dataset. Its primary goal is to organize, summarize, and present the data in a meaningful way, allowing for easier interpretation and understanding. Descriptive statistics provide simple summaries of the sample and the observations within it. Common measures and techniques in descriptive statistics include.

1. **Frequency Distributions & Graphical Representations:** It involves summarizing the number of times each value or category occurs in a dataset, in the form of tables, histograms, or bar charts. Visualizations such as histograms, box plots, scatter plots, and pie charts help in illustrating the characteristics of the data.
2. **Measures of Shape:** These statistics describe the distributional shape of the data and include skewness (asymmetry) and kurtosis (peakedness).
3. **Measures of Central Tendency:** These statistics indicate the center or average of the data and include measures such as the mean (average), median (middle value), and mode (most frequent value).
4. **Measures of Variability:** These statistics describe the spread or dispersion of the data and include measures such as the range, variance, standard deviation, and interquartile range.

Descriptive statistics are useful for gaining insights into the basic properties of a dataset, identifying patterns, detecting outliers, and making initial interpretations. They form the foundation for more advanced statistical analyses and inferential statistics, which involve making predictions or inferences about a population based on sample data.

8.2 Levels of Measurement

There are four types of measurement scales: nominal, ordinal, interval and ratio scale. Each scale is characterized by the amount of preciseness carried by the scale due to its characteristics. These scales are used to measure different types of variables in your data. The detail of these scales is given in the table below:

Table 8.1: Types of measurement scales and the data variables

Type of Scale	Characteristics Description	Numerical Data	Examples	Descriptive Statistics
Nominal	Classification but no order, distance, or origin	Counting	Gender (1=Male, 2=Female)	Frequency in each category Percent in each category Mode
Ordinal	Classification and order but no distance or unique origin	Rank ordering	Academic status (1=Freshman, 2=Sophomore, 3=Junior, 4=Senior), Age group, Income level, Employment Status	Median Range Percentile ranking
Interval	Classification, order, and distance but no unique origin	Arithmetic operations that preserve order and magnitude	Temperature in degrees Satisfaction on semantic differential scale	Mean Standard deviation Variance
Ratio	Classification, order, distance and unique origin	Arithmetic operations on actual quantities	Age in years Income in Saudi riyals	Geometric mean Coefficient of variation

8.3 Frequency and Frequency Tables

In statistics, frequency refers to the number of times a particular value or category occurs in a dataset. It's a fundamental concept used to summarize and analyze data. Frequencies are often presented in tables, histograms, or other graphical representations to show the distribution of values within a dataset. For example, if you have a dataset consisting of exam scores for a class of students, the frequency of each score would indicate how many students achieved that particular score. This information can help you understand the distribution of scores and identify patterns or outliers within the data.

A basic frequency table has two columns. The first column lists all the categories of data. The second column lists the frequency of each category, which is the number of data values in the category. For example, a teacher made the following list for the grades she awarded to her 25 students on an essay: A C C B C D C C F D C C C B B A B D B A A B F C B

This list contains all the grades, but it is not easy to read. A better way to display these data is with a frequency table—a table showing the number of times, or frequency, that each grade appears (as shown in the table). The five possible grades are called the categories for the table.

Table 8.2: Frequency Table

Grade	Frequency
A	4
B	7
C	9
D	3
F	2
Total	

8.3.1 Cumulative and Relative Frequency

The relative frequency of any category is the fraction (or percentage) of the data values that fall in that category:

$$\text{Relative frequency} = \frac{\text{Frequency in category}}{\text{total frequency}}$$

The cumulative frequency of any category is the number of data values in that category and all preceding categories.

The relative frequency for a category expresses its frequency as a fraction or percentage of the total. For example, (check the table given above) 4 of the 25 students received A grades, so the relative frequency for A grades is $4/25$, or 16%. The total relative frequency must always be 1, or 100% (though rounding may sometimes cause the relative frequencies in a table or chart to add up to slightly more or less than 100%). The cumulative frequency is the number of responses in a particular category and all preceding categories. For example, the cumulative frequency for grades of C and above is 20, because 20 students received grades of either A, B, or C.

Example:

Consider the following set of 20 scores from a 100-point exam:

76 80 78 76 94 75 98 77 84 88 81 72 91 72 74 86 79 88 72 75

Determine appropriate bins, and make a frequency table. Include columns for relative and cumulative frequency, and interpret the cumulative frequency for this case.

Solution: Please check the following table for the required calculations.

Table 8.3: Relative and Cumulative Frequency of the exam scores

Scores	Frequency	Relative frequency	Cumulative frequency
95-99	1	0.05 = 5%	1
90-94	2	.10 = 10%	3
85-89	3	0.15 = 15%	6
80-84	3	0.15 = 15%	9
75-79	7	0.35 = 35%	16
70-74	4	0.20 = 20%	20
Total	20	1.00 = 100%	20

8.4 Measures of Central Tendency: Mean, Median and Mode

Measures of central tendency are statistical measures that describe the central or typical value in a dataset. They provide a single value that represents the "center" of the data distribution. The three most common measures of central tendency are:

- 1) Mean:** The mean, also known as the average, is calculated by summing all the values in the dataset and then dividing by the total number of values. It is easy to calculate, and it takes into account each and every value of a given dataset. However, it is sensitive to extreme values (outliers) and is affected by the distribution of the data. The formula for the mean is given below:

$$\text{Mean} = \frac{(\text{Sum of all values})}{(\text{Number of values})}$$

Example 5, 10, 12, 16, 8, 42, 25, 15, 10, 7

Solution:

$$\begin{aligned} &5+10+12+16+8+42+25+15+10+7 \\ &= 150 / 10 \\ &\text{Mean} = 15 \end{aligned}$$

- 2) **Median:** The median is the middle value in a dataset when the values are arranged in ascending or descending order. If there is an even number of values, the median is the average of the two middle values (add the middle values and divide it by 2). The median is less affected by extreme values and is often used when the data is skewed or contains outliers. It does not take into account each and every value of a given dataset, only the midpoint of the data. It is used when the calculation of the average is not possible.

Example:

12, 15, 10, 20, 18, 25, 45, 30, 26

By arranging the data in ascending order

10, 12, 15, 18, 20, 25, 26, 30, 45

So,

Median = 20

- 3) **Mode:** The mode is the value that occurs most frequently in a dataset. A dataset can have one mode (unimodal), two modes (bimodal), or more than two modes (multimodal). The mode is useful for categorical data like shoe size, average yield of crops etc. It can also be used for numerical data. However, it does not take into account each and every value of a given dataset but only the one that is frequently occurring.

Example

12, 24, 15, 18, 30, 48, 20, 24

So,

Mode = 24

These measures provide different insights into the central tendency of the data and are used in various contexts depending on the nature of the dataset and the purpose of the statistical study.

8.5 Shapes of Distribution: Symmetry, Variation and Normal Distribution

8.5.1 Skewness

Skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable about its mean. In simpler terms, it indicates whether the data is skewed to the left or right relative to the mean.

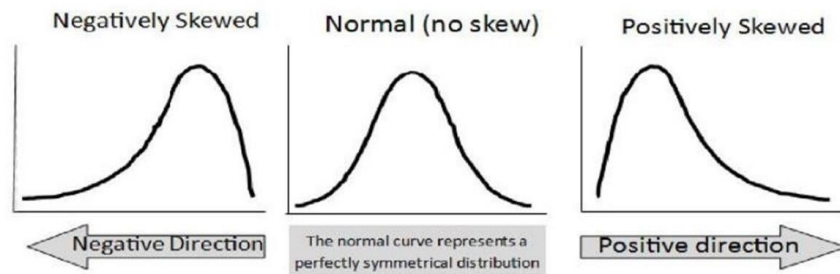


Figure 8.1. Skewness

If the distribution is skewed to the left (negatively skewed), it means that the tail on the left side of the distribution is longer or more spread out than the tail on the right side. In other words, the majority of the data points are concentrated on the right side of the distribution, and there are relatively few data points on the left side. If the distribution is skewed to the right (positively skewed), it means that the tail on the right side of the distribution is longer or more spread out than the tail on the left side. In this case, the majority of the data points are concentrated on the left side of the distribution, and there are relatively few data points on the right side.

Skewness is often measured using the skewness coefficient or skewness statistic. A skewness value of 0 indicates a symmetrical distribution, while positive and negative skewness values indicate right and left skew, respectively. The magnitude of the skewness value indicates the degree of skewness.

8.5.2 Normal Distribution

If the data is evenly distributed, our graph will come across a curve (as shown in the image below). In statistics, this curve is called a normal curve and in social sciences, it is called the bell curve. A standard normal curve has a mean of 0 and a standard of 1. The larger the standard deviation, the flatter the curve will be and vice versa.

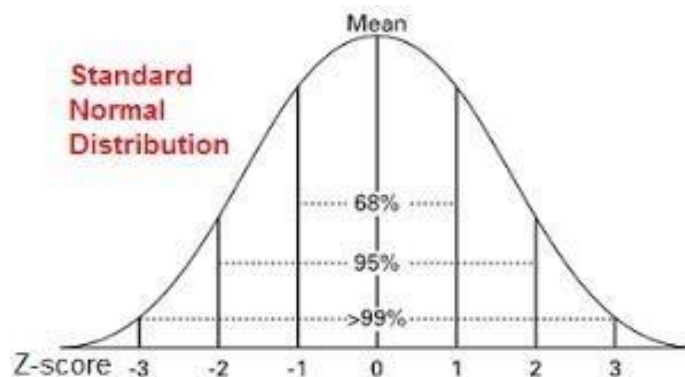


Figure 8.2. Normal Distribution

For a normal distribution curve of a dataset, the score of mean, median or mode are equal. The curve is symmetric at the center (i.e. around the mean). Exactly half of the values are to the left of the center and half to the right. The total area under the curve is 1.

8.5.3 Kurtosis

Kurtosis is a statistical measure that describes the shape of the probability distribution of a real-valued random variable. It measures the "tailedness" or the degree of peakedness of a distribution relative to the normal distribution.

If a distribution has positive kurtosis (leptokurtic), it means that it has fatter tails and a higher peak compared to the normal distribution. This indicates that the distribution has heavier tails and more extreme values than the normal distribution. If a distribution has negative kurtosis (platykurtic), it means that it has thinner tails and a lower peak compared to the normal distribution. This indicates that the distribution has lighter tails and fewer extreme values than the normal distribution. If a distribution has a kurtosis value close to zero (mesokurtic), it means that it has similar tail behavior and weakness to the normal distribution.

Kurtosis is typically measured using the kurtosis coefficient or kurtosis statistic. The kurtosis of the normal distribution is 3. Therefore, excess kurtosis is often calculated as the kurtosis coefficient minus 3. A positive excess kurtosis indicates leptokurtosis, while a negative excess kurtosis indicates platykurtosis. Kurtosis is important in statistics because it provides information about the shape of the distribution and helps to identify the presence of outliers or heavy-tailed distributions. It can also affect the performance of statistical tests and models, particularly those that assume a normal distribution.

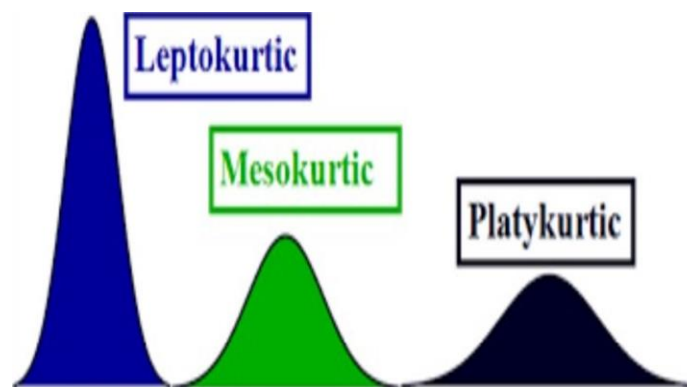


Figure 8.3. Kurtosis

1.6 Measures of Variation: Range, Quartile, Standard Deviation, Standard Scores, Percentiles

The measure of central tendency informs us about the central value of a dataset. However, it does not tell about the spread of values in the dataset because any two sets of data may have the same value of the central tendency but have vastly different magnitudes of variability. Consider the following two types of datasets having same the mean value but different variability.

10, 12, 11, 14, 13

10, 2, 18, 27, 3

In such cases, additional information regarding how the data are dispersed around the average is required. It is called the measures of dispersion. By dispersion, we mean the degree to which data is spread about an average value. The measures of dispersion are given below:

i. Range

The Range is defined as the difference between the largest score (L) in the set of data and the smallest score (S) in the set of data.

$$\text{Range} = L - S$$

Let us calculate the range of the following data:

4 8 1 6 6 2 9 3 6 9

The largest score (L) is 9; the smallest score (S) is 1; the range is:

$$L - S = 9 - 1 = 8$$

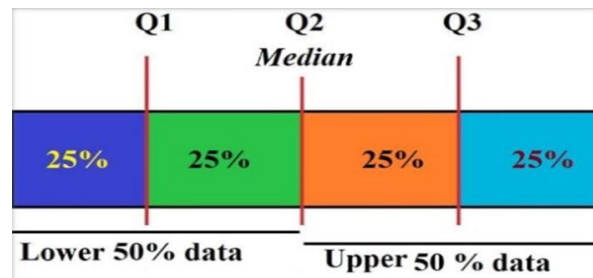
It gives a rough idea of the spread of the data but is sensitive to outliers. The range is used when you have ordinal data or you are presenting your results to people with little or no knowledge of statistics. The range is rarely used in scientific work as it does not a very detailed information about the dataset. It depends only on two scores in the given dataset i.e., the highest number and lowest number. Two very different sets of data can have the same range. For example, “1, 6, 1, 4, 9” and “1, 3, 5, 7, 9” are two datasets with the same range.

ii. Interquartile Range (IQR)

The interquartile range is the range of the middle 50% of the data and is calculated as the difference between the third quartile (Q3) and the first quartile (Q1). It is less

sensitive to outliers compared to the range and provides a measure of the spread of the central portion of the data.

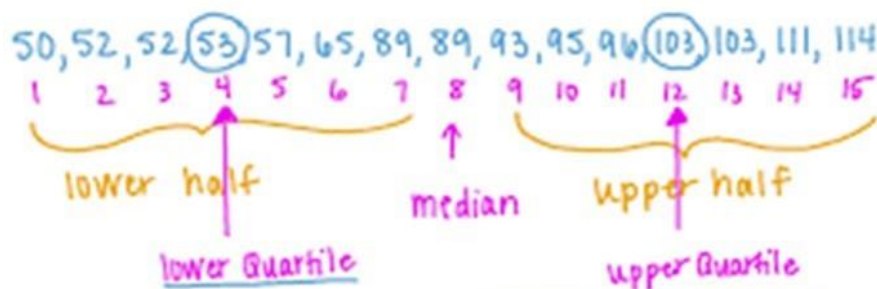
$$IQR = Q3 - Q1$$



- Q1. It is the midpoint of the lowest 50% of data and is also known as the lowest quartile or first quartile.
- Q2. It is the median of the data or the middle point of the given data set and is also known as the second quartile.
- Q3. It is the middle point of the highest 50% of data and is also known as the highest quartile or third quartile.

Therefore, quartiles measure the dispersion of scores above and below the median by dividing the entire data set into 4 equal groups.

Determine the upper and lower quartiles of the following set of data:
~~114, 103, 50, 52, 95, 103, 95, 53, 65, 57, 52, 89, 101, 89 and 96.~~



Lower Quartile is 53 and Upper Quartile is 103

iii. Mean Deviation

It is the mean of the distances of each value from their mean. It helps us get a sense of how "spread out" the values in a data set are. There are three steps involved:

- Find the mean of the value
- Find the distance of each value from the mean
- Find the mean of those distances

Example

Find the mean deviation of the following data set

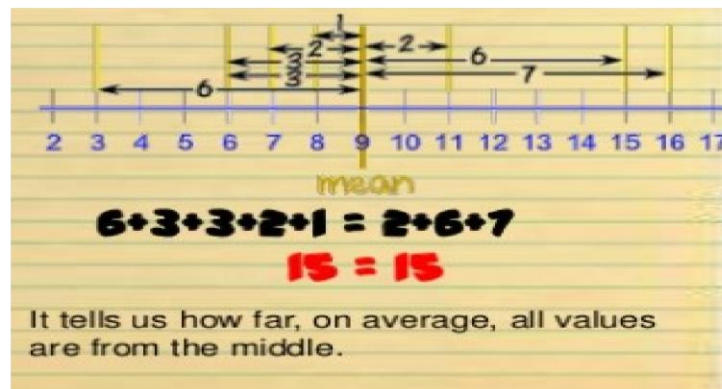
3, 6, 6, 7, 8, 11, 16, 15

Step 1: Calculate the mean score of the given dataset. (sum of values/number of values)

$$3+6+6+7+8+11+16+15 / 8 = 72/8 = 9$$

Step 2: Find distance of each value from the mean score.

Value of Dataset	Distance from Mean (i.e., 9)
3	6
6	3
6	3
7	2
8	1
11	2
15	6
16	7



Step 3: Find mean of these distances [Mean Deviation (MD)].

$$\begin{aligned} \text{MD} &= 6+3+3+2+1+2+6+7 \\ &= 30/8 = 3.75 \end{aligned}$$

So, mean of the given dataset is 9 and mean deviation is 3.75

iv. Variance

Variance measures the average squared deviation of each data point from the mean of the dataset. It provides a measure of the spread of data around the mean. However, since it involves squaring the deviations, the variance is not in the same units as the original data, making it less interpretable. Although, it can manually be calculated by using a formula, MS Excel and SPSS can help to compute the variance of a given dataset.

v. Standard deviation

Standard deviation is the square root of the variance. It measures the average deviation of data points from the mean score and is often preferred over variance because it is in the same units as the original data and is easier to interpret. Standard deviation was reported with the mean score. Although, it can manually be calculated by using a formula, MS Excel and SPSS can help to compute the variance of a given dataset.

SUMMARY

In this unit on Descriptive Statistics, we have gained the essential knowledge and skills in understanding and summarizing data. Descriptive statistics serves as a foundational tool in the field of statistics providing us with the means to explore and interpret datasets in a meaningful and informative manner. Throughout this unit, we have delved into the fundamental concepts of descriptive statistics including measures of central tendency, variability, and distribution. We have learned how to calculate and interpret summary measures such as the mean, median, and standard deviation.

By mastering these techniques, we have gained the ability to uncover patterns, trends, and relationships within datasets, enabling us to make informed decisions and draw meaningful conclusions. We have also developed critical thinking skills to evaluate summary statistics. Furthermore, we have recognized the limitations of various techniques of descriptive statistics. As we move forward, let us carry forward the knowledge and skills gained from this unit, recognizing the importance of descriptive statistics in various fields, including scientific research, business analytics, and social sciences. By applying descriptive statistics techniques effectively, we can continue to analyze and interpret data with confidence and communicate findings to stakeholders or audiences in a clear and concise manner.

EXERCISES

Multiple Choice Questions

1. Which measure of central tendency is most affected by extreme values?
 - a) Mean
 - b) Median
 - c) Mode
 - d) Range

Answer: a) Mean

2. What is the main characteristic of a normal distribution?
 - a) It is skewed to the right.
 - b) It is skewed to the left.
 - c) It is symmetric and bell-shaped.
 - d) It has multiple peaks.

Answer: c) It is symmetric and bell-shaped.

3. Which level of measurement involves categories that can be ranked?
 - a) Nominal
 - b) Ordinal
 - c) Interval
 - d) Ratio

Answer: b) Ordinal

4. The difference between the highest and lowest values in a dataset is called the:
 - a) Mean
 - b) Range
 - c) Quartile
 - d) Standard deviation

Answer: b) Range

5. Which percentile corresponds to the median?
 - a) 25th
 - b) 50th
 - c) 75th
 - d) 90th

Answer: b) 50th

6. What does a standard score (z-score) indicate?
 - a) The sum of all values in a dataset.
 - b) The number of standard deviations a value is from the mean.
 - c) The highest value in the dataset.
 - d) The central point of the dataset.

Answer: b) The number of standard deviations a value is from the mean.

7. In a skewed distribution, which measure of central tendency is the best indicator of the center?
- a) Mean
 - b) Median
 - c) Mode
 - d) Range

Answer: b) Median

8. Which measure of variation divides a dataset into four equal parts?
- a) Range
 - b) Standard deviation
 - c) Quartiles
 - d) Percentiles

Answer: c) Quartiles

9. A distribution of the data where most values are concentrated on one side is called:
- a) Symmetric
 - b) Skewed
 - c) Normal
 - d) Bimodal

Answer: b) Skewed

10. What is the purpose of a frequency table?
- a) To calculate the mean and median of data.
 - b) To display how often each value occurs in a dataset.
 - c) To identify the standard deviation of a dataset.
 - d) To compare two datasets.

Answer: b) To display how often each value occurs in a dataset

Short-Answer Questions

1. Define the term "standard deviation" and explain its importance.
2. What is the difference between a symmetric and a skewed distribution?
3. Provide an example of data that would be measured at the nominal level.
4. How do you calculate the range of a dataset?
5. Why might the median be a better measure of central tendency than the mean in some cases?

Subjective Questions

1. Describe the three measures of central tendency (mean, median, and mode) with examples. Explain situations where each measure is most appropriate.
2. Explain the characteristics of a normal distribution. Why is it important in statistics?
3. Discuss the steps involved in creating a frequency table with the help of an example.
4. What are quartiles, and how are they used to measure variation in a dataset? Provide an example.
5. Explain the calculation and interpretation of measures of variability, including the range, variance, and standard deviation.
6. What are the implications of descriptive statistics techniques to analyze and interpret data in various contexts, including scientific research, business analytics and social sciences?
7. Reflecting on your learning journey in this unit, what are some key insights or takeaways you've gained about descriptive statistics, and how do you plan to apply this knowledge in the future?

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Unit–9

USING TABLES AND GRAPHS FOR DATA

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INTRODUCTION

In data analysis, tables and graphs serve as powerful tools for organizing, summarizing, and visualizing data in a meaningful and interpretable way. They provide a clear and concise representation of complex datasets, enabling us to uncover patterns, trends, and relationships that may be hidden within the numbers. In this unit, we explore the fundamental concepts and techniques of creating and interpreting tables and graphs of data. We will begin by understanding the importance of data visualization and its role in enhancing our understanding of data. We will learn how to construct and interpret tables to organize and summarize data in a systematic manner. Building upon this foundation, we will delve into the world of graphical representations, exploring various types of graphs such as bar graphs, line graphs, and histograms. We will learn how to choose the appropriate type of graph to effectively communicate different types of data and relationships within the dataset. We will learn techniques for designing clear and informative tables and graphs, avoiding common pitfalls and misleading visualizations.

By the end of this unit, you will have developed a solid understanding of tables and graphs of data equipping you with the skills and knowledge necessary to analyze and interpret data effectively. Whether you are a student, researcher, or professional in any field, the ability to create and interpret tables and graphs will be invaluable for communicating findings, making informed decisions, and driving meaningful insights from data.

LEARNING OUTCOMES

After completing the unit, you would be able to:

1. Understand the importance of data visualization and the role of tables and graphs in organizing and summarizing data effectively.
2. Interpret tables to organize and present data in a clear and systematic manner.
3. Gain insight into the principles of graphical design, including clarity, accuracy, and honesty, to ensure that visual representations accurately convey the underlying data.
4. Understand the techniques for choosing the appropriate type of graph based on the characteristics of the data and the message to be communicated.

5. Develop critical thinking skills to evaluate the effectiveness of tables and graphs in conveying information and identifying potential biases or misleading visualizations.
6. Explore advanced techniques for enhancing the visual appeal and clarity of tables and graphs, such as labeling, scaling, and incorporating color and visual elements.
7. Apply tables and graphs effectively to analyze and interpret data, identify patterns and trends, and communicate findings to stakeholders or audiences in a clear and concise manner.
8. Develop a deeper appreciation for the power and versatility of tables and graphs as tools for data analysis and communication, and recognize their importance as essential skills for success in the modern data-driven world.

9.1 Tables and Graphs for Data

The source of the statistical knowledge lies in the data. Once we obtain the data values from the sample in a study, one way to become acquainted with them is to display them in tables or graphically. Tables, Charts and graphs are very important tools in statistics because they communicate information visually. These visual displays may reveal the patterns of behavior of the variables being studied. The most common graphical displays are the frequency table, pie chart, bar graph, Pareto chart, and histogram.

9.2 Tables

Tables play a fundamental role in statistics and data presentation by providing a concise and organized way to present information. One of the key advantages of using tables is their ability to summarize large datasets into a compact format, making it easier for researchers, analysts, and decision-makers to interpret the data. Tables allow for the systematic arrangement of data into rows and columns, enabling comparisons, trend identification, and pattern recognition. Whether it's presenting demographic information, experimental results, or financial data, tables offer a clear and structured way to convey complex information effectively.

Tables provide a visual representation of numerical data, enhancing understanding and facilitating analysis. By organizing data into rows and columns, tables allow for straightforward comparisons of different variables or categories. This comparative analysis is essential for identifying relationships, trends, and patterns within the dataset. Additionally, tables can accommodate a variety of statistical measures such as means, standard deviations, percentages, and correlations, providing valuable insights into the distribution and characteristics of the data. This structured format enables researchers to draw conclusions, make informed decisions, and communicate findings with clarity.

Tables offer flexibility in data presentation, allowing users to customize the layout and format according to their specific needs and preferences. Whether it's adjusting column widths, adding descriptive titles, or highlighting important data points, tables can be tailored to enhance readability and emphasize key findings. With the advent of digital technology, interactive tables enable users to sort, filter, and manipulate data dynamically, empowering them to explore the dataset in-depth and extract relevant information efficiently. Overall, tables serve as a versatile tool in statistics and data presentation, facilitating comprehension, analysis, and communication of complex datasets across various fields and disciplines. Look at the table given below with a table caption at the top, containing four examples of tables summarizing a large amount of information in a concise space.

Table 9.1: Examples of Tables for Data Presentation

Example 1

Marks obtained	No. of students
80 – 89	8
70 – 79	22
60 – 69	35
50 – 59	25
40 – 49	10
Total	100

Example 2

Marks Obtained	No. of students		Total
	Girls	Boys	
80 – 89	2	6	8
70 – 79	4	18	22
60 – 69	10	25	35
50 – 59	5	20	25
40 – 49	2	8	10
Total	23	77	100

Example 3

Marks Obtained	Girsl			Boys			Grand Total
	B	NB	T	B	NB	T	
80 – 89	0	2	2	5	1	6	8
70 – 79	2	2	4	10	8	18	22
60 – 69	6	4	10	10	15	25	35
50 – 59	4	1	5	8	12	20	25
40 – 49	2	0	2	6	2	8	10
Total	14	9	23	35	42	77	100

B = Boarders ; NB = Non-Boarders

Example 4

Table 8 Showing Average Scores on Tests of Mathematics of Urban and Rural Boys and Girls belonging to Different Socio-Economic Levels

	Sex	Socio-Economic Levels					
		Lower		Middle		Upper	
		Maths	Engl.	Maths	Engl.	Maths	Engl.
URBAN	B	120	110	150	140	135	145
	G	105	95	155	130	140	125
RURAL	B	100	80	140	120	125	115
	G	80	55	125	110	110	90

B = Boys

G = Girls

Important Labels for Graphs

Title/caption: The graph should have a title or caption (or both) that explains what is being shown and, if applicable, lists the source of the data.

Vertical scale and title: Numbers along the vertical axis should clearly indicate the scale. The numbers should line up with the tick marks—the marks along the axis that precisely locate the numerical values. Include a label that describes the variable shown on the vertical axis. **Horizontal scale and title:** The categories should be clearly indicated along the horizontal axis; tick marks are not necessary for qualitative data but should be used with quantitative data. Include a label that describes the variable that the categories represent.

Legend: If multiple data sets are displayed on a single graph, include a legend or key to identify the individual data sets.

9.3 Bar Graph

A graph of bars whose heights represent the frequencies (or relative frequencies) of respective categories is called a bar graph.

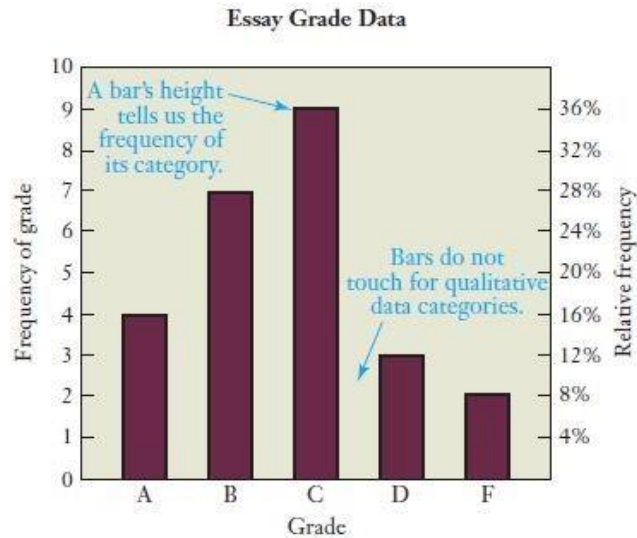


Figure 9.1. Bar Chart for Essay Grades

9.4 Pie Chart

Pie charts are used primarily for relative frequencies, because the total pie must always represent the total relative frequency of 100%. Figure 9.2 shows a pie chart for the essay grade data. The size of each wedge is proportional to the relative frequency of the category it represents.

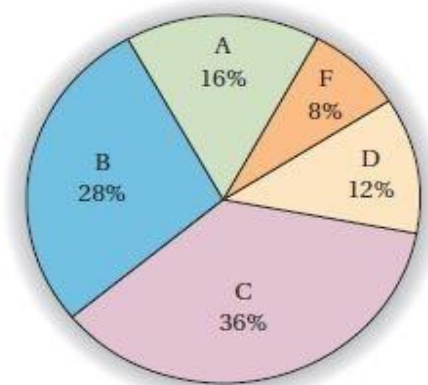
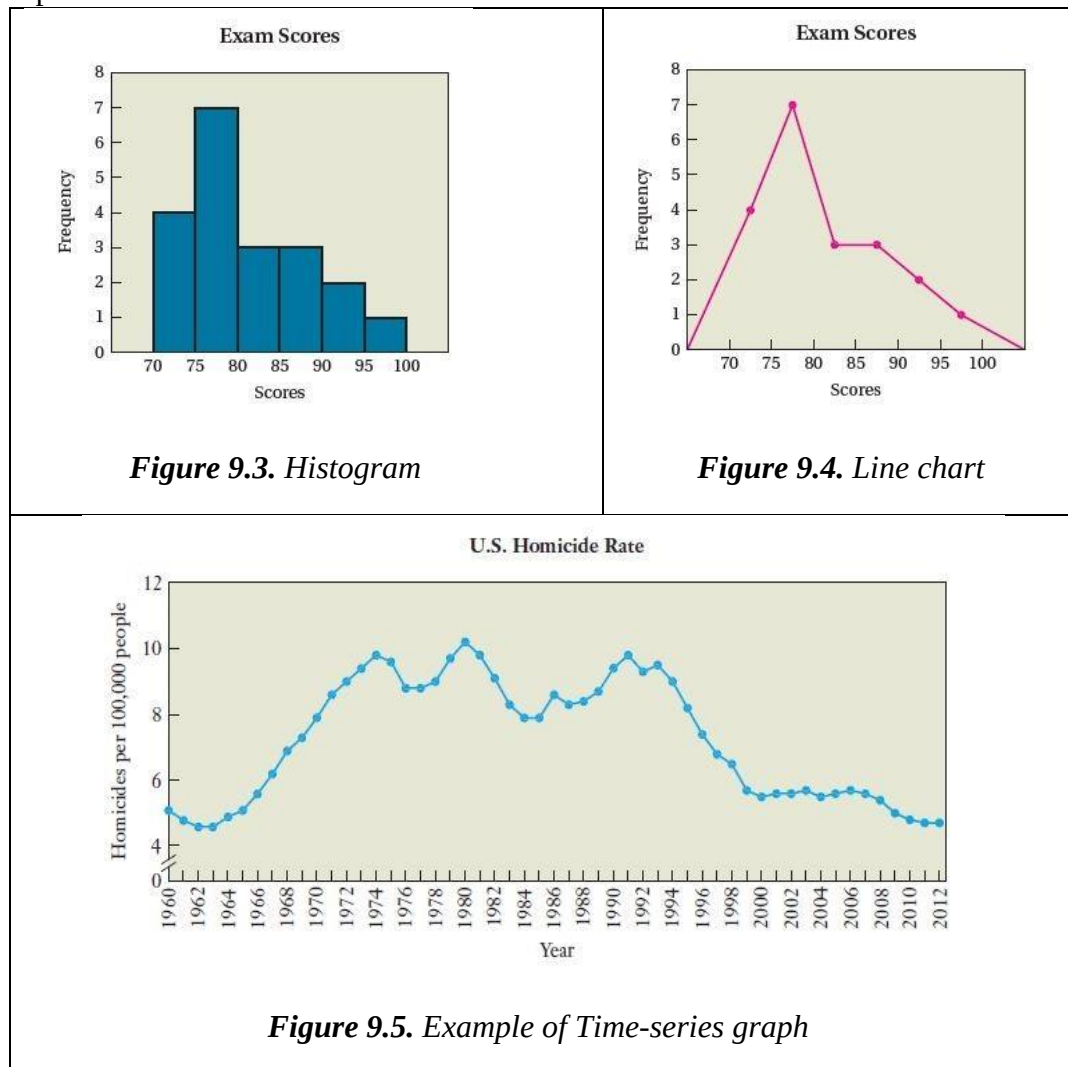


Figure 9.2. Pie Chart for Letter Grades

9.5 Histograms and Line Chart

A histogram is a bar graph for quantitative data categories. The bars have a natural order, and the bar widths have a specific meaning. A line chart shows the data value for each category as a dot, and the dots are connected with lines. For each dot, the horizontal position is the center of the bin it represents and the vertical position is the data value for the bin.

A time-series graph is a line chart or histogram in which the horizontal axis represents time.

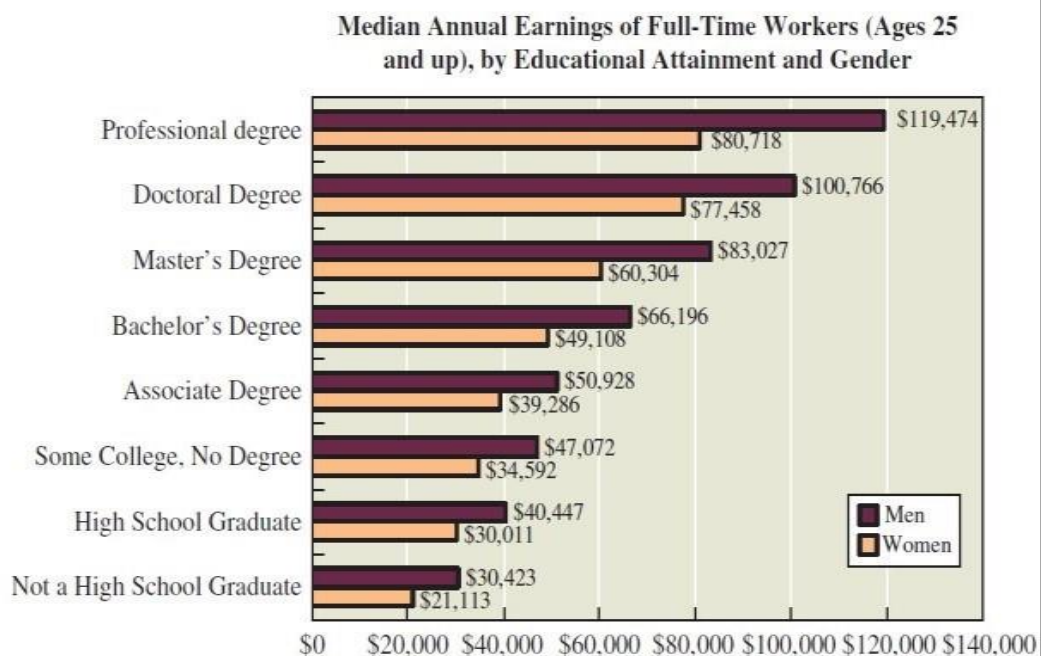


9.6 Multiple Bar Graphs and Line Charts

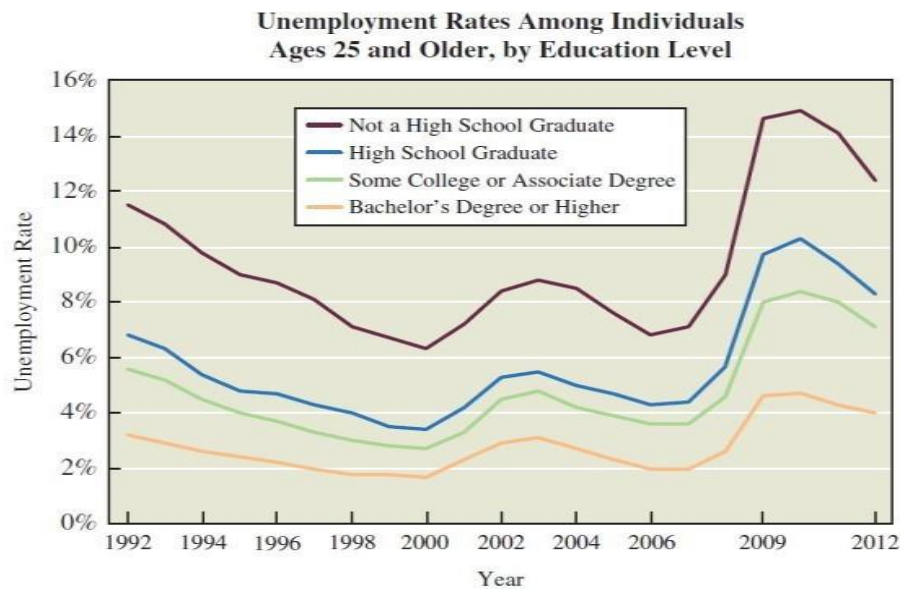
A multiple bar graph is a simple extension of a regular bar graph. It has two or more sets of bars that allow comparison of two or more data sets. All the data sets must have the same categories so they can be displayed on the same graph. The example given below is a multiple bar graph with two sets of bars, one for men and one for women.

The data categories (levels of educational attainment by gender) are qualitative, which makes a bar chart the best choice for display. In cases in which data categories are quantitative, a multiple line chart is often a better choice. The example given below shows time-series data using four different lines for four different data sets, each based on a different level of educational attainment. The data are quantitative in this case because the categories (on the horizontal axis) are years and the data values are unemployment rates, both of which are measured quantitatively.

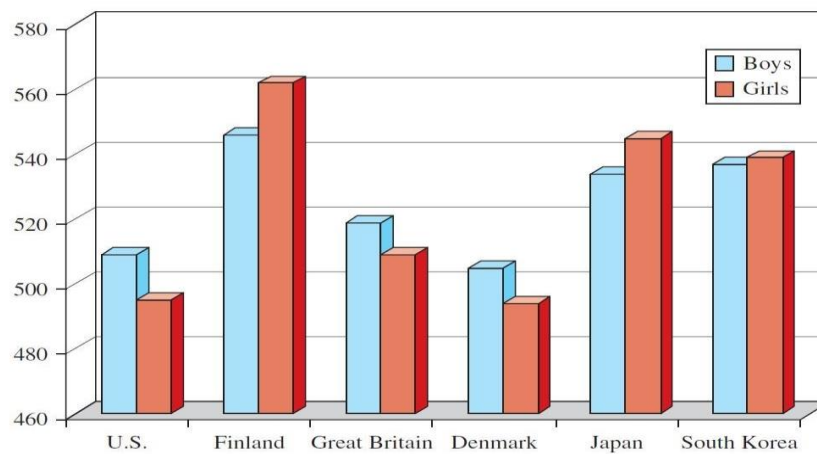
Example of Multiple Bar Chart



Example of Multiple Line Chart



Example of Multiple Bar chart



9.7 Scatter Plot

A scatter plot (aka scatter chart, scatter graph) uses dots to represent values for two different numeric variables. The position of each dot on the horizontal and vertical axis indicates values for an individual data point. Scatter plots are used to observe relationships between variables.

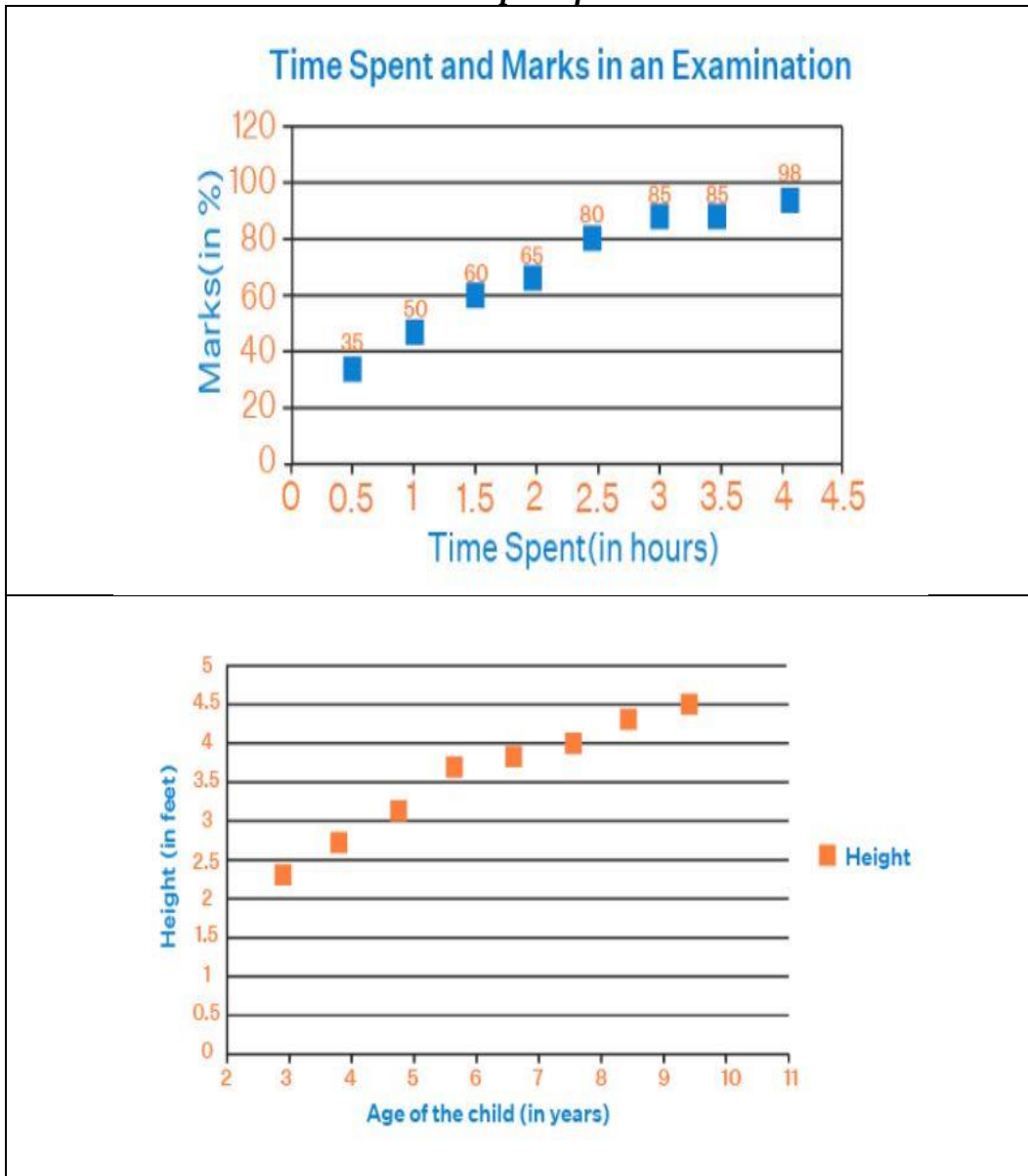
There are three simple steps to plot a scatter plot.

Step-1: Identify the x-axis and y-axis for the scatter plot.

Step-2: Define the scale for each of the axes.

Step-3: Plot the points based on their values.

Table 9.2: Examples of Scatter Plots



9.8 Stack Plots

Another way to show two or more related data sets simultaneously is with a stack plot, which shows different data sets stacked upon one another. Data can be stacked in both bar charts and line charts. Figure 5.14 shows a stack plot using stacked bars laid out horizontally. Each bar is divided into sections, which are color-coded according to the legend at the top. For example, the top bar shows that the total budget for the average commuter student at a two-year public college is \$15,584 and the light blue segment in that bar shows that this average student spends \$7,419 on room and board. Stack plots can also be made with stacked lines, which are particularly useful for showing trends over time.

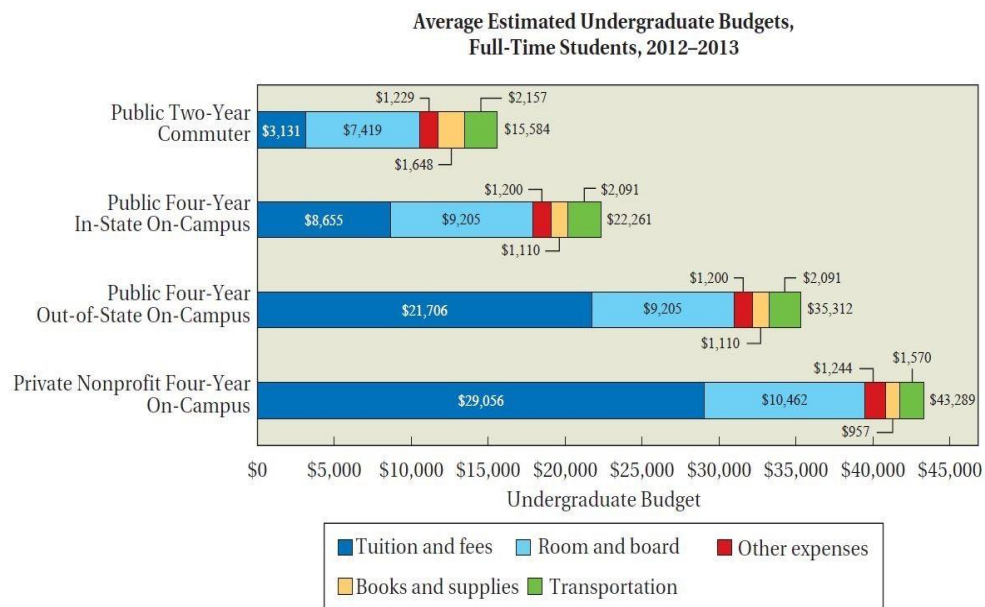


Figure 9.6. Example of Stack Plot

9.9 Graph for Geographical data

We are often interested in geographical patterns in data. The figure given below shows one common way of displaying geographical data. In this case, the map shows trends in energy use per capita (per person) in different states. The display in the Figure given below works well because each state is associated with a unique energy usage per person. For data that vary continuously across geographical areas, a contour map is more convenient.

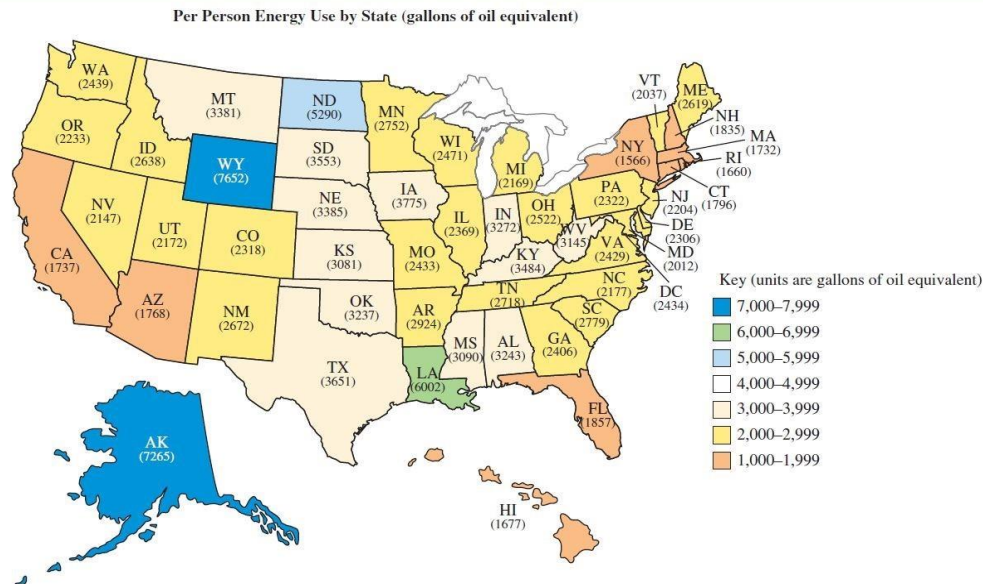


Figure 9.7. Example of Graph for Geographical Data

9.10 Three-Dimensional Graph

Nowadays, computer software makes it easy to give almost any graph a three-dimensional appearance. Its three-dimensional effects are purely cosmetic. It carries distinct information along its three axes, making it a true three-dimensional graph. The advantage of three-dimensional graphics is that they allow us to show richer data sets. The drawback is that it can be difficult to read the data precisely. Three-dimensional graphics therefore tend to work best when they are interactive (online), so that figures can be rotated or viewed from different perspectives.

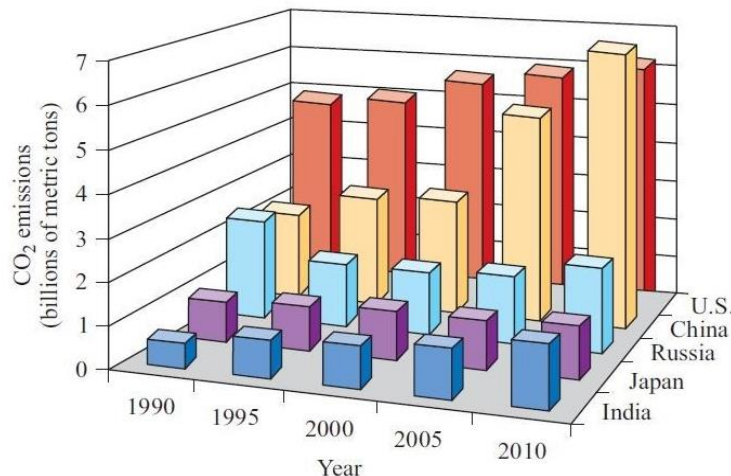
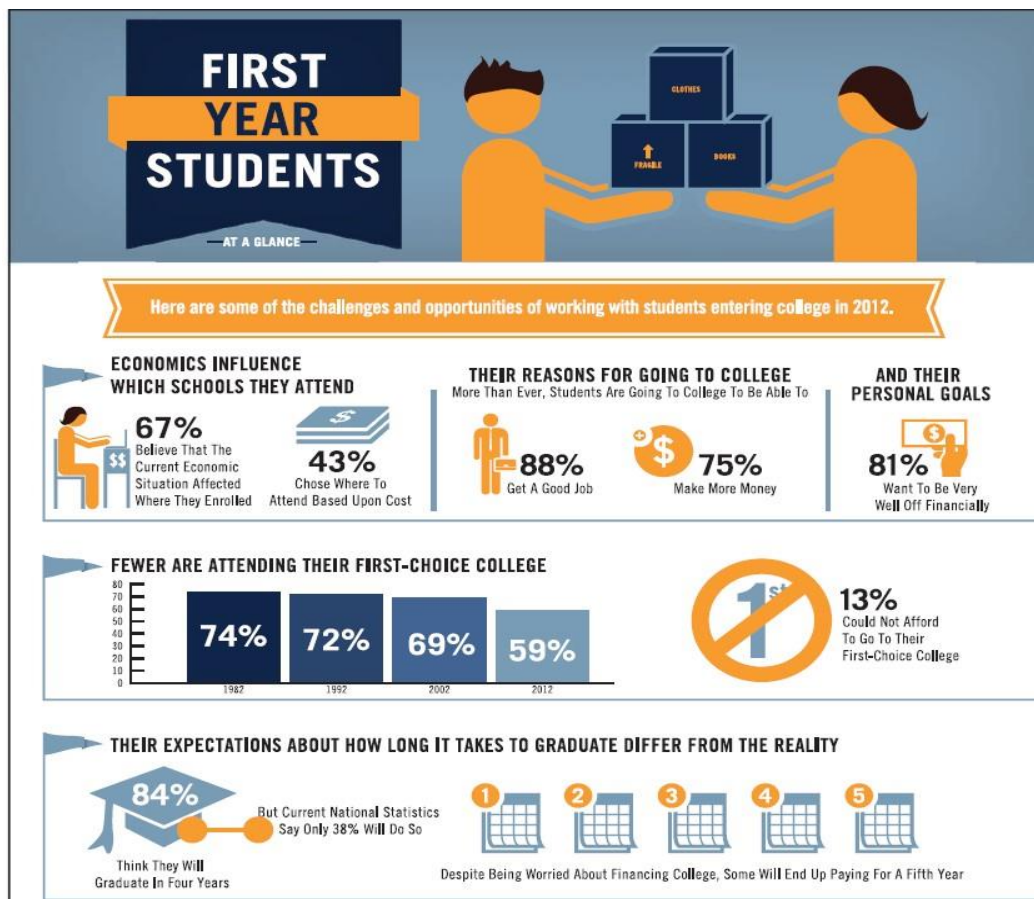


Figure 9.8. Example of 3-Dimensional Graph

9.11 Infographics

All of the graphic types, we studied so far, are common and fairly easy to create. However, the availability of sophisticated software has made more complex graphics increasingly common. These graphics come in many types, with one of the most general types being what has come to be called “information graphics,” or infographics for short. The goal of an infographic is to present a large, interrelated set of information in a visual way that can be interpreted clearly and easily.

The figure given below shows an infographic summarizing the characteristics of first-year college students. Notice that it contains a combination of numerical data values (such as the percentages in the top row), two bar graphs (in the second and third rows), a line chart (in the bottom row), and words and icons designed to help explain what is going on. It is certainly a case of a picture being worth far more than a thousand words. Many online infographics go even further, adding interactive or animated features that add to their wealth of information.



9.12 Pictographs

Pictographs are graphs embellished with additional artwork. The artwork may make the graph more appealing, but it can also distract or mislead. The figure given below is a pictograph showing the rise in world population from 1804 to 2040 (numbers for future years are based on United Nations intermediate-case projections). The lengths of the bars correspond correctly to the world population for the different years listed. However, the artistic embellishments of this graph are deceptive in several ways. For example, your eye may be drawn to the figures of people lining the globe. Because this line of people rises from the left side of the pictograph to the center and then falls, it might give the impression that future world population will be declining. In fact, the line of people is purely decorative and carries no information. The more serious problem with this pictograph is that it makes it appear that the world population has been rising linearly. However, notice that the time intervals on the horizontal axis are not the same in each case. For example, the interval between the bars for 1 billion and 2 billion people is 123 years (from 1804 to 1927), but the interval between the bars for 5 billion and 6 billion people is only 12 years (from 1987 to 1999). Pictographs are very common, but as this example shows, you have to study them carefully to extract the essential information and not be distracted by the cosmetic effects.

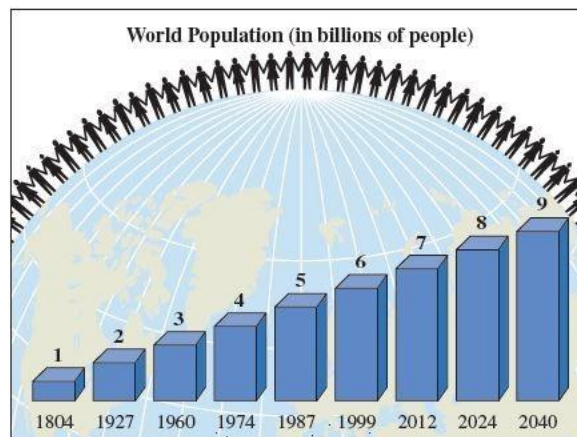


Figure 9.9. Example of a Pictograph

9.13 Guidelines for Graphics

Take care of the following points while looking at and interpreting the graphs.

1. Many graphics are drawn in a way that distorts our perception of them.
2. Take care of the scale and the distribution.
3. Notice the values given at the vertical and horizontal axis.

SUMMARY

Tables and graphs serve as powerful tools for conveying complex information in a clear and concise manner, enabling us to uncover patterns, trends, and relationships within datasets. Throughout this unit, we have explored various types of graphs, including bar graphs, line graphs and histograms to visually represent different types of data and relationships. Moreover, we have gained insight into the principles of graphical design, emphasizing clarity, accuracy, and honesty in data representation. By applying these principles, we can create visualizations that accurately convey the underlying data and enhance our understanding of the information presented. Furthermore, we have recognized the importance of critical thinking in evaluating the effectiveness of tables and graphs in conveying information. By developing the ability to assess visualizations for biases or misleading representations, we can ensure that our analyses are based on reliable and trustworthy data. Whether in academic research, business analytics, or public policy, the ability to create and interpret tables and graphs will be invaluable for making informed decisions and driving meaningful insights from data.

EXERCISES

1. Which type of graph is best suited for showing the proportion of categories in a dataset?
- a) Bar graph
 - b) Pie chart
 - c) Line chart
 - d) Histogram

Answer: b) Pie chart

2. A histogram differs from a bar graph because:
- a) Histograms use percentages instead of counts.
 - b) Histograms represent continuous data, and bar graphs represent categorical data.
 - c) Histograms use vertical bars, while bar graphs use horizontal bars.
 - d) Bar graphs do not show any data distribution.

Answer: b) Histograms represent continuous data, and bar graphs represent categorical data.

3. What does a scatter plot display?
- a) Relationships between two variables.
 - b) Trends over time.
 - c) Proportions of a whole.
 - d) Comparisons between categories.

Answer: a) Relationships between two variables.

4. What is the primary purpose of a line chart?
- a) To display frequency distribution.
 - b) To represent parts of a whole.
 - c) To show trends over time.
 - d) To compare multiple datasets.

Answer: c) To show trends over time.

5. Which type of chart is commonly used to display geographical data?
- a) Pie chart
 - b) Line chart
 - c) Map-based graph
 - d) Stack plot

Answer: c) Map-based graph

6. What does a stack plot visually represent?
- a) Changes in individual variables over time.
 - b) Accumulation of data across categories over time.
 - c) Frequency distribution.
 - d) Correlation between variables.
- Answer:** b) Accumulation of data across categories over time
7. What is the main advantage of using a pictograph?
- a) It displays data in a three-dimensional format.
 - b) It uses symbols to represent quantities, making data visually appealing and easy to interpret.
 - c) It shows trends over time.
 - d) It is suitable for precise numerical analysis.
- Answer:** b) It uses symbols to represent quantities, making data visually appealing and easy to interpret.
8. Which of the following graphs is best for visualizing relationships between three variables?
- a) Pie chart
 - b) Three-dimensional graph
 - c) Scatter plot
 - d) Line chart
- Answer:** b) Three-dimensional graph
9. In an infographic, the key purpose is to:
- a) Represent data visually using minimal text.
 - b) Display data using only pie charts.
 - c) Analyze trends over time.
 - d) Perform detailed numerical calculations.
- Answer:** a) Represent data visually using minimal text
10. Which guideline is most important when designing graphics for data?
- a) Use as many colors as possible.
 - b) Ensure simplicity and clarity to avoid confusion.
 - c) Avoid labels and legends.
 - d) Use only one type of graph per presentation.
- Answer:** b) Ensure simplicity and clarity to avoid confusion.

Short-Answer Questions

1. Define a bar graph and give an example of when to use it.
2. What is the difference between a pie chart and a stack plot?
3. Explain the importance of guidelines in designing effective graphics.
4. What is a scatter plot, and how does it show relationships between variables?
5. Why are infographics increasingly popular for presenting data?

Subjective Questions

1. Explain the importance of data visualization and the role of tables and graphs in summarizing and presenting data effectively.
2. What are the various ways of organizing the data in a systematic manner and presenting it clearly and concisely?
3. What are the advantages of creating various types of graphs such as bar graphs, line graphs, and histograms, to visually represent different types of data and relationships within the dataset?
4. What are the principles of graphical design, including clarity, accuracy, and honesty? How it can be applied to create visualizations that accurately convey the underlying data?
5. Describe the procedure of choosing the appropriate type of graph based on the characteristics of the data and the message to be communicated.
6. Reflecting on your learning journey in this unit, what are some key insights or takeaways you have gained about tables and graphs of data, and how do you plan to apply this knowledge in the future?

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